

NOT A POTATO

THE QUIET MEN



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Chapter 1 — Wrong, or Fraud, and That's the End of It

The woman on the podcast had a voice like a bath you didn't want to get out of, and she was using it to bury a man.

“—because here's the thing about the two-hundred-mile-per-gallon carburetor,” she said, warm, unhurried, a little amused, the way you'd tell a friend a story you both already half knew the end of. “It's a *great* story. A broke mechanic in Depression-era Canada builds a magic box, drives a Ford clear across the prairie on a thimble of gas, and then — poof. The oil companies get to him. The patents vanish into a vault. The man dies in obscurity, selling, of all things, oil filters.” A pause, perfectly weighted. “And it's all true. Except the part where it isn't. Which, spoiler, is most of it.”

Rain came down on the Beast in handfuls. They were parked on the concrete apron of a service station an hour and a half out from the city, the only vehicle under the canopy, the pumps to their left lit a hard sodium yellow and going to streaks on the glass. The heater laboured. It was June, and it had no business being this cold in June, and the prairie out past the canopy lights had gone the colour of a wet road in every direction, which was to say no colour, which was to say it had simply quit and left the rain to it.

Naila had her knees up on the seat and the phone propped on the dash and a service-station coffee going cold in the cupholder because she always forgot the coffee once a voice got interesting, and this voice was interesting. It was doing the thing well. It wasn't

sneering. That was the part people got wrong about debunking — they thought it was a sneer, and the good ones never sneered, the good ones loved the story first, out loud, so that when they took it apart you felt the loss of it and trusted them for being sorry.

“So let’s do the physics,” the voice said, “because the physics is short, and the physics is brutal, and I promise it doesn’t hurt.” Naila felt herself nod at the dark glass. “A gallon of gasoline holds a fixed amount of energy. Fixed. About thirty-three and a half kilowatt-hours, and not one watt-hour more, no matter what you bolt onto your engine. A carburetor doesn’t *make* energy. It mixes air and fuel and hands the mix to the cylinder. That’s its entire job. It is a bartender. And you cannot get a bartender to pour you more whisky than is in the bottle, no matter how nicely you’ve decorated the bar.”

“She’s good,” Yusuf said, from the bench seat behind. He said it the way one tradesman watches another work. He’d have his feet up if there’d been anywhere to put them.

“She’s right,” Naila said.

“That too.”

“—so two hundred miles a gallon out of a 1933 V8,” the voice went on, “would mean getting *more* energy out of the fuel than the fuel contains. Which is the universe’s one unbreakable house rule. You don’t get out more than you put in. Ever. The carburetor can stop you wasting fuel — it cannot conjure it. And here’s my favourite part, the part the documentaries leave out because it ruins the ghost story—” the smile audible now, the kindly knife “—*Pogue himself said so*. Nineteen fifty-three. A reporter asks him straight: two hundred miles a gallon, true or false? And Charles Pogue says no. Says he never claimed two hundred. Says he never claimed *half* that. The man at the centre of the world’s most famous fuel miracle told a journalist, on the record, that the miracle was newspaper talk.” A beat. “Wrong, or fraud. That’s the whole menu. And honestly? On the machine, it’s not even a close call.”

Tomi made a small wounded noise from the very back, where he’d folded himself in among the cases with a soldering iron he wasn’t using and a circuit board he wasn’t fixing, just holding, the way some people held a worry stone. “She doesn’t have to say it like *that*.”

“Like what,” Naila said.

“Like it’s settled.”

“It is settled. About the machine.” She picked up the cold coffee, found it cold, put it down. “She’s not wrong, Tomi. That’s the annoying thing about her. There’s no seam in it. A gallon is thirty-three point four kilowatt-hours whether you’re a fraud or a saint. You can’t mix your way to two hundred. You can’t mix your way to a hundred. You can’t mix your way to *doubling* it. The combustion’s the ceiling and the ceiling doesn’t negotiate.” She heard her own voice doing the flat thing it did, laying it down clean, and didn’t soften it, because softening it would have been the lie. “If a man tells you he built a carburetor that does two hundred, he is telling you he built a thing that breaks the one rule that doesn’t break. So no. The machine can’t do what they said it did. Full stop. End of that sentence.”

Rain. The heater. Somewhere under the apron a drain was working hard and losing.

“That’s a funny way to put it,” Edith said.

She was up front, passenger side, and she hadn’t said anything in twenty minutes, which from Edith was conversation. She had the window cracked an inch against the fug — she always wanted a way for the air to leave a room — and the rain came in cold off the gap and she didn’t close it. Her camera sat in her lap with the cap on. She’d taken one frame since they’d parked, of the pumps gone to yellow streaks, and then put it down, and Naila had clocked it and said nothing, because Edith putting the camera down was Edith’s whole language and you didn’t translate a person’s language back at them.

“What is,” Naila said.

“*End of that sentence.*” Edith turned her head. In the sodium light half her face was the colour of the pumps and half was the dark. “Like there’s another one.”

And there it was — the thing Naila had been circling since the drive-through last night, since she’d stood in the cold pulling the patents up on her phone one after another, four of them, real, stamped, free to anyone, while the order-board crackled and a teenager asked twice if she wanted the combo. She’d read them in the dark with the engine

running and felt the shape of it bother her, the way a column of figures bothers you when the figures are all correct and the *sum* is still wrong.

"There's another one," Naila said slowly, because she was working it out in her own mouth, which was the only place she could ever work anything out. "She's done one sentence perfectly. *The machine can't do what they said*. True. Beautiful. I'd sign it. I'd notarise it." She watched the rain. "And then she put a full stop after it and called it two sentences. *Wrong, or fraud*. But that's—" her hands moved, found the edges of it "—that's not the same sentence. 'The machine can't do what they said it did' and 'the man was a fraud' — those are different sentences. They feel like one. She *needs* them to feel like one, it's tidier, you can close the file. But you can build a thing that doesn't do the impossible and still not be lying about it. You'd just be — wrong, about how good it was. Or right about how good it actually was, and have a press that wasn't listening."

"Hm," said Yusuf, and it was the *hm* of a man who has run enough cons to know exactly where the seam in one is, and has just been shown a place where there might not be a con at all. "Say the second one again."

"Right about how good it actually was," Naila said, "and a press that turned it into a thing it never was, and wouldn't take the correction." She frowned. "There's a real version of this rig. A vapour carburetor that warms the fuel off the engine's own exhaust — that's not fairy dust, that's *thrift*, that's a man clawing back the gasoline a 1933 engine was throwing up the pipe half-burned. On a car that thirsty? It could nudge the thing. High twenties, maybe. Not two hundred. But high twenties out of low teens is real, and it's worth being proud of, and—"

On the phone, the beautiful voice was wrapping up, lowering, doing the *you decide* turn that all of them did at the end because it was the only honest thing they had left after an hour of being certain. "So that's the two-hundred-mile-per-gallon carburetor," she said. "No cover-up. No vault. No murdered genius. Just physics, doing what physics does, which is not care what we want. The man had real patents — and the machine in them was real, and modest, and never came close to the legend. Wrong, or fraud, and that's the end of it." The pause. The wink you could hear. "Unless you know something I don't. *You decide*."

The episode ended. An advert started, a mattress, aggressively cheerful. Naila killed it.

Quiet, then. The heater. The drain. Frik, who had been so still in the corner of the back seat that you could forget he was there, which Frik would tell you was the entire problem with Frik in a held car, said it to the streaming window, not loud:

“Stopped.”

A breath. His shoulder rolled, the small managed thing.

“She stopped.”

His hand came up by reflex, the old half-apology, and went down again, because nobody in this van had ever once needed him to apologise for the thing his mouth did before the rest of him caught up.

“Frik’s right,” Naila said, and turned it over, because Frik was always right and only ever early, and the trick was hearing what he’d heard a half-second before you’d have heard it yourself. “She stopped. She did the machine — perfectly — and the *second* the machine ran out, she stopped. *Wrong or fraud, end of it.*” She looked at the dead phone. “But that’s exactly where it gets interesting. She closed the file at the machine. And the man’s still in the room.”

“How d’you mean,” Tomi said, leaning forward now between the seats, too fast, the way he did when something elegant might be about to happen, already three steps down a road nobody’d built yet.

“I mean—” Naila pressed the heel of her hand to the bridge of her nose, where the cold had got in, and made herself slow. “Fifty-three. The reporter asks him. Two hundred — true or false. And Pogue says no. He says, I never claimed two hundred. He says, I never claimed half that.” She let it sit. “Good. Honest. The man refuses the lie the world hung on him. So the reporter asks him the obvious next thing, the *only* next thing — *all right then, what did it really do?* He’s sitting right there. It was his rig. He built it with his hands, he ran it, he knows the number better than anyone alive will ever know it.” She stopped. “And he won’t say.”

The rain.

“He won’t say,” she said again. “Not won’t *remember* — won’t say. He had it. The one true modest number, the thing he could’ve handed

over in a single breath and been believed and been *done* — and he took it down into the quiet on purpose and made oil filters for the rest of his life and died with it still in his mouth.” She turned in the seat. “She debunked the two hundred. Fine. The two hundred was always going to fall over, it breaks the house rule. But she stopped *exactly* where the man got strange. The hole isn’t the machine. We *know* the machine. The hole is — there’s a man who knew a number that would’ve cost him nothing to say, and he’d rather die than say it, and nobody who’s ever told this story has once asked him *why*.”

Nobody said anything. The heater ran. Out past the canopy a transport went by on the highway with its lights smeared and its spray standing off it like a wake, and was gone, and the prairie took the dark back.

Jakobus had not said a word.

He was in the driver’s seat, which was the only seat that fit him, both hands off the wheel and resting on his thighs, the shades he’d had no use for in this weather pushed up into his hair, watching the rain come down on the glass the way another man might watch a fire. He was a very long way from anywhere his hands knew the shape of, and he wore it the way he wore everything, without a word of complaint about it, the cold of a foreign June soaking up out of the floor of his own truck into his own boots. There was a square canvas water bag lashed to the wing mirror outside, weeping its slow film down the door even now, in the rain, out of pure habit, looking like neglect and being the opposite — and it was the only thing on the whole vehicle that behaved exactly as it would have on a different continent, under a sun none of them could feel from here.

He didn’t look at any of them. He kept his eyes on the rain, and when he spoke it was low and flat and not to the question Naila had asked but to one underneath it, in the plain trade-shop English he found for rooms like this.

“A man builds a good thing,” he said, “and the world tells him it’s a different thing. Bigger. A miracle.” A pause, the kind he spent like a coin down to his last. “And then it can’t be that, because nothing’s that. So they call him a liar.” He turned the water off in his own thought a moment, then back. “Some men, you take their true thing and dress it up in a thing it can’t be, and the dressing-up is what kills

it — and after a while they'd sooner say nothing than hear you make their good thing into your big one." He moved his hand, just once, to the key, and didn't turn it. "I'd want to know what he made. Not the miracle they hung on it. The thing he actually built in a cold shop with his hands. And then I'd want to know who he stopped talking to."

He let that be all of it. He was not, Naila had learned, ever asking *is it real* — the man could look at a made thing and skip clean past *is it real* to the only question that ever moved him, *what is it for*, the way some people skip the small talk because they've buried too many people to spend the time. He was asking it now of a dead mechanic he'd never meet, of a box he'd never get to put his hands on, and the asking sat in the cab with the rain and the heater and didn't need an answer yet.

"So we go and find his keeper," Yusuf said, into the quiet, because somebody had to be the one who turned a feeling into a plan, and it was always him. "Whoever's got the rest of him. The part that isn't on your phone."

"There's a museum," Naila said. "An hour up. A woman keeps it." She'd found her in the dark too, last night, the way she found everything. "She has the file. I don't think she gives it out."

"They never do," Edith said quietly, "when they've watched it get used." She closed the window the last inch at last, and the cold stopped coming in, and the glass began to fog where she'd breathed on it.

Tomi was already somewhere up ahead of them all, vibrating, the soldering iron forgotten in his fist. "If he had the exhaust-jacket *and* the atomising, the real architecture, you could actually—"

"Tomi."

"—I'm just saying we could rebuild the *real* one, the modest one, and just *measure* it, and then we'd have his number, the one he wouldn't—"

"That's not the number," Naila said, gentle, and put it down between them the way you put a flat hand on a dog's chest, kindly, before it bolts. "His number isn't a measurement we can take. It's a thing he *chose* not to say. Even if we built it and measured it perfectly, we'd

have a number. We still wouldn't have *why he wouldn't*." She looked back out at the rain. "That's the one we're going for."

Jakobus turned the key. The big diesel came up out of sleep all at once, that low even thunder it made, blueprinted and built past sense and absurd under a service-station canopy in the Canadian wet, and the cab filled with the good honest racket of it, and the water bag on the mirror shivered and went on weeping.

"And the rest of them?" Tomi said, because Tomi could never not reach for the rest of them. "After the carburetor. There's the — the vacuum thing, the energy that isn't anything. And the Russian. The one with the mirror."

"The astronomer," Naila said. She'd read about the astronomer too, last night, deeper than she'd meant to, the way you fall down a stair you only meant to look at. A man the camps couldn't kill. A gold medal. A crater on the Moon with his name on it. And a coil of polished aluminium standing in a cold room in Siberia, named for him, that he'd had no hand in building at all — a question shaped like a mirror, with a man's whole grief folded up inside it. "Yeah. There's the astronomer."

"Is the mirror real?"

"The man is real," Naila said, which was not an answer to what Tomi had asked and was the only true answer she had, and she heard Friko go very still in his corner, the way he went still when a story was also, somehow, his.

Jakobus put the Beast in gear. She came off the apron heavy and sure, out from under the lights and into the rain proper, and the wipers took it in slow patient sweeps, and the prairie opened up dark and flat and enormous on every side, going nowhere, going on.

It was an hour to the museum. After that, if the woman gave them the man and not just the machine, it was a long way further to everything else — to a basement where they'd weighed the emptiness of space and found it pushing back, to a stolen word for sale in a tent, to a magnet that was only ever a belief. And further still, past all of it, at the cold top of the map, the institute waited with the heating off and the medal in its box and the spiral standing in the dark like something asked and never answered.

It was a long way north.

Chapter 2 — The Lock-Up in Winnipeg

The four patents Naila already had. Anyone had them — one million, seven hundred fifty thousand, three hundred fifty-four, and three more after it, scanned and sleeved and free to the world since the day the office stamped them, pullable on a phone in a drive-through, which she had done the night before. Public record was the cheapest thing in this story. What Margaret Dolch had, and would not give them, was the rest of him.

The museum sat on the main street of a prairie town an hour out from the city, between a credit union and a shut bakery, the kind of museum a town keeps the way a family keeps a drawer: a threshing photograph, a switchboard, a cabinet of arrowheads labelled in a hand dead for forty years. The reading room was at the back, past the furnace. Margaret had let them as far as the table. She had not let them past it.

“You want the carburetor,” she said. She said *carburetor* the old way, four syllables, the way a town that had built things said it. “Everybody wants the carburetor.”

“We want the man,” Naila said.

It was the wrong thing to say, and she heard it land wrong. Margaret Dolch was somewhere past seventy, in a quilted vest and reading glasses on a chain, with a face that had decided things and would decide more, and the thing it decided now was *no*. Not a door slammed. A door held at exactly the width it had been at.

“Everybody wants the man, too,” Margaret said. “I’ve had the man

wanted at me for thirty years.”

She did not sit. The table between them was Formica, the speckled grey-and-gold of a church-hall fold-out, and on it she had set, with no ceremony, the four patents in their plastic sleeves and a clothbound museum register turned to a page of donor names. The published things. The shop window. Behind her, against the cinderblock, a beige microfiche reader the size of a small television held its breath, and beyond that a roller door she had not opened and a padlock she had not touched.

Naila read the room the way she read everything, all at once and then in pieces. She let her eye go down the donor register’s right-hand column, the dates, and the dates were a shape, and the shape had a wrong note in it the way a column has a wrong note when one row went in in a different pen. She filed it. Not now.

Jakobus was by the furnace, out of the light, where he put himself in rooms like this, which was every room. The shades he’d taken off against the prairie glare hung from his collar. He’d said three words to Margaret at the door — *good of you* — and then her name, the way the town said its own names, and then gone to the wall and become a part of it. He was a long way from anywhere he knew, and he let that show the way a man lets weather show: without comment.

“You drove up from where,” Margaret said. It was not curiosity. It was the start of the test.

“Down,” Yusuf said. “We came down. Long story involving a shipping container and a customs broker in Thunder Bay who I now consider family.” He had the warm easy front he put on the world, the front that had once moved money it shouldn’t have, and Margaret looked at it for a second longer than the front liked and then looked away from it, unbought.

“And you do what, exactly,” she said. “You make a programme.”

“We tell stories about things that didn’t get told straight,” Yusuf said.

“That’s what they all say.” She folded her arms. “I had a fellow up here in—” she paused, and the pause was her sorting which year out of several “—the nineties. Polite. Tape recorder. Bought a coffee at the legion. And do you know what he made of three weeks in this room? *The Secret the Oil Barons Buried*. Front cover. My grandfather—” she

stopped on the word, reset it, the way you reset a thing you've decided not to spend in front of strangers "—a man who fixed combines so the McKenzies could get the crop in, with a *cape* on him, near enough. Suppressed genius. Murdered by Standard Oil, the implication, never quite said so they couldn't be sued for it. They sold a great many copies." She let that sit. "He was a quiet man who made a good thing and got told a lie about it until he stopped talking. That's the story. It doesn't sell."

Naila felt the sentence go into her like a clean number goes in. *Until he stopped talking*. She held very still around it.

"The patents are real," she said. "I'm not here about the cape."

Margaret looked at her properly for the first time.

"Go on," she said.

So Naila told her the true thing, because the true thing was the only currency she trusted and she was bad at the other kind. She put her finger on the second sleeve. "Exhaust-heated vaporiser. The fuel doesn't go in as a spray, it goes in as vapour — he runs the air up through the liquid and warms the chamber off the engine's own exhaust, so it comes to the cylinder already gas, already wanting to burn. That's good engineering. On a 1933 V8 that was drinking like a barn fire, a rig like that claws back fuel that was going up the pipe unburned. It's real. It nudges the thing. High twenties, maybe. From the low teens that car was doing, high twenties is a small miracle a sober person can stand behind."

"And the two hundred," Margaret said. Flat. Testing.

"Is impossible." Naila said it the way she'd say a kettle was at a hundred degrees. No cruelty in it; physics had no opinion of Charles Pogue. "A gallon of gasoline holds about thirty-three and a half kilowatt-hours. Fixed. It's chemistry, it doesn't care about your nozzle. A carburetor only mixes — it can't add what isn't in the can. You can stop wasting it. You can't conjure it. Two hundred miles on a gallon out of that engine would be getting more energy out than the fuel contains, and the universe declines. So no. Not two hundred. Not, as he himself said, even half that."

Something moved in Margaret's face. It was not agreement. It was recognition — the particular ease of a person who has spent thirty

years braced against the cape, hearing someone refuse the cape unasked.

“You know about fifty-three,” she said.

“He denied it in fifty-three,” Naila said. “Two hundred, or even half that, he said. And then—”

“And then nothing.” Margaret’s mouth went thin. “That’s the part they cut. He didn’t replace it with a better number. A reporter sat where you’re sitting — well. Where the *table* is. And asked him what it really did, and he had it. He knew. It was *his*. And he wouldn’t say.” She picked up the church-hall register and squared its edge against the table, twice, a tidying that wasn’t about the register. “You want to know what it did. The real figure. So do I. So does everyone who ever came up that road with a coffee. And the one man on this earth who could have told us took it down the way you take a thing you’ve decided is yours.”

The furnace ticked. Frik, who had been quiet by the cabinet of arrowheads, doing the thing he did in held rooms — a slight roll of the shoulder, a breath managed — said, not loudly, to the glass case in front of him: “Liar.” A beat. His jaw worked. “*Pepper*.” And then his hand came up, the old reflex, half-apology, and Naila saw Margaret’s head come round, sharp, ready to take offence on the dead man’s behalf —

“The editor,” Naila said, fast and even, laying it flat on the table between them so it couldn’t be mistaken. “He means the editor. Pepper, at *Canadian Automotive Trade*, ran the original report, the one that started the two hundred. And then he called his own piece *entirely misleading*. In print. The man who lit the fire put it out, and the fire kept burning anyway.” She looked at Frik. Frik looked at the arrowheads, breathing. “He’s not wrong,” she said to Margaret. “He’s only ever early.”

Margaret held the offence she’d loaded a moment, and then, Naila watched it happen, set it down. “Pepper,” she said. “Yes. He was the liar. Not my grandfather.” She looked at Frik with something that wasn’t pity, which was the only thing that would have lost her. “Your friend says it cleaner than I manage, and I’ve had thirty years.”

Edith had not said a word.

Edith had come in last and gone to the window, where the prairie light came flat and grey off the street, and she had her camera, because she always had her camera, the way Jakobus always had the shades and the clean hanky and the tin of cayenne nobody asked about. And there had been, for a moment, a frame. Naila had seen Edith see it: Margaret Dolch at the speckled table, the four patents fanned, the dead reader's hand in the register, the donor light, the whole grief of it composed and waiting — *Suppressed Genius's Granddaughter Guards the Last File*, the photograph that book in the nineties would have killed for, the photograph that would do the man in all over again, dressed up as honouring him.

Edith lowered the camera. She did it without being asked and without making it a thing — capped the lens, let the strap take the weight, and went back to the street like a woman who'd decided the street was the better picture. She'd read which frame would steal him, the way Naila read a column and Frik read a lie, and she'd put it down before it cost the man anything. Margaret saw her do it. She was meant to.

"You didn't take that," Margaret said to her.

"No," Edith said.

"Why not. It was a good one. I could feel it being a good one." There was an edge to it; she was testing this too, looking for the angle.

Edith was slow with words, the way people are slow who have seen what words get used for. "Because you'd have to live with it," she said. "And he would. Forever. And he can't say no to it anymore." A one-shouldered shrug, a person setting down something heavy without wanting it remarked on. "You can. So it's yours. I'll shoot what you say I can shoot."

For a long moment Margaret Dolch did not say anything at all, and in the not-saying Naila could feel the held door move a half-inch on its hinge. Not open. Just considered.

Jakobus spoke from the furnace then, once, and Naila had learned to listen for it because he spent his words like a man down to his last coins. He didn't move into the light. He said it to the table and the patents and the register, in the flat trade-shop English of a town that had said *carburetor* in four syllables and meant it: "He built a good box." A pause. "I'd have liked to put my hands on it." That was all

— a maker's sentence about another maker, claiming nothing, asking nothing — and Margaret heard a man who wanted to *read* the thing rather than *take* it.

She uncrossed her arms.

She went to the roller door. She didn't lift it. She put one hand on the cinderblock beside the padlock the way you put a hand on the flank of an animal you're not yet going to move, and she stood with her back to them, and she said: "There's a unit. Out by the highway. I pay for it monthly, which my sister thinks is madness. His shop notebooks are in it. The ones with the real numbers, if the real numbers got written anywhere, which I am not telling you they did. And the letters. There are letters that would break your heart and I have read them so you don't have to." She turned around. "That's not in the museum. That's not in the patents. That's not on your phone. That's *him*."

"We'd like to read it," Naila said. "Only read it."

"I know you would." Margaret almost smiled, and didn't, and it was kinder than a smile. "You've about half convinced me, the lot of you. Your friend out-honested me with one word. The photographer put down the only picture I'll be remembered by. The big quiet one wants to hold the machine like it's a baby." She looked at Naila last. "And you won't lie to me, even to be nice. I noticed. You'd rather hand me a fact that hurts."

"Yes," Naila said, because it was true, and because saying the kind thing instead would have been the first lie.

Margaret crossed back to the table. She bent to a cardboard banker's box on the lower shelf that Naila had clocked on the way in and made herself not look at, and lifted out a single folder, buff and soft-cornered, a pencilled date on the tab too faint to read upside down, and set it on the speckled Formica.

Face-down.

She laid her hand flat on top of it, as if it might decide to open itself.

"This is one of them," she said. "Not the notebooks. One folder. I'll decide tomorrow whether you see what's in it." Her hand stayed where it was. "The file's mine. My grandfather's, and then mine, and a man came up that road once already and made it a lie I've spent

thirty years living down. So you'll forgive me if I don't open it tonight on the strength of one good word and one picture not taken." She looked at the roller door, then back. "Come to the legion at eight. I'll have decided. If it's no, you'll have driven a long way for nothing, and I'll buy the coffee."

"And if it's yes?" Yusuf asked.

"Then I'll be trusting you with a quiet man," Margaret said, "and you'll be the first I ever have." She took her hand off the folder, finally, and left it lying there face-down on the table between them — the date hidden, the man hidden, the door not shut and not open, ajar exactly as far as she had decided to hold it. "Eight," she said, and turned off the reading-room light, so that the last of the prairie came grey through the one window and the buff folder went the colour of everything else.

Outside, the wind came across the flat with nothing to stop it, the way it must have the year a broke mechanic in a town like this built a box that clawed a little fuel back from the fire and never, to his grave, said how much. Naila stood by the Beast with the cold getting into the steel of it and thought about the donor register — the wrong note in the column, the row in a different pen. Somebody recent had come asking after these patents too. She filed it beside the rest of the evening, which was a folder lying face-down on a church-hall table, in a town nobody drives to, where the one true number had been taken down into the quiet on purpose, and the only person left who got to decide who heard it had said *tomorrow*, and meant it.

It was a long time until eight.

Chapter 3 — Real Engineering, Modest Truth

She'd said yes at the legion at eight, the way she'd said she would, and then made them earn it anyway — drove out ahead of them to a row of storage units off the highway, roller doors the colour of weak tea, and stood in the cold a moment with the key in her fist before she'd open it.

Now it was nine the next morning, the unit open, the heater Tomi had carried in off the Beast losing to the prairie, and Naila had the four patents out of their sleeves and laid on a folding table, granted-date left to right, the way you lay out anything you mean to actually read.

People said *the patent* like there was one. There were four, and the difference between them was the whole thing nobody bothered with, because the difference was work and the legend was easy.

She started at the left. Nineteen-thirty. One million, seven hundred fifty thousand, three hundred fifty-four — filed in twenty-seven, three years in the office before it stamped, which told you the man had been at this before the crash, before there was a Depression to be broke in. The drawings were the good kind, the kind a man does himself, because the lines knew things a paid hand wouldn't bother to know. A chamber. Air drawn down and *up* through the liquid fuel, so the petrol didn't sit and wait to be sprayed at — it got walked through, bubbled, every drop given a surface. She put her finger on the bubbler and left it there a second. That was the idea the other three were arguments about.

"He's not inventing the carburetor," she said, to whoever was listening: Tomi crouched at the heater, Margaret in the doorway with her arms

folded, Jakobus somewhere behind in the cold he wasn't complaining about. "Everybody had a carburetor. He's inventing a way to get the fuel *dry* before it burns."

"Dry," Tomi said, from the floor, already wanting the rest of it.

"Liquid burns wrong." She said it the way the patent said it, because the patent had said it well. "A droplet doesn't explode, it *flames* — burns from the skin in, and the engine's gone past it before the middle of the drop's caught. So you're firing fuel that half-burns and goes up the pipe still wanting to. He says it right here." She didn't read it aloud; she'd have had to do the voice, and it wasn't hers to do. She moved to the second sheet instead.

Thirty-three. One million, nine hundred thirty-eight thousand, four hundred ninety-seven. Filed while the first was still warm — a man who'd built the first and run it and found where it was thin. The bubbler was still there. But now there were nozzles, fine ones, a Venturi throat, the fuel not just walked through air but torn fine, atomised, the droplets made small enough that *small* started to mean *gas*. She traced the road with one nail, intake to throat to chamber, and it held; a thing could travel it.

"Each one fixes the last one," she said, and didn't fight the small lift in her own voice, because this was the part she was built for, the part that was clean. "Not one idea sold four times to four clerks. A man with a *problem* who keeps coming back at it. You can read the failures in the gaps — the first one bubbles and it's not enough, the droplets too fat, so" — she touched the second — "nozzles. Tear them finer." She moved on. "And then—"

Thirty-five. One million, nine hundred ninety-seven thousand, four hundred ninety-seven, and here was the one the legend lived in without knowing it. A jacket: a chamber inside a chamber, the inner one holding the vapour, the outer wrapped close around it like a hand around a cup. And the line ran from that outer jacket back to the engine's exhaust manifold. The heat that went out the pipe as pure waste came back and warmed the fuel on its way in.

She stopped on it. It was, she would have said if anyone had pushed her and got the truth instead of the kind version, beautiful, in the way good bookkeeping was beautiful — a man counting the heat the engine spent and finding a way to spend it twice.

"Okay," Tomi said, up off the floor now and over her shoulder, fast, the soldering-iron tone, *but if you just—* "okay, so if the exhaust is free, and you stack that with the atomising, your margins start compounding, and if he'd had a way to recover even the—"

"Tomi."

"—I'm just saying the *architecture's* there, you could—"

"You could get into the high twenties," she said, flat, and put it down on the table between his reaching and the thing he reached for, the way you put a hand on a dog's chest. "That's the top of it. A 1933 V8 ran rich and dumb and threw fuel up the pipe by the cupful; a rig like this gives some of it back. From the low teens that car was doing — high twenties is real. A thing a sober person stands behind." She let that be the gift it was, hers to give him clean. "A better-run version of this universe. Not a different one."

Tomi looked at the jacket on the drawing, at the exhaust line going home, and she watched him want the next rung and not get to take it.

"The fourth one's plumbing," she said, gentler now, moving them along. Thirty-six. Two million, twenty-six thousand, seven hundred ninety-eight, the last one before the man went quiet. A vacuum-driven pump. A bypass to hold the vapour at a steady pressure so the engine didn't gulp and starve. "He's not chasing the big number here, he's making the thing *behave*. A man who'd run it on a real car in real weather and watched it stumble and come back to fix the stumble. A fraud quits at the patent that makes the headline." She squared the four sheets back into their row. "He kept going after the headline. He went and worked on the *idle*."

Margaret had not moved from the doorway. "You sound like the file," she said.

"I haven't read the file."

"No." Margaret came in then, two steps, into the losing warmth. "You read the patents the way he'd have wanted somebody to. The papers got the cape on him before the ink was dry and never once asked what the thing actually *did*." She looked at the four sheets in their honest row, and something in the look was thirty years old and tired and, for a second, set down. "You asked."

That was when Jakobus came in.

He came the way he came into rooms, taking the cold off himself at the door first, the shades already hung at his collar against a glare that wasn't here, and he didn't go to the table. He never went to the table first. He looked at the third patent from where he stood, the one with the jacket, the way a man looks at another man's work across a yard before he's earned the touch of it.

"Read me the third one," he said — to Margaret, not to Naila. He'd worked out whose it was.

Margaret didn't read it. She carried the sleeved sheet to him and held it where he could see, and didn't let go of it, and he didn't ask her to, and the two of them stood over a 1935 drawing of a heating chamber the way you'd stand over a cot.

He looked at it a long time. Long enough that Tomi started to vibrate with wanting to fill the silence and Naila put two fingers on his sleeve.

"That'll work," Jakobus said.

He said it the way he said the few things he said — flat, in the trade-shop English he found for rooms like this, no music in it, a mechanic's verdict and not a host's line. He put one finger near the glass over the jacket, not on it, tracing the exhaust line back to the manifold without touching the sleeve.

"He's giving the fire its own breath back," he said. "Engine throws the heat out the pipe, it's gone, you paid for it and you threw it out the window. He catches it on the way out and lays it under the fuel coming in. Warms it. Dries it. So the cylinder gets gas, not soup." A pause, the kind he spent like a coin. "Costs him nothing. He just wouldn't let it leave for free."

"That's what I—" Tomi started.

"It's a good thing," Jakobus said, over him, gentle, final, and the word *good* in his mouth was not nothing; Naila had learned its weight. He gave it to the dead man across ninety years the way one maker hands another a tool, handle-first. "A man stood in a cold shop and built a way to keep the heat. That's a whole life in a box." He took his hand back. "What's it for, then." He asked it the way he always asked it, of the made thing, not the legend — and he asked it not of Naila or

the air but of Margaret, stepping back from the patent so the room's small space opened toward her and not toward him.

And Margaret, who had been waiting thirty years for somebody to want the answer to *that* question instead of the other one, did not give it yet. She set the sleeved patent back in its place in the row, careful, granted-date in order, and she said, "There's the part you came for," and went to the banker's box.

What she lifted out was not a notebook. It was a folder, and inside it a yellowed clipping gone soft as cloth at the folds, which she laid on top of the four honest patents like a stone on a grave.

A page of *Canadian Automotive Trade*. The masthead, and under it, in the dense grey column-type of a magazine that had once set a continent's newspapers on fire with a number, a later editor's note. R. Pepper. The man who'd run the original report — the two hundred miles, the Ford on a pint, the run to Vancouver on fourteen and a half gallons — coming back into his own pages, in his own ink, to say it.

Entirely misleading, the column said. The carburetor *never was a practical success*. No expert who had tried it, the editor wrote, had got anything near the mileage the magazine had suggested.

Naila bent over it. A thing you could hold and photograph and check — and that was the part that always undid the easy story for her: not that the dismissal was wrong, but that it was *right*, on the record, signed by the very man who'd lit the fire. The magazine that broke the miracle had un-broken it in print, and the world had kept the miracle and thrown the retraction in a box in a town nobody drives to.

"He printed that," she said.

"He printed that," Margaret said. "The man who made my grandfather famous wrote it with his own name on, and you'd think it would have ended things." She tapped the clipping once. "It ended nothing. The two hundred had legs by then. The two hundred didn't need Pepper anymore. The two hundred had the whole country."

Frik, over by the roller-door track, doing the small managed business of the shoulder and the breath, said to the clipping, not loudly, "*Says so himself.*" His jaw worked. "*His own paper.*" The hand half came up and wasn't needed; Margaret had already turned to look at him

without flinching, because they'd done this dance once and she'd learned the step. She gave him a small nod, byline to byline, and let it go.

So there it sat, the two things touching on the one table. Four patents that described a machine that really, modestly worked — warmed the fuel on the engine's own thrown-away breath and clawed a thirsty V8 into the high twenties, no more, a good box built by a man who went back afterward to fix the idle. And on top of them a yellowed column in which the very editor who'd launched the legend called the legend entirely misleading.

The same root. That was the thing Naila kept coming back around to. The high-twenties that was real and the two hundred that was impossible weren't two stories that got mixed up; they were the *same* story, told twice — once by the man who'd built the thing and knew exactly what it did, once by a press that took his modest true number and ran it up a flagpole until it meant a Ford crossing the prairie on a thimble. The believable kernel and the impossible claim grew off the one stem, and you could spend ninety years pulling at the impossible one and never notice you'd let go of the real one.

She straightened, moved Pepper's retraction gently off the patents and back to its folder, and laid her flat hand on the third one, the jacket, the exhaust coming home.

"Here's the real figure," she said, and it came out of her the way the clean numbers always did, an offering. "High twenties. Maybe thirty on a good day, downhill, tail wind — but high twenties is the honest middle of it. That's what he built. Not two hundred. Not, as he said himself, even half that." She looked at Margaret, because the answer was Margaret's and she was only handing back the arithmetic of it. "A man built a good box that did a real thing worth being proud of, and they put a cape on him for a thing the box never did, and—"

She stopped.

Because the patents were there, all four, public to anyone with a phone. Pepper's retraction was there, dated and signed and checkable. The file Margaret guarded was there, with letters that broke a heart she'd read so you didn't have to. Every document in the whole story was on this table or in this box or on a server somewhere — and you could weigh every one of them, and not one was the number.

The one figure it all turned on — what the box *actually did*, the modest true thing a man could have said in a breath to the reporter sitting where the table was — that was the one document that didn't exist. He'd had it. It was his. He'd taken it down into the quiet on purpose and made oil filters and died, and the record kept everything except the thing the record was about.

She kept her hand flat on the patent, the four sheets in their honest row, and at the centre of all of it the number-shaped hole where a sober man's true answer should have been and never would be — because the only person who could fill it had decided, ninety years ago, that nobody had earned the hearing.

"High twenties," she said again, quieter, to no one, because it was the closest the world would let her get, and it wasn't close at all.

Behind her, Jakobus put his shades back on against the prairie coming up white and flat through the open roller, and he didn't say anything either, and Margaret reached past Naila and squared the four patents one more time, granted-date in order, into their row.

Chapter 4 — The Refused Number

The space heater in the corner of the back room ticked like it was counting something down. Naila had been watching it instead of watching Connie Pogue's face, because the heater was honest about what it wanted and the face was not.

Three patents lay open on the table under a sheet of museum glass weighed at the corners with washers — 1,750,354; 1,938,497; 1,997,497 — and a fourth, 2,026,798, still in its acid-free sleeve because Connie hadn't decided yet whether they were the kind of people you let touch the fourth. Naila respected that. You earned the sleeve.

"It's a real progression," Naila said. She kept her hands flat on the glass, fingertips only, because the wanting-to-turn-the-page feeling lived in the hands and she'd learned to leave it there. "He's not throwing claims at the wall. Thirty: heating chamber off the exhaust manifold. Thirty-three: he's added the bubbling — air drawn up through the liquid fuel, here, this column. By thirty-five he's worrying about the atomizing nozzles fouling. That's a man who *ran the thing*. You don't refine a problem you never had."

Connie Pogue did not say thank you. She was perhaps seventy, in a cardigan the colour of weak tea, and she had the particular stillness of someone who has told the same true story to forty people and watched thirty-nine of them not hear it.

"Everyone who comes here can read a patent," she said. "Half of them came with a printout already highlighted."

“The highlighted line being two hundred,” Naila said.

“The highlighted line being two hundred.”

Across the table Yusuf had gone quiet in the way he went quiet when he was costing something out. He’d come up reading rooms like this in a different life, the kind where you smiled at a custodian until they handed you the box that made somebody rich. Naila could see him not doing that on purpose. It was a visible effort, like a man holding a door against wind.

“Two hundred miles to the gallon,” Bashir said gently, from where he sat with the camera dark in his lap. Not Edith — Edith was by the window with the lens *down*, deliberately, both hands around a paper cup of the museum’s terrible coffee, photographing nothing. That had been a decision in the van and Connie had clocked it the second they walked in. *No camera up until you say*. It was the first thing they’d offered her and the only thing so far she’d seemed to weigh.

“A Ford V8,” Connie said. “Winnipeg, summer of ’thirty-three. The papers said two hundred. Then there was the run to Vancouver — fourteen and a half gallons, they printed.” She lifted her chin a half-degree. “You know what fourteen and a half gallons gets a 1933 flathead V8 to Vancouver on?”

“No,” Naila said, though she did, roughly, and Connie knew she did, and the *no* was a courtesy. The number lived in Naila’s chest like a stone she’d swallowed years ago. A gallon of gasoline held about thirty-three kilowatt-hours and not one watt more, fixed, chemical, indifferent to genius. A carburetor mixed. It did not manufacture. You could be the cleverest mechanic in Manitoba and the gallon still held thirty-three.

“It doesn’t get there,” Connie said. “I worked it once, on graph paper, when I was angry. It doesn’t get there on fourteen and a half gallons or twenty-nine.”

“No,” Naila agreed. “It gets there on something closer to forty, if the rig was good and the run was downhill and somebody coasted the Fraser canyon and called it a record.” She turned her hand over on the glass, palm up, an old gesture that meant *here is the empty thing I’m holding*. “A vapor rig nudging a thirsty engine into the high

twenties — that's real. That's *plausible* and it's *good* and a man could be proud of it. Two hundred is impossible. The same bench produced both stories. That's the part nobody sits with."

For the first time Connie Pogue looked at her instead of at the room behind her.

"You know who said it was misleading?" Connie asked.

"Pepper," Naila said. "*Canadian Automotive Trade*. He ran the original report and then he called it 'entirely misleading' in print. The man who started the legend tried to kill it himself and it wouldn't die."

"And in fifty-three?"

This was the one. Naila felt it come up the table like cold off a window.

"Your grandfather said he never claimed two hundred. 'Or even half that,' he said." She paused. "And then they asked him what the real figure was."

Connie's hands folded.

"And he wouldn't say," Naila finished.

The heater ticked.

Here was the hole, and it was not the hole the highlighter people came for. The highlighter people wanted the suppression: the oil men, the bought-up patent, the threatening phone calls in the night. There was no evidence for any of that — Naila had read it down to the studs, Snopes and the Straight Dope and the dry encyclopaedia entries all agreeing in their bored way that the suppression story was air. That file closed clean. She'd closed it herself in the van, almost relieved.

This file would not close. A man stood up in 1953, with nothing left to gain, and said *I did not claim two hundred*. A debunker should love that sentence — it's a confession of the legend's falseness, from the legend's own mouth. And then, in the same breath, the same man declined to say what he *had* done. He took the impossible number off the table and refused to put the real one in its place. He left the space empty on purpose and carried whatever lived there to his grave in 1985.

Naila knew the shape of a hole where a number should be. The world had been handing her those her whole life. *What's it like inside your head, Naila* — and no number ever fit, so she'd learned to leave the space honest. Pogue had left his honest too. She felt something for a dead Winnipeg mechanic she had no business feeling, and she let herself feel it, flat-handed, fingertips only.

"You think he was protecting it," Yusuf said. Carefully. "The figure. You think he knew it was good — the real one — and didn't want it dragged down into the two-hundred circus."

"I think," Connie said, "you don't get to decide what he thought. That's the whole — that's the *thing*. You all want to give him a reason. The reason is the prize. If I tell you he was protecting a good honest number, you write SUPPRESSED, and if I tell you he was ashamed he'd let the papers run wild, you write FRAUD, and either way you've got your story and he's got nothing. He's been dead forty years and he can't say *no, that's not it* anymore. So I say it for him." She set her flat palm on the sleeved patent, the fourth one. "I say no, that's not it. To all of you."

Edith, by the window, very quietly photographed the coffee cup.

It was Frik who broke.

He'd been good for nearly an hour, which for Frik in a room this charged was a marathon, the tic riding low and contained, a few throat-clicks, a shoulder roll, the small private weather of him. Naila had felt it building the way you feel a sneeze in someone else. He'd gone still — that bad still, the still that wasn't calm — and his jaw worked once, and then it came out of him sideways and loud, in a bright tabloid bark that wasn't his voice at all:

"TWO HUNDRED MILES A GALLON — THE CARBURETOR BIG OIL BURIED!"

It hit the little room like a dropped tray.

Frik's hand was already coming up to his mouth, his face folding into the particular wretchedness Naila hated to see on him — not the tic itself, never the tic, but the half-second after, when the world decided what it thought of him. She got her words ready, the ones she kept loaded for rooms exactly like this, *that's Tourette's, it's involuntary, it is not what he believes* —

Connie Pogue laughed.

It came out of her short and surprised, one sharp note, and she put her own hand to her mouth almost the same way Frik had, as if to catch it.

"That's it," she said. "That's the headline. Word for word, near enough. They had it on a sandwich board outside a gas station in 1936." She looked at Frik, and something in her cardigan-coloured stillness gave way an inch. "You said it like you hated it."

"I *do* hate it," Frik said, low and his own again. "Sorry. It picks the worst thing in the room. Always has. I can't —" He gestured, the gesture meaning *the inside of me*.

"No," Connie said. "Don't be sorry. You said the lie in the lie's own voice." She studied him. "Most people that come in here, they *believe* the sandwich board and they whisper it like a secret. You shouted it like it was an insult to him." She nodded slowly. "It is. It always was."

And that, Naila understood, was the thing that turned. Not the patents — anyone could read a patent. Not Naila's careful thermodynamics — Connie had done her own on graph paper, in anger, years before Naila was born; the keeper out-read them all on the only ground that mattered, which was the man. What turned Connie Pogue was a blurt nobody could fake, a thing that came up out of Frik with no strategy in it, naming the tabloid myth as the obscenity it was. You couldn't perform that. You couldn't grift it. It was the truest thing said in the room and the man who said it hadn't meant to say it.

Connie pulled the fourth patent out of its sleeve and laid it under the glass with the others.

"There," she said. "Now you've all four."

—

The corporate man had called on a Tuesday.

Connie told them about it while she was making fresh coffee in the museum's galley kitchen, talking with her back to them, which Naila thought was the only way she could say it.

"Polite," Connie said. "They're always polite. A foundation, he said.

Preserving the legacy of overlooked Canadian innovators.” She said *legacy* the way you’d hold something out of a drain. “They’d digitise everything. Climate-controlled archival storage. A scholarship in Granddad’s name. All they’d need is the original materials. For *conservation*.”

“Did he name the foundation,” Yusuf said, and his voice had changed — not louder, just *aimed*.

“He left a card.” Connie found it among the magnets on the little fridge, held it out without turning. Yusuf took it like it might still be warm.

Naila watched his face do the thing it did. He’d sold words like *legacy* and *conservation*; he’d been the polite man on the phone, in the life before the van. He read the card the way Naila read a patent — for the progression underneath the surface, the bench-work of a con.

“They don’t want it preserved,” Yusuf said. “They want it *gone*. Originals into a climate-controlled box nobody can subpoena, and then a press release, and then quiet. A man who never claimed two hundred turns into a foundation’s tax write-off and stops being able to say *no*, *that’s not it* — same as you said. Forever, this time.”

“Why,” Connie said, finally turning. “Why would anyone *pay* to bury a carburetor that doesn’t even —”

“Because the carburetor’s not the asset.” Yusuf set the card down. “The *quiet*’s the asset. There’s people whose whole product is making sure the suppressed-genius story stays exactly the right amount of true — alive enough to sell, dead enough to never have to deliver. Your grandfather’s silence is worth more to them than any number he could’ve named. They’re not buying the bench. They’re buying the *hole*.”

Connie Pogue stood in her tea-coloured cardigan in the fluorescent galley light and looked, for a moment, exactly her age.

Then she went to the back room, and they heard the photocopier wake up — an old one, the kind that warmed up with a smell — and the soft repeated clunk-and-flare of it going through the folder, page after page, the four patents and the press clippings and Pepper’s retraction and a typed sheet Naila hadn’t seen, and whatever else

was in there that Connie had never decided to show the highlighter people.

She came back with a banker's box, not heavy, and held it out to Naila, not to Yusuf.

"Copies," Connie said. "Good ones. You read the pattern, you said. So go read it where they can't get at it. The originals stay with me till I'm in the ground and then they go to the provincial archive with a deed of gift so tight no foundation on earth can pry it loose. I called a lawyer Wednesday."

Naila took the box. It weighed about what a box of paper weighs. It felt like more.

"There's a name on the typed sheet," Connie said. "Near the back. Granddad kept up with the patent journals his whole life — it was a kind of hobby, who else was working the edges. He'd underline things. Last page, he's underlined a fellow in a university physics department, decades after, going on about the vacuum." She almost smiled. "Empty space that isn't empty. Granddad wrote in the margin — you'll see it — *they'll call him a crank too*. He'd know."

"The vacuum that isn't empty," Naila said, and the stone in her chest shifted, because she knew that one, knew it was *real*, knew it had been weighed to within a percent and still wouldn't light a bulb, and she felt the descent open under her feet like a stair going down into a dark she'd already half-agreed to take.

"He underlined it twice," Connie said.

Outside, past the window where Edith finally, gently, with permission given by a nod, raised the lens — the prairie went flat and gold to a horizon that held no number at all, and the heater in the back room ticked on, counting down to nothing anyone in that room would say out loud.

Chapter 5 — The Vacuum That Isn't Empty

The lab was in the basement, because the good vibration isolation always is.

Naila had read the building's name off a brass plate by the door — some donor, some decade — and then forgotten it before the lift, because the brass plate was not the thing. The thing was a granite slab the size of a dining table floating on six pneumatic legs, and on the slab, under a bell jar, two plates of gold-coated glass held a hundred nanometres apart by a piezo stack, and a torsion balance so fine that the woman minding it spoke in a near-whisper, not for drama but because her own voice, at the wrong pitch, would show up in the data.

Her name was Dr. Aoife Mbeki. She had a postdoc's salary and a professor's lab and the careful flatness of someone who had been asked to explain the impossible to credulous visitors before and had learned to make her face say *I will give you the real thing, and only the real thing*, before she said a word.

"You can't feel it," she said. "You can barely measure it. Lamoreaux did it at Washington in 1996 — agreement with theory to about five percent. People have since pushed under one. It is one of the best-measured tiny forces in physics." She looked at them, mild. "Which is why I find it tiring when men in my inbox say it powers their generator."

"How many men," said Yusuf, "in your inbox."

"This week?"

Edith had the camera down at her hip, not up. She'd asked Mbeki at the door — *may I* — and Mbeki had said *the equipment, yes; me, sparingly*, and Edith had nodded like that was the most reasonable sentence she'd heard in a month. The red light was off. Naila had clocked that. Edith turned the red light off when she meant it.

—

Here is the official story, and Naila held it the way she held all the true ones: flat in the palm, weighed, no thumb on the scale.

Empty space is not empty. This is not fringe. This is not a *yet*. This is the floor of quantum field theory, the thing the textbooks teach the second-years, the thing the Nobel committee has decorated more than once. Pull every atom out of a box. Cool it toward absolute zero. Shield it from every field you can name. The box is still seething. The electromagnetic field — every field — has a lowest possible state, a ground floor, and that ground floor is not *off*. It jitters. Zero-point energy: the energy a system keeps when it has given up everything it is allowed to give up. The vacuum hums at the bottom of the well.

Push two uncharged metal plates very close together in that humming vacuum, and the hum between them gets *quieter* than the hum outside them — because only certain wavelengths fit in the gap, like only certain notes fit a short organ pipe. More modes outside than in. More push from the outside. The plates get shoved together by nothing. By the absence of *enough* nothing. Hendrik Casimir wrote it down in 1948 on the back of an argument with Niels Bohr, and for fifty years it was a beautiful sum nobody could weigh, and then Lamoreaux weighed it, and there it was, to five percent, exactly where the maths put it.

“Watch,” said Mbeki.

She didn't touch anything. She talked them through a voltage trace on a monitor, a line that was mostly flat and then, as the piezo crept the upper plate down through a hundred and forty nanometres, bent — a small, certain, attractive bend, the plates leaning into one another across a gap a thousand times thinner than a hair.

“That's it,” said Mbeki. “That is the vacuum doing work on those plates. That number agrees with a calculation about empty space.”

Nobody said anything for a second. The slab floated. The jar held its

breath.

“That’s the most haunted graph I’ve ever seen,” said Edith, quietly, and Mbeki — for the first time — almost smiled.

The vacuum is not empty. Naila turned the sentence over and found no crack in it. It was true. It was extraordinary. It was peer-reviewed and replicated and load-bearing under half of modern physics. And it was about to become the worst thing in the room, because it was *also* the exact true sentence that a thousand garage frauds had nailed to the front of a thousand machines that did not work.

—

“So,” said Yusuf, gently, the way he opened every con he’d ever taken apart, “if the vacuum’s full of energy. Walk me through why I can’t have some.”

Naila felt the question land in her own chest first. Because she *wanted* it too — she always wanted the world to be allowed to be generous — and the wanting was a thing she had to set down on the bench, deliberately, beside the patents, every single time.

Mbeki nodded. She’d been walked here before, and she walked it kindly.

“Two reasons. The first is the gentle one. Energy isn’t the thing you spend. *Difference* in energy is the thing you spend.” She turned a pen end over end. “A boulder at the top of a hill has energy. A boulder at the bottom of a deep valley has *less*. You can run a mill on the boulder rolling from hill to valley. You cannot run a mill on the boulder sitting in the valley — even though it still ‘has energy,’ even though there’s loads of it — because there’s nowhere lower for it to go. The zero-point field is the valley floor. The *bottom*. The clue is in the name. It’s the lowest state. You can’t extract energy by letting a thing fall to a place it’s already at.”

“The plates fall,” said Naila.

She heard her own voice come out level and quick. Mbeki turned to her, interested rather than corrected.

“The plates fall,” Mbeki agreed. “Once. You get a tiny pulse of work — femtojoules — as they snap together. And then they’re together, and to do it again you have to *pull them apart*, and pulling them apart

costs you back every joule you got, plus losses. It's a spring, not a spring you wound for free. The vacuum lent you the energy and the vacuum takes it back at the door." She set the pen down. "There's a real paper — Pinto, around 1999 — on whether you could even cycle it. The honest answer is: you'd spend more recharging the cycle than the cycle ever gives. It's a closed account. It always balances."

"And the not-gentle reason," said Yusuf.

"The not-gentle reason is that if it didn't balance, you'd have a perpetual motion machine, and the universe has been very clear with us on that point for two hundred years, and it has never once blinked." Mbeki said it without heat. "Conservation of energy isn't a rule we're hoping holds. It's the most tested statement humans have ever made. Every machine that claims to beat it is wrong, and we know it's wrong *before* we open the box, the way you know a man who's sold you a triangle with four sides is wrong before you count."

Frik had been very still by the bell jar, his hands folded into his own sleeves the way he did when he was managing himself. The blurt came out small, almost apologetic, aimed at no one.

"Free," he said. Then, lower, on a tighter beat: "*free energy — sold separately.*"

A beat. Mbeki looked at him — not the look people gave Frik, the *what's wrong with him* look. The other one. The one that had heard the joke land *exactly on the seam* and was a little impressed.

"Sold separately," she repeated. "Yes. That's — yes. That's precisely the move."

Frik's shoulders came down a quarter inch. Naila saw it. Banked it. Said nothing.

—

Here is the potato, and the potato in this book was always the dismissal straining against the residue — except this time the residue and the dismissal were *the same true sentence wearing two coats*, and that was the whole horror and the whole beauty of the node.

The dismissal: *zero-point energy machines are perpetual-motion frauds*. True. Plainly, provably, conservation-of-energy true. Every "ZPE generator" on every crowdfunding page fails the first reason

and the second reason and usually a third reason about where the heat went. The word *zero-point* on a product is a tell, the way *quantum* on a wellness patch is a tell, the way *as seen on* is a tell. The con men did not steal a *fake* word. That was the genius of the theft. They stole a *real* one — the realest, most decorated, most beautiful word in the room — and they nailed it to the front of the boulder-already-in-the-valley and sold the valley as a waterfall.

“That’s what makes it almost impossible to fight,” said Mbeki, and now she did sound tired, the specific tiredness of a person guarding a true thing from people who love it for the wrong reasons. “If I say *the vacuum is empty*, I’m lying — it isn’t, and I’ve spent my life measuring that it isn’t. So I can’t take the word off them. I can’t say *there’s no energy there*. There *is*. I have to say the harder thing, the thing that doesn’t fit on a placard. *There’s energy there and it’s at the bottom and you can’t have it*. And the man with the machine just nods and says *that’s what they want you to think*, and I have no shorter sentence than the truth, and the truth has never once fit on a placard.”

Naila felt that one go in deep, where the patents lived. *I have no shorter sentence than the truth*. She knew that sentence in her own body. The world ran on placards and she had never once been able to make one.

—

Yusuf, who read money the way Naila read patents, asked the question that was really the chase.

“Who buys it?” he said. “Not the marks. The marks I get. Who *funds* it — at the top. Who writes the cheque to keep a ZPE outfit breathing past the point any engineer would’ve walked?”

Mbeki considered him. Recalibrated. Understood, maybe, that this crew had not come to be cured of their wonder.

“Two kinds,” she said slowly. “There’s the believer with money, who’s just a bigger mark. And then there’s the other kind. The kind that doesn’t care if it works.” She watched Yusuf’s face. “The kind that wants the *patent filing*, or the *press release*, or the *thirty months of plausible R&D* — for reasons that have nothing to do with electricity. I’ve consulted for due-diligence people twice. Both times the ques-

tion wasn't *does it work*. Both times the question was *what is this man's silence worth, and to whom.*"

The slab floated. The graph held its haunted bend.

"I told them it doesn't work," Mbeki said. "Both times. And both times they thanked me very warmly, and I got the strong impression it changed nothing they were going to do."

Yusuf didn't write anything down. Naila watched him not write it down, which meant he'd file it somewhere the file couldn't be subpoenaed. The name they were chasing — the outfit with the clean hands and the deep pockets, buying up the papers of dead quiet men — had not been said in this room. But it had walked through it, in the shape of *both times*, and they all felt the draft of it pass.

—

They didn't stay long after. You don't, in a room where your own breath is data.

At the door Mbeki paused, one hand on the light switch for the granite slab, and looked back at the bell jar like a person checking a sleeper.

"People want it to be a door," she said. "The vacuum. They want it to be the place we haven't looked, with the free thing behind it. I understand the want. I have the want." Her voice stayed flat and that flatness was the most generous thing in it. "But it's not a door. It's a *floor*. It's the most beautiful floor we've ever found, and it holds up everything, and you can stand on it your whole life, and it will never once open."

She turned the light off over the slab. The bend in the graph went on glowing on the monitor in the dark — the vacuum still leaning two plates of gold together by the strength of all the notes that wouldn't fit between them.

In the lift, nobody talked. Frik had his hands back in his sleeves. Edith's camera was still down. Naila stood with the floor-and-not-a-door turning in her, and underneath it, quietly, the next thing — because the descent only went one way, and the next man down the ladder had not measured a force to one percent. The next man down had aimed a telescope at the Moon on the third of November, 1958, and written in his log that he had seen it *breathe* — and the State

that buried him had given him, before and after, a gold medal and a Gulag and a crater with his name on it, and Naila already knew, the way she knew which true sentences wouldn't fit on placards, that this floor was about to stop being a floor at all.

The lift opened on the lobby and the brass donor plate she'd already forgotten, and Yusuf held the door, and they went out into the grey afternoon to find a Russian.

Chapter 6 — A Word That Got Away

The expo had taken over a curling rink.

You could still smell the ice under the carpet they'd rolled across it, that mineral cold rising through the synthetic pile, and the fluorescent tubes overhead buzzed at a pitch that Naila had been hearing for forty minutes and would go on hearing in her teeth for an hour after she left. Banner cloth sagged between the support posts. ABUNDANCE TECHNOLOGIES EXPO, it said, the letters cut from a font that wanted to be expensive. Folding tables ran in rows. A man two booths down was selling orgone pyramids and a sleep mask that "harmonised your pineal."

"Don't make the face," Yusuf said, not looking up from his phone.

"I'm not making a face."

"You're making the whole face." He pocketed the phone. "I need you to read the room, not correct it. There's a difference and you know it."

She did know it. She'd been reading rooms since before she could spell the word, because nobody had ever let her stop — the trick of watching the thing everyone else's eyes slid off, the exit, the tell, the place where the story didn't hold its own weight. She read this room now. Two hundred people who wanted the world to be more generous than it was. A scatter of skeptics with lanyards, there to laugh. And, somewhere in here, a man with a chequebook buying things nobody else wanted.

"Stage's filling," Yusuf said. "That's our guy."

The man on the little raised platform was named, his slide said, DR. RAYMOND COLE, and underneath that, ZERO-POINT ENERGY SYSTEMS, and underneath *that*, the word PATENT-PENDING in a colour engineered to feel like a promise.

He was good. That was the first thing, and Naila held onto it because it was true and because it would matter later. He didn't rant. He had a soft Maritime voice and a way of pausing before the big claim, as though he too were a skeptic and had been won over against his will. He stood beside a thing the size of a bar fridge — anodised aluminium, a clear acrylic window, inside it a wheel of magnets and copper, and an LED strip that glowed a steady, honest blue.

"The vacuum," Cole said, "is not empty."

And here was the thing that made Naila's jaw unclench by half a degree: *that part was true*. She felt Yusuf glance at her, because he'd felt her not-react.

"The space between the stars," Cole went on, "the space inside this room, the space inside *you* — physicists will tell you it's seething. Boiling with energy. They have a word for it." He let the pause land. "Zero-point energy. The energy that's left when you take everything else away. When you cool the universe to absolute zero and there is, by rights, *nothing* — there's still *this*."

A murmur went through the folding chairs. Naila watched the blue LED. It was a real LED. It was really lit.

"They've measured it," Cole said. "A scientist named Lamoreaux, 1997. Two plates, micrometres apart, in a vacuum. And the vacuum *pushed them together*. Pushed them, with a force you can weigh. Within a few percent of what the equations predicted, and the refinements since have it under one percent." He spread his hands at the bar-fridge. "All I have done — all this device does — is *tap* it."

Naila's hand found Yusuf's sleeve and held it. Not a signal they'd agreed on. Just the place her hand went.

Because every word up to *tap* was a thing she could have said herself, in a reading room, with the papers open. The Casimir force was real. It was measured. Lamoreaux at Los Alamos, then Mohideen and

Roy tightening it to better than a percent, and the vacuum genuinely is not nothing — it has a floor energy you cannot drain because it is the floor, it is the lowest the field can go, it is the *definition* of as-empty-as-empty-gets. She loved that fact. She loved it the way some people loved a piece of music. It was one of the strangest true things in the whole catalogue of true things, and it sat there in the literature, weighed and peer-reviewed and *not hidden by anyone*.

And he had reached out and picked it up and put it in a bar fridge with an LED, and the LED was lit, and the two of those were not connected.

“Run the numbers in your head,” Yusuf murmured. “You’re doing it anyway.”

She was doing it anyway. The Casimir force at a useful gap is real and it is also *tiny* — micronewtons over square centimetres, and worse, to get the energy back out you have to let the plates snap together, and then they’re together, and now you have to pull them apart again, and pulling them apart costs you exactly what you got. The floor is the floor. You cannot stand below it to push off. There is no down from down.

“It doesn’t store,” she said, very quietly. “There’s nothing to discharge. He’s calling the basement a waterfall.”

The fish, if there’d been a fish, would have had something to say about the LED.

“Sir. SIR. Your miracle box has a wall plug.”

There wasn’t a fish. There was Yusuf, who had spent a previous life selling words very like *zero-point* to men with more money than physics, and who now leaned a careful inch sideways and breathed, “Look at the floor behind the table.”

She looked. A power strip. A grey cable, gaffer-taped flat and snaking back behind the black drape, doing its honest grey best to be invisible.

“Could be just the slide projector,” she said, because she was fair to a fault and the fault was load-bearing.

“Could be.” Yusuf almost smiled. “Watch the demo.”

The demo was that Cole disconnected something. He made a show of it — held up a barrel plug, a little theatrical, “*and now, on its own resources*” — and unplugged a lead from the side of the fridge, and the blue LED stayed blue. The crowd made the sound a crowd makes. Naila watched the LED and watched the cable behind the drape and understood, with the cold clean click of a thing fitting its socket, that he had unplugged the *charger* and the device was running on a battery he had charged from the wall this morning, the way you’d run a torch, and the only mystery was how long before the blue went out, and she’d have put money it would outlast the talk.

“He’s not lying about the physics,” she said. The unfairness of it crawled on her. “That’s what’s — the physics he says is *right*. Lamoreaux is right. The vacuum is right. He’s just—”

“He’s wearing it,” Yusuf said. “Like a stolen coat. Fits him fine. Doesn’t make it his.”

She found the keeper by accident, which was the only way you ever found the people who mattered at these things.

She’d gone to the side, away from the buzzing tubes, to stand near the wall and let her ears stop ringing, and there was a woman there in a quilted vest doing exactly the same thing — back to the cinderblock, arms folded, watching Cole’s stage with an expression Naila recognised because it lived on her own face: the look of someone correcting the room in their head and being too tired to do it out loud.

“He’s quoting Lamoreaux,” the woman said, to nobody, to Naila. “He always quotes Lamoreaux. He never quotes the part where Lamoreaux says you can’t get net work out of it.”

Naila turned her head. The woman had a lanyard, but it was tucked into the vest, the name flipped. Retired-something hands. Reading glasses pushed up.

“You’re a physicist,” Naila said.

“I was. Solid state.” She unfolded one arm to gesture at the rink, the banners, the orgone pyramids, Cole’s blue light. “I come every year. I don’t know why. To grade the homework, I think.” She glanced at Naila properly now, took her in — the noise-cancelling buds looped

at her collar, the stillness, the way Naila was watching the cable and not the man. “You saw the power strip.”

“I saw the power strip.”

“Good.” The woman seemed to decide something. “You’re not one of his.”

“No.”

“And you’re not one of *theirs* either.” She tipped her chin toward the far corner, and Naila followed it.

The far corner had a different kind of table. No banner. No pyramids. Two men in good quiet clothes and a laptop, and a steady, unhurried line of expo people going to them and coming away looking faintly pleased and faintly fleeced at once. One of the good-quiet men was shaking Cole’s hand now, having materialised at the side of the stage the moment the talk ended.

“They’ve been here every day,” the physicist said. “They’re not buying the boxes.” A dry sound that wasn’t quite a laugh. “They’re buying the *paper*. The patent-pendings. The provisionals. The dead ones, especially — the guys who filed in 2009 and ran out of money. They buy the application, the rights, the lot. Forty of them, fifty. Pennies. The inventors are *thrilled*. Somebody finally believes in them.”

Naila felt Yusuf arrive at her shoulder before she heard him; he had a sense for the corner of a room where the actual transaction was happening. He’d gone still in a way she knew. Watching the good-quiet men the way you watch someone doing a thing you used to do.

“Why,” Naila said. “There’s no device. You said it yourself — none of it works, the physics forbids it. Why buy fifty patents on machines that can’t exist?”

The physicist looked at her, and at Yusuf, and something in the way Yusuf was holding his face must have told her the rest, because she answered him and not Naila.

“You’d know better than me, son.”

Yusuf didn’t deny anything. “A portfolio,” he said. “You don’t sell them as machines. You sell them as a *position*. We hold the world’s largest

collection of zero-point energy intellectual property.’ To investors who can’t tell Lamoreaux from a lamp. It’s not a lie you can sue. It’s a sentence that’s true and means nothing.” He exhaled. “And every one of those is also a man’s whole life he sold for a handshake and a feeling.”

The physicist took them out the fire door, into the loading bay, where the cold was real and not synthetic and the tube-buzz finally died.

She had a folder in her car. She’d brought it, she said, the way you bring a thing you’re not sure you’ll show anyone. Manila, swollen, the elastic gone slack. She didn’t hand it over right away. She held it against the quilted vest with both arms, the way you hold a thing.

“My husband’s,” she said. “And four of his friends’. And two students of mine, before I retired, who I should have warned harder.” She looked at the folder, not at them. “Prototypes. Every one of these is a prototype that didn’t work. Magnetic motors. A ‘Casimir cell.’ A water thing. A man in Sudbury who genuinely believed — *genuinely*, I sat at his table — that he’d get free electricity out of the gap between two gold-plated wafers.” Her jaw worked. “He wasn’t a grifter. That’s the part nobody wants. He was a sweet, careful man who read one true thing and built his whole heart on the part of it that isn’t there.”

She held the folder out. Naila took it, and it was heavier than paper.

“You want to know what I learned, doing this for thirty years?” The physicist pulled her glasses down off her head and folded them, a thing to do with her hands. “The word is real. That’s the cruelty of it. *Zero-point energy* is a real thing in a real paper, weighed to within a percent, and it sits there shining, and it pulls people in from two directions at once. The dreamers, who read the real part and can’t hear the *no*. And men like your friend used to be” — a nod at Yusuf, not unkind — “who read the dreamers and price them.”

A truck reversed somewhere out beyond the bay door, beeping.

“The grifters I can stand,” she said. “You can fight a grifter. It’s the dreamers that finish you. Because you have to tell a man who loved one true sentence that the sentence didn’t love him back, and watch him decide you’re the enemy. I stopped going to the funerals.” She put the glasses in her vest pocket. “So. There it is. Open it carefully.

The man in Sudbury's drawings are near the bottom and I can't look at them anymore."

Naila held the folder against her own chest, the same two-armed hold, mirroring her without meaning to. Inside her, the LED was still glowing blue in the rink, running on this morning's wall current, and the Casimir force was still real, still measured, still sitting in the literature being one of the strangest true things there is, and a sweet careful man in Sudbury had built his whole heart on the half of it that isn't there, and Naila could not for the life of her find where to put any of that down.

"The word got away from them," Yusuf said quietly. To the physicist. To the loading bay. "Out of the journals and into the rink. And once it's loose you can't ever get it back, you can only watch what it does to people."

The physicist looked at him for a long moment.

"You hold that folder like you might hand it to somebody worse," she said. "Don't. That's all I'm asking. You read it, and then you give it back to me, or you give it to someone who'll grieve it. Not someone who'll *price* it."

Behind them, through the propped fire door, the buzz of the tubes and the soft Maritime voice starting its next talk. *The vacuum is not empty.* True, every word. The blue light burning down its borrowed charge in the cold over the curling ice.

Chapter 7 — The Man Who Watched the Moon Breathe

The heating was off in the institute, and Edith had stopped pretending it would come on.

She kept her camera under her coat to spare the battery, the way she'd learned in colder places than Pulkovo, and watched the breath come out of the man across the table in slow grey ropes. His name was Lev Antonovich Vyrubov. He had been Nikolai Kozyrev's last graduate student, and he was eighty-one years old, and he had not yet decided what to do with them.

"You came a long way," he said, "to look at a man the Academy already finished looking at."

"We did," said Naila.

He waited for more. Edith had seen the keepers do this — Pogue's granddaughter had done it, the long silence held open like a door on a chain. Vyrubov had something better than suspicion. He had patience. He had outlasted a State; he could outlast a film crew from a van.

The room was a reading room the way a cell is a room: a table, a window, shelving that climbed into the dark above the reach of the one working lamp. Frost ferned the lower corner of the glass. Outside, the great refractor's dome sat against a sky the colour of an old coin, and below it the observatory grounds ran down to graves — the astronomers of the siege, Edith had read, buried where they fell defending the instruments. People had died here for lenses. It changed the way you held a camera.

She did not raise it.

The official story is clean, and Yusuf had laid it out in the van so they'd arrive knowing it cold.

Nikolai Aleksandrovich Kozyrev: real astronomer, no asterisk required. Gold-medalled. A lunar crater named for him on the far side, voted in by the people who name craters, which is the International Astronomical Union and not a man's mother. Co-discoverer, in his youth, of work on stellar atmospheres that still has his fingerprints on it.

And then the second life. The life that got the asterisk.

Kozyrev decided time was not a passive river. He decided it had a *flow* — a direction, a density, a thing that could be measured, that did work, that leaked into systems and changed them. *Causal mechanics*, he called it. He built mirrors to focus it. He pointed telescopes at where stars truly are *now*, not where their old light says they are, and reported he was getting signals.

"The physics is no," Naila said, in the van, before they'd ever come up the hill. She said it the way she'd said it of Pogue's two hundred miles to the gallon — not unkindly. A statement of weather. "There's no torsion field. The Novosibirsk work on it never replicated. A mirror doesn't focus time. The signal-from-the-real-position result has never come back when anyone else points the dish."

"*Crank*," said Frik, from the back, the word jumping out of him sideways, and then his face did the thing it did, the apology that lived behind his eyes. "Sorry. Sorry. It's not — I don't mean—"

"It's the word," said Edith. "It's the right word for the file. You're just saying the file out loud."

Frik nodded, miserable and grateful at once.

That was the file. *Wrong, then mad, then quiet*. The Academy looked, and named the crater anyway — you can put a man on the Moon and still not let him into the room — and stopped looking.

The potato was that the file had a number in it the file couldn't explain.

“You know the year they took him,” Vyrubov said. Not a question. He’d decided to give them this much and see what they did with it.

“Nineteen thirty-six,” said Naila. “The Pulkovo affair. The arrest sweep of the observatory.”

“Ten years.” Vyrubov set his hands flat on the cold table. They were big hands, an old worker’s hands, and Edith found herself reading them the way she read everything, the way she’d never been able to stop — the missing flexion in the left little finger, the way the knuckles sat. “He did them in the north. Norilsk. Dudinka. You understand what that is, the north?”

“Yes,” said Edith. She had photographed gulag survivors once, a long time ago, for a paper that ran two of her frames and spiked the rest as too quiet. *Nothing happening in them*, the editor said. There had been everything happening in them.

“They take an astronomer,” Vyrubov said, “and they put him underground. In the mine. They give him a sentence, and then in the camp another tribunal gives him a second sentence on top of it, for nothing, for words, and the second one is death.” He paused. “It was not carried out. This is the only mercy in the story and it is an accident of paperwork. He sat in a cell waiting to be shot and he did astronomy.”

“In a cell,” said Naila. Very flat. The flatness was how Edith knew it had reached her.

“He had no instruments. He had no sky — they kept him from the window. So he kept the field in his head. He worked stellar atmospheres on the wall of his memory because the alternative was to wait to die and feel it.” Vyrubov’s mouth moved. “When they wanted to break a man fully in those places, they gave him cold and they gave him nothing to think about. The thinking was the resistance. The Moon, later — all of it — you cannot read it if you do not first understand that the man learned to do physics *as the thing that kept him alive*. It was never a hobby. It was the last warm room he had.”

Edith’s hand was on the camera under her coat. She did not move it. She knew the shape of the photograph that was sitting in the room.

Suppressed Genius. The Astronomer They Couldn't Silence. You could sell that frame; outfits did, every day, the same outfit that was — Yusuf had heard — already on a flight, already making offers for the archive, the clean-handed kind that called itself a foundation and bought silence by the box. The frame would be true in every fact and a lie in its whole body. It would turn a man who survived into a man who triumphed, which is a different and crueller thing, because it tells the dead that the cold was *worth it*, that it *built character*, that the State did him a favour by feeding him to the north.

She let the camera stay cold.

He came back in 1947. Rehabilitated, eventually. Given a telescope again, by which point part of him had decided what time was, down there in the dark where time had been the one thing they couldn't take.

And then, the thing the file can't quite eat.

"Nineteen fifty-eight," said Naila. "November the third. Alphonsus."

Vyrubov looked at her for the first time as though she were a person and not a problem.

"He was at the great reflector," he said. "Watching the crater Alphonsus. And in the central peak he saw a — a reddening. A patch, gone slightly red, slightly bright, and then a return to grey. He had the spectrograph running. He caught the spectrum."

This was the part Naila had not been able to say *no* to in the van, and Edith had watched her try.

Because the Moon is dead. Everyone knows the Moon is dead — airless, cooled, geologically finished, a stone that keeps the same face turned to us out of tidal courtesy and does *nothing*. That is the textbook, and the textbook is mostly right. And on the night of 3 November 1958 Nikolai Kozyrev, the disgraced mystic, the man with the time-mirrors, the crank, pointed a real instrument at a real crater and recorded a spectrum of gas — carbon, the molecular signatures — venting from the central peak of Alphonsus. A *transient lunar phenomenon*, caught, for once, not by an eyeball that could be fooled but by a slit and a plate that could not.

The dead Moon had exhaled, and the one man watching was the one man nobody would believe.

"They didn't believe him," said Naila.

"At first, no. A mystic sees the Moon breathe — of course. Convenient." Vyrubov's voice had no bitterness in it, which was somehow worse. "And then others looked, with care, and the spectrum held. It is in the record now. Outgassing. It is not woo. The Moon is not so dead as the book says, in the small ways, sometimes, and he saw it." He spread the big hands. "The same man. The same eyes. The breathing Moon is real and the time-mirror is nothing, and they came out of one head, in one life, and the world decided which one to remember by deciding what to do with the *man*."

The lamp buzzed. Far off in the building a pipe knocked, hoping.

"*Couldn't pick*," Frik said softly, the blurt gone gentle this time, almost finishing the old man's sentence. "Sorry — the world. It couldn't pick which one of him was the real one, so it threw out both."

Nobody corrected him, because he had said the file and the hole in the same breath, and Vyrubov was nodding, slow.

Yusuf had been quiet against the shelves, which was Yusuf doing his job. Now he said, lightly, "Someone else has been asking to see the archive."

It was not quite a question.

"A foundation." Vyrubov pronounced the word like a coin he'd bitten and found soft. "From the west. They wish to *preserve*. They have come twice. They photograph the catalogue and offer — assistance. They are very polite. They want the mirror most of all." He looked at Edith. "You want the mirror too. Everyone wants the mirror. The mirror is the thing you can put in a photograph."

The mirror was in the room.

She'd clocked it when they came in and made herself not look at it, the way you make yourself not look at the one face in a crowd you've been sent to find. A shrouded shape on the lowest shelf, draped in grey cloth gone stiff with cold — an aluminium thing, she'd been told,

a focusing geometry, a *causal* mirror, built to gather the flow of time onto a detector and prove the future leaned on the present.

“Does it work,” said Naila. She had to. It was the question she carried for all of them, the one she could hold in one hand while holding the man in the other.

“No,” said Vyrubov.

He said it without flinching, and it cost him nothing, and that was the whole dignity of him, that he could love the work and bury the claim in the same sentence, which is a thing the foundation would never do and the tabloids would never do and Edith was not certain the crew could yet do, all five of them, all the way down.

“It focuses nothing,” he said. “I have spent thirty years not lying about this, which is harder than you think when men with money keep arriving to be lied to. The mirror is what a great mind built when it decided the universe owed it more than the universe owes anyone. It is the most human object in this room and it does not function.” He laid one finger on the edge of the cold table. “The Moon, though. The Moon he caught.”

The light was failing in the window and they had not been thrown out, which Edith understood was the test passing, slowly, in real time. She had not raised the camera once. The mirror sat shrouded. The dead astronomers slept under the snow down the hill, having given their lives for lenses and got, in return, a footnote and a frost.

Vyrubov stood. His knees did not want it and he did it anyway, the way he had probably done everything for sixty years.

“You did not photograph him,” he said to Edith. “The man who came before you — the foundation’s man — he photographed everything before I had finished the sentence. He had the picture in his head when he walked in. You walked in with no picture.” He almost smiled. “This is rare. Or you are very patient liars.”

“We might be,” Edith said. “But not today.”

He went to the deep shelving, to a place the lamp didn’t reach, and Edith heard the slide of a long flat box, the particular shush of archival

card, of something kept the way you keep a thing you have decided to outlive the State for.

“He kept the plates,” Vyrubov said, into the dark. “The originals. The spectrograms of the third of November. Not the published prints — the plates from the slit, his hand on the labels, the night he saw it breathe.” The box came into the light. It was older than anyone in the room except him. “The foundation has never seen these. I tell them the originals are lost.” He set the box on the cold table, between the four of them and the shrouded mirror, and rested both worker’s hands flat on the lid as if warming them. “They are not lost.”

He looked at Naila, and then, longer, at Edith and the dark shape of the camera she had not used.

“You will not sell him,” he said. It was not a question and it was not quite a request.

“No,” said Edith.

He lifted his hands from the lid.

Chapter 8 — The Mirror and the Crater

The institute kept its cold the way some houses keep their dead — with a kind of formality, as if the temperature were a guest who had outstayed everyone and could no longer be asked to leave. Edith counted four radiators down the corridor, each one a flaking grey accordion bolted under a window, and not one of them gave off so much as the memory of heat. Her breath stood in front of her. She had photographed colder rooms. She had never photographed one that felt this *deliberate*.

“They turn it off in May,” said the keeper. “On paper, May. In practice, whenever the budget runs out, which this year was March.”

His name was Arkady Vasilyevich, and he was not what the dossier had let Edith expect. She had half-built him in the van — the stooped survivor, the watery-eyed believer guarding a shrine. He was none of that. He wore a good wool coat done up to the throat and a watch that worked, and he moved through the building like a man doing inventory of his own grief and finding it all present and accounted for. He had been Kozyrev’s last graduate student. He was sixty-one. He had outlived the man by forty years and the country they had both worked for by a little less.

“You can take your gloves off in here,” he said, opening a door. “There’s nothing fragile. He didn’t believe in fragile.”

The room had been a laboratory and was now mostly an archive, which is to say it was a room where work had stopped and paper had been allowed to accumulate in its place. Long benches under fluorescent tubes, half of them dark. A wall of card-index drawers.

And on the central table, laid out the way Arkady must have laid them before they came, three notebooks, bound in the pebbled black board the Soviet stationery combines had turned out by the ten million, the spines repaired with cloth tape gone the colour of weak tea.

Naila was already standing over them. She had not touched them. She had her hands behind her back, the way she did at the patent table in Winnipeg, reading first, wanting later.

"These are his," Arkady said. It was not a question and not quite a permission. "1958 to 1963. The astrophysics. Before the rest of it."

Here is the official story, and it is a true one, and it is a story the men who tell it tell with their chests out.

Nikolai Aleksandrovich Kozyrev was a real astronomer — not a garage tinkerer, not a man with a typo for a patent number, but a working observational scientist of the first rank. The State arrested him in 1936 and the State was wrong to, and the State buried him for ten years in the camps of the far north, and when he came back he did real work that real astronomers cited. On the third of November, 1958, at Crimea, he turned the spectrograph on the central peak of the lunar crater Alphonsus and recorded an emission — a gas, a glow, the Moon exhaling after four billion years of holding its breath. Lunar transient phenomenon, the textbooks call it now, dry as the surface itself. In 1969 the International Academy of Astronautics gave him a gold medal for it. There is a crater on the far side of the Moon named Kozyrev. All of this is in the record. None of it is in dispute.

And then there is the other thing. The thing that came after.

Kozyrev decided that time was not a coordinate but a *substance* — that it flowed, that it had density and a direction, that it could be reflected and focused, that a star burned not because it fused hydrogen but because time poured into it. He built mirrors. Aluminium spirals, helical baffles, instruments to concentrate the current of time the way a dish concentrates radio. He claimed his detectors registered the *true* position of stars — not where the light says they are, but where they are *now*, the signal arriving instantaneously because it travelled not through space but through time.

The official solve handles this cleanly. The mirror experiments were

never independently replicated. Not once, under controls anyone outside Novosibirsk accepted. The “time flow” has no mechanism that survives contact with thermodynamics. The whole second life is filed, gently or not, under *the tragedy of a great man’s decline* — the genius the camps had cracked, doing brilliant work and then beautiful nonsense, and the kindest thing is to remember him for the Moon.

That file closes. Edith had read it three times on the flight. It closes very well.

What it does not do is account for the room she was standing in.

“Show her the medal,” Naila said, without looking up from the notebooks. “She’ll want the medal first.”

Edith glanced at her. Naila had not asked what Edith wanted; she had simply known, the way she knew the grammar of a patent, that the photographer would go for the object that proved the man before the object that strained him. It was not unkind. It was just accurate, and being read that accurately by someone who would never weaponise it was one of the few things in Edith’s life that did not cost her anything.

Arkady went to a steel cabinet, worked a key that did not want to turn, and brought back a flat case the size of a hardback book. Inside, on velvet that had once been blue, the gold medal sat heavier than it looked. Edith could tell it was heavier than it looked because of the way he held it — two hands, from underneath, the carry of a man who knows the true weight of a thing the camera will lie about.

“You may photograph this,” he said. “People believe the medal. It is the rest they do not believe.” He set it down on the velvet, under the working half of the fluorescents, and stepped back to let her work.

She shot it the way she had taught herself to shoot the things that were *meant* to settle an argument — flat, even, no drama, no rim light making it heroic, no shadow making it sinister. Just the object, the embossing, the year. 1969. A medal is a fact. She had spent a career learning that a fact, photographed with the slightest lean in the light, becomes an opinion. She kept the light honest.

“He wore it once,” Arkady said. “At the ceremony. Then it lived in that cabinet. He said the Academy gave it to him for the part of his work

that had stopped surprising him.”

Naila had opened the first notebook now, with the back of one finger, the way you'd open the wing of something you were not sure was dead.

“This is the Alphonsus run,” she said. Her voice had gone level and fast, the register Edith thought of as Naila *downhill* — when the pattern caught and she stopped having to translate herself for the room. “Look. He's not claiming anything. This is just an observing log. Plate numbers, exposure times, the seeing, the airmass. Here — he's annoyed the guiding slipped on the second plate.” She turned a page. “He's *bored*, half of this. It's a working night. And then —” her finger stopped over a line of Cyrillic and a sketch, a small dark crater drawn with a child's care and an expert's hand. “There. The emission. He doesn't even underline it. He notes it like a man noting the milk's gone off.”

“That is what I cannot make people understand,” Arkady said. He had come up beside her without Edith hearing him cross the room. “The Moon was the easy part. He saw a thing nobody believed the dead Moon could do, and he was right, and they gave him gold for it. So when he saw the next thing —” He stopped. “He had already learned that the world being certain meant nothing. The camps taught him the world's certainty meant nothing. He had a medal that proved the world's certainty meant nothing. You understand the problem this creates in a man.”

It was, Edith thought, the most dangerous education a person could receive: to be disbelieved, and buried for it, and then proven right *with a medal*. It would teach you the wrong lesson with the authority of the right one.

She did not say this. The themes file in her own head — she had her own — forbade it. She lifted the camera instead and shot Naila's hands over the open log, the breath fogging between her fingers and the page, the small drawn crater. One frame. She did not caption it, even to herself.

Frik had been quiet by the card-index for a long time, which with Frik was its own kind of weather. Now he said, to nobody, to the drawers,

in the flat helpless way the words came when they came: “*Wrong. He was wrong.*” And then, lower, his own voice climbing back over the other one, mortified: “Sorry. Sorry, that’s not — I don’t —”

“You are not wrong to say it,” Arkady said, mildly. “He may have been wrong. About the time. I think, on the bad days, he may have been wrong.” He let that sit in the cold air, and did not rush to take it back, and Edith found she respected him enormously for the not-rushing. “That is allowed to be true.”

Frik’s shoulders came down an inch.

“Tell us about the mirrors,” Naila said. The downhill voice had a hunger in it now, and Edith watched Arkady hear the hunger and decide what to do with it.

He went to the wall of drawers and pulled one — D-tray something, Edith couldn’t read the label — and brought out not a mirror but a photograph of one. A black-and-white print, institutional, a man’s hand for scale beside a coiled aluminium spiral the size of a dinner plate, helical, like a flattened spring, like the inside of a shell.

“He believed time was a current,” Arkady said. “Not a measure. A *substance* that flows, like water flows downhill, and that flowing does work — it is what makes the difference between cause and effect, the asymmetry, why you cannot unbreak the cup. He believed a process that *creates* order somewhere must shed time somewhere, and a process that destroys order absorbs it, and that with the right mirror you could *aim* the current. Concentrate it. Read with it.” He set the photograph beside the notebooks. “His detectors, he said, found the star three times. Where it was. Where its light says it was. And where it will be. The middle one is the only one physics admits.”

“And nobody got the same numbers,” Naila said. Not a challenge. The potato, stated plainly, the way she always made herself state it before she let herself want anything.

“Nobody outside got the same numbers.” Arkady did not flinch from it. “The replications failed. Or were never run properly. Or were run by people who had already decided. I have read every objection. Most of them are correct.” He looked at the spiral in the photograph the way Edith had once seen a war widow look at a folded flag — fully,

without illusion, and without letting the lack of illusion touch the love. "The mirror is not science any more, you understand. Science is the part other people can do again. No one could do this again. So it is not science."

"But it's still *here*," Naila said quietly.

"It is still here," Arkady agreed.

Edith lowered the camera. She had not decided to. The body did it — the old reflex she was supposed to be unlearning, the lens coming down off the eye when a room turned into something that should not be photographed yet, or perhaps ever. The light through the dead fluorescents and the dirty windows was lying down across the bench, across the spiral in the print and the bored brilliant log and the medal that proved the world's certainty meant nothing, and she let it lie there unphotographed.

Hecklefish would have said something. Somewhere in the back of Edith's skull a voice that was not Arkady's and not hers observed, dryly, that *the mirror found the star in three places and the grant committee in none*. She let that go too.

Arkady straightened. He looked at the four of them — Naila over the log, Frik by the drawers with his hands jammed in his pockets, Yusuf in the doorway who had not come all the way in and who Edith knew was already pricing what someone else would pay for these three notebooks, and Edith herself with her camera at her hip like a holstered thing.

"There is one more room," he said. "It is not in the catalogue. The institute does not list it, because the institute is embarrassed by it, and the institute is right to be embarrassed, and I keep the key anyway." He produced it — a long iron thing, older than the steel cabinets, older maybe than the cold. "The mirror chamber. He sat in it. He had his students sit in it. They were supposed to feel the current — a lightness, a heaviness, the hair on the arm. Most felt nothing. Some felt something and could never say what, and spent the rest of their lives not saying it." A pause. "I felt something. I have never decided what. That is my whole honest report, after forty years: *I felt something and I have never decided what.*"

He turned the key over in his fingers.

“You came to read him,” Arkady said. “You can read all of this from out here, and write it down, and you will be accurate, and it will be true, and it will close the file the way the file always closes.” He held the iron key out, flat on his palm, toward the room, not toward any one of them — leaving it for them to take or not. “Or you can come and sit where he sat. I cannot tell you what it is for. He could not tell me what it is for, and I loved him, and I asked him every week until he died.”

The cold breathed in the corridor behind them.

“It is the one thing,” Arkady said, “I have never let anyone photograph. So leave the camera, and bring yourselves, and we will go and sit in the dark in a mirror that does nothing, that a man who caught the Moon breathing believed would let us feel the current of time go by.”

Edith looked at the camera at her hip. Then she set it down on the bench, beside the medal, lens to the wood, and stepped toward the door he was holding open onto the dark.

Chapter 9 — A Silence in the Academy

The chamber was a coffee can.

That was the first thing, and Edith made herself hold it, the way she held everything now before she let the camera near it: not a portal, not a resonator, not the throat of time. A length of aluminium gutter-stock rolled into a cylinder, maybe forty centimetres tall, the inside lapped to a brightness with what looked like a thousand hours of hand-polishing, set on a wooden cradle a carpenter would have called rough. The Pulkovo people had built better. This one Kozyrev's student had built in a kitchen in 1971, and Lidiya Arkadyevna — the keeper, the niece, the woman who owned the silence in this room — had kept it on a shelf above the radiator that did not work for fifty years.

"You may put your hand in it," Lidiya said. "Everyone wants to. He let them. He was not precious about the hands."

The institute corridor outside ran cold and fluorescent, the heating off in great economical stretches, Saint Petersburg pressing grey at every window. Inside the little side-room it was no warmer, but the cold felt curated, like a thing being preserved. Bashir had set the tripod and stepped back. Naila stood with her arms folded so tight it read as a held breath, reading the cradle, the seam, the polish — reading it as a *made thing*, which it was, which is all Edith trusted any of them to say about it.

Yusuf put his hand in first. Of course he did.

"Cold," he reported. "Aluminium cold."

“And?” said Lidiya.

“And nothing.” He grinned, not unkindly. “I want to be honest with you. I’ve stood in a room in Sedona where a woman charged forty dollars to feel a vortex. I felt forty dollars.”

“Good,” Lidiya said. “Kozyrev would have liked you. He hated the forty-dollar men more than he hated the Academy. The Academy at least had the decency to be wrong on the record.”

Edith lifted the Leica and did not press the shutter. This was the part the series kept asking her to do — *lower the camera* — and the part she kept refusing, and the refusing had become its own small daily war. A hand in a polished can. Frame it tight, you got mysticism. Frame it wide, you got an old woman’s kitchen craft on a broken radiator shelf and the sneer wrote itself. Neither was true. Both were checkable. She let the shutter sit cocked under her thumb and felt the want in it like a pulse.

“What did *he* say it did?” Naila asked. Her voice came flat and exact, the question stripped of everything but the question, and Edith watched Lidiya’s face change, soften by a degree, the way the keepers always softened at Naila — they could hear she wasn’t performing interest.

“He said,” Lidiya answered carefully, “that the properties of a body change when a process happens nearby. That when something *becomes* — when ice melts, when a man dies, when sugar dissolves — time, he said, does work, and the work is not nothing, and a sufficiently sensitive instrument set near the becoming would feel it.” She turned the cylinder a quarter on its cradle. “He weighed gyroscopes. He measured the column of mercury near dissolving salt. He pointed a telescope mirror not where a star *was* but where it would be, and where it had been, and said he caught a signal at all three.” She paused. “He published it. The Academy of Sciences read it.”

“And the Academy,” said Yusuf, gently, because someone had to play it straight, “said?”

“The Academy said *no*.”

The Academy was right.

The book had a rule about this and Edith carried it now like a stone in the pocket — *play the official story dead straight, and with respect*. So: in 1978, a commission of the Soviet Academy of Sciences reviewed Nikolai Aleksandrovich Kozyrev's causal mechanics, his torsion of time, his asymmetric gyroscopes, and found no there there. The effects sat at the edge of his error bars. The mercury column twitched in ways a draught explained. The gyroscope's weight change was milligrams on instruments that drifted milligrams. Nobody outside Pulkovo reproduced the column of mercury or the breathing of the stars, and inside Pulkovo it was always Kozyrev's own hands on the apparatus, Kozyrev's own eye at the scale. The verdict, translated and folded into the Western literature, was the verdict the whole world now repeated and could repeat with a clean conscience: *mysticism in a lab coat*. Time does not have a density. Time does not do work. A gallon of grief does not turn a gyroscope.

Naila said all of this. She said it standing in the cold side-room with her arms still folded, and she said it without cruelty because to Naila it was simply load — the physics is against him, the physics has always been against him, conservation does not flinch for anyone's sorrow. Edith had learned to listen for the moment Naila reached the edge of what she'd defend, and there was no flinch in her. The mechanics were not real. Naila could not make them real and would not pretend.

"He was a crank," Naila said, finally. Not a verdict. A test of the word in her mouth, holding it up to the light.

And Lidiya, who had let them say all of it, who had stood very still while a woman she'd met an hour ago dismantled her uncle's life's work in level, exact Russian, said:

"No. He was a man in a crater."

She made them sit for the next part. There were three kitchen chairs and a stool, and Frik took the stool because Frik always took the worst seat as a courtesy nobody had asked for, and Lidiya stood, because the standing was part of it.

"Let me tell you what is true and not in dispute," she said, "and then you may keep your physics, and I will keep the rest."

Edith filmed this. Lidiya had said she could film *this* — the words, not

the can.

It was true that Kozyrev had been one of the most gifted astrophysicists of his generation; that as a young man he had done foundational work on the energy sources of stars; that in 1936 the State had arrested him; that he had spent ten years in the camps, in Norilsk, in the dark above the Arctic Circle, sentenced for things that were never things. It was true — Lidiya said this slowly — that during the famine in the camp they had taken away his books and that he had recited the night sky from memory to keep his mind, and that he had decided, in the dark, that the energy of the stars could not come only from where everyone said it came, and that *something* fed them, and that whatever fed them might be the thing the rest of physics had forgotten to weigh: the passing of time itself.

“He built his entire heresy,” Lidiya said, “to keep himself alive in a hole the State had put him in. You understand. The theory is the rope he climbed out on.”

It was true that he was released and rehabilitated, that he returned to Pulkovo, that the State that had buried him later put his name on a crater on the Moon. And it was true — this one was in the published record, *Sky and Telescope*, November 1958, and Naila had it on her phone before Lidiya finished the sentence — that on the third of November, 1958, Nikolai Kozyrev pointed a spectrograph at the lunar crater Alphonsus and recorded an emission, a gassy reddish glow, evidence of outgassing on a Moon that everyone knew, *knew*, was a dead stone. That for years the West called it impossible, a plate fogged, a man seeing what he needed. And that decades later an orbiting probe found the Moon does, in fact, exhale — transient gas, radon, a slow geology nobody had granted it.

“On the *Moon*,” Lidiya said, “the dead thing, the thing everyone wrote off — he caught it breathing. And nobody believed him for thirty years, because of who he was, and because of the other thing, the time thing, the thing you have correctly told me is wrong.” She let that sit in the cold air. “Do you see the cruelty of it? The same reaching that made him wrong about the gyroscope made him *right* about the Moon. You cannot have the one without the other. He was not a careful man. He was a man who could not stop reaching, because reaching was the only thing the camp could not take.”

Frik had gone still on his stool. Edith caught it in the corner of the frame and did not turn the camera toward it, because there were things you did not frame, and a man recognising his own wiring in a dead stranger's was the chief of them. Naila uncrossed her arms. Across the little room a thing passed between the two of them that had no words and needed none, and Edith felt the old ache that was not envy and was not pity, watching the family she'd been let into find its reflection in a crater on the Moon.

"I do not ask you to believe his physics," Lidiya said. "His physics is dead and I have made my peace with the death of it. I ask you only to write down that the grief was real. That is all anyone ever buried — not the machine. The man."

The offer came at the door, on the way out, the way they always came — softly, with a coat already on.

His name was Crane. He had been at the institute that morning "as a private scholar," which meant nothing, and he had a card from a foundation in Geneva that funded "the preservation of marginalised scientific heritage," which meant something. Yusuf had clocked him in the corridor before any of them and gone quiet in the particular way that meant *money in the room*. Crane was clean-handed. That was the thing about Crane. He genuinely loved the archive. He wanted, he said, to save it — the Kozyrev papers, the camp letters, the spectrograms, the unpublished pages — from the damp and the dead radiator and the indifference of a State that had killed the man twice and would happily let his paper rot. The foundation would catalogue, climate-control, digitise. The foundation would pay. The figure he named to Lidiya, low and private, made Yusuf exhale through his nose.

"It is a kindness," Crane said, and almost meant it. "You are an old woman in a cold room. This belongs to the world."

Lidiya looked at him for a long moment.

"It belongs," she said, "to the quiet."

She did not explain it. Edith understood that *she* understood it — that the foundation in Geneva would do exactly what it said, would preserve every page beautifully, would also own the right to decide

which pages were *interesting*, and that the breathing Moon would be lovely on a website and the camp letters would be a footnote and the grief, the actual grief, would be conserved out of existence by people who could not bear to leave a thing unframed. Crane would save the man by finishing the burial. And he would never know he had done it.

“No,” Lidiya said. To the money. To the kindness. To Geneva.

But she did not let them leave empty.

She crossed to the shelf above the dead radiator and took down a folder Edith had not seen her touch all day — oilcloth, tied with a bootlace — and laid it on the kitchen table and untied it. Inside, one page. Cyrillic, in a hand that shook, dated in pencil, the last thing.

“His last paper,” Lidiya said. “Not published. He withdrew it. He decided, at the end, that he had not earned it — that the measurements were his hands and his hope and nothing a stranger could lift.” She smoothed it flat. “You may copy it. Photograph it, every word. Take it with you. Read it to whoever will listen. The reaching should not be lost — only the *claim*.”

Naila photographed it. Bashir lit it flat and shadowless and Edith stood back and let Naila do the work because it was Naila’s to do, the close exact passes, the corners, the pencil date. Edith took one frame of her own and it was not of the page. It was Lidiya’s hands at the table’s edge, flat on the oilcloth, not holding the paper, just near it.

When the copying was done Lidiya retied the folder with the single page back inside, and carried it to the stove — a real stove, gas, the one warm thing in the apartment — and lit the ring, and held the corner of the page to the flame.

Yusuf made a sound. Naila did not. Naila had the photographs. Naila understood the photographs were the point; that the original was the thing Crane could buy and the copy was the thing he could not; that a man who withdrew his own last paper because he had not earned it deserved to have the *paper* honoured by being let go and the *reaching* honoured by being carried.

The page took the flame and curled and went. Lidiya watched it the whole way, until there was nothing in her fingers but a corner too small to hold, and she dropped that in the sink.

“Now,” she said, “no one owns it. Now it is only what he reached for, and you have that, and they cannot buy what does not exist.” She turned off the gas. The ring ticked, cooling. “He went quiet because the Academy told him to, and he obeyed, because the camp teaches you to obey the men who can take your books. I am the only one left who knows what the silence cost.” She looked at them, at the camera Edith had finally, quietly, lowered. “Write that down. Not the can. The cost.”

Outside, the light had gone the colour of pewter and the heating in the corridor was still off and somewhere down it Crane was making a phone call to Geneva that Edith did not need to hear to know the shape of.

She still had the want in her thumb. The frame she hadn’t taken. The hand in the can.

She let it go.

Chapter 10 — The Paper That Dissolves

The institute kept its archive in a room the heating had forgotten.

Edith felt it through her boots first, the cold rising out of a concrete floor that had been poured in some decade when the State still believed in poured concrete. The radiator under the window was the colour of old teeth and just as warm. Anastasia Volkova — the keeper, Kozyrev's last graduate student's daughter, the woman who had not let go of the boxes when the institute tried three times to absorb them — wore fingerless gloves and a coat indoors, and she watched the crew arrange themselves around the reading table the way you watch people you have decided, provisionally, not to throw out.

"You may photograph," she said. "Not with flash. The flash is worse than the years."

Edith lowered the camera to her chest by its strap, a small surrender she had been making all book and was getting no better at. The official story of Nikolai Kozyrev was, by now, one the crew could recite among themselves like a grace before a meal. A real astronomer. A real gold medal from the Royal Astronomical Society. A real crater on the Moon with his name on it. A real eight years in the Gulag for the crime of being an astronomer when astronomy had become inconvenient, and then — this was the part the textbooks left out and the cranks left in — a real night in November of 1958, at the Crimean Astrophysical Observatory, when the man pointed a spectrograph at the crater Alphonsus and caught, on a photographic plate, what looked for all the world like gas venting from a dead world. *The Moon breathing*. Carl Sagan had cited the plates. Sagan, of all people, the patron

saint of the cold eye.

And then the second half of Kozyrev's life: the part where he decided time had a flow, a density, a *current* you could weigh — and spent forty years building mirrors and gyroscopes to prove it, and the Academy weighed him instead, and found him wanting, and the file closed.

"The torsion stuff is nonsense," Naila had said in the van, flat, the way she said things that were true and cost her nothing to say. "Unrepliated. The Novosibirsk lot ran with it and it never goes anywhere." She'd paused. "The 1958 plates are real, though. The crater's real. The medal's real." She always laid the real out like that, in a row, so nobody could pretend she'd been soft on the woo.

Tomi had said nothing in the van. Tomi was saying nothing now, leaning over the box Anastasia had set on the table, hands flat on the wood, not yet touching anything.

The plates were there. That was the gift of the visit, and Anastasia gave it without ceremony — a folder of glass-plate scans, the 1958 spectra, the dark line of the venting smeared faint across the emulsion. Naila bent over them with the particular stillness she got, the one that meant a thing was holding up to her. "There," she said. "That's the emission band. That's not nothing." She looked up at Anastasia, and something passed between them, two people confirming a fact in a cold room, neither needing the other to be impressed.

"It is published," Anastasia said. "Sky and Telescope, 1959. You can find it anywhere. People find it and then they write that he proved the Moon is alive." Her mouth went thin. "He proved a spectrograph caught something on one night. That is a smaller and better thing."

The fish, who had ridden in on Yusuf's shoulder and was currently using Tomi's coffee as a reflecting pool, surfaced. "*Smaller and better. Put it on a t-shirt. Sell it next to the SUPPRESSED GENIUS mugs.*"

"Nobody's selling anything," Edith said, before Anastasia's eyes could narrow further. It was true. It was the whole reason the woman had let them past the door. They'd shown her the Pogue chapter, the way Edith had photographed a dead man's patent folder like a body, with respect, not appetite. Trust was a currency you spent and spent and never quite owned.

Then Anastasia reached into the box and took out the last folder, and held it a moment longer than the others before she set it down.

"This one," she said, "I cannot give you."

It was the late paper. The torsion work — the time-flow manuscript, the one with the diagrams of the mirrors, the columns of measurements from the gyroscope rigs. The one the believers wanted and the skeptics dismissed and the Academy had, in its careful institutional voice, declined to take seriously. The crew leaned in to look, and saw why she had held it back.

It was rotting.

Not torn, not burned, nothing so dramatic. Damp. A pipe had gone in the seventies, Anastasia said, in the year after Kozyrev died, when the boxes were in a basement nobody had budget for. The water had found the bottom folder and sat in it. The pages had stuck and then, when someone finally pried them apart, taken each other's faces off — the ink lifting, the emulsion of the diagrams gone to a brown bloom, whole paragraphs reduced to a topography of stain. Mould had drawn its own galaxies across the margins. You could see, here and there, where a column of figures had been, a ghost of verticality, the comb-tooth rhythm of numbers — but the numbers themselves had dissolved into the paper that held them, ink and pulp becoming one undifferentiated ruin.

Tomi made a small sound.

"May I," Tomi said, and then, when Anastasia hesitated, "I won't touch it. I won't. Just — let me look."

She nodded.

Edith knew what was about to happen and stepped back to give it room, lifting the camera now because this was the thing worth keeping. Tomi read the way Tomi always read — the want out in the open, the hunger to make the broken thing whole. Tomi found the ghost of a column and started building it. "That's a decimal, see the dot's survived, so this is — point something, and the digit above it has a flat top, that's a five or a seven —" leaning closer, breath held so as not to even *breathe* on it. "If I had a raking light. Anastasia, do you

have a lamp, just a desk lamp, low angle — the indentation, the pen pressed through, you can read the impression even where the ink's gone —”

Anastasia got the lamp. She wanted it to work as much as Tomi did, Edith could see; the keeper's hand shook a little setting it down. They raked the light across the dead page together, the keeper and the reader, and for a moment there was something — a furrow, a stroke, the bones of a 4. Tomi wrote it down. Then another character, and the next furrow crossed it, two passes of the pen from two different lines pressed into the same fibre, and you could not tell which stroke belonged to which thought, and the 4 became a smear of maybes.

Tomi kept going. Found a word that might have been *plotnost* — density — and then the word beside it was gone, simply gone, a brown nothing, and a word with nothing on either side of it is not a sentence, it is a stone with one letter on it.

The fish, very quietly, did not heckle.

Yusuf was the one who said it, because Yusuf was the one who had spent a former life knowing exactly what a missing number was worth. “Tomi.” Just the name. Tomi didn't look up. “Tomi. There's no copy.”

“There's always a copy,” Tomi said. “He'd have made a fair copy, a man like that, he'd have — institutes keep things, there's a carbon somewhere, a draft, the journal he submitted to —”

“He never submitted it,” Anastasia said.

The room got, if possible, colder.

“He stopped.” She folded her gloved hands. “After the Academy. After the verdict. He had the mirror results, the gyroscope results — the numbers you are looking for, the actual measurements, the thing that would say yes *he was a crank* or *no there is something here* — and he did not publish them. He wrote them down, in this.” She did not touch the ruined folder. “And then he was quiet. For the rest of his life, on this work, he was quiet. Some say he was afraid. Some say he was finished, in the heart, after what they said to him. I think —” and here she stopped, and looked at the brown bloom of the pages, and chose her words in a second language with great care. “I think

he kept them for himself. Because they were the only proof he had that the years had not been for nothing. And he did not want to give the Academy the satisfaction of weighing them again.”

“And the pipe took them,” Edith said.

“And the pipe took them.” Anastasia’s voice did not crack. It had had decades to learn not to. “So now no one will ever weigh them. Not the believers. Not the doubters. Not me.” She looked at Tomi, not unkindly. “You cannot reconstruct it. I have tried, for nineteen years. There is nothing under the stain. The number is in the paper, and the paper is gone.”

Tomi stopped reading.

It was the stopping Edith photographed, in the end — not the page. Tomi’s hands coming off the table, settling into Tomi’s own lap, the pen set down with a click that was loud in that concrete room. The face doing the thing Edith had seen it do exactly once before, in Winnipeg, over a folder that had a hole in it where a number should be. The world keeps handing this one holes, Edith thought, and did not say.

Naila said the true thing, because Naila could always be trusted with the true thing when the room needed it and no one else could lift it. “The torsion work was probably wrong anyway.” Quiet. Not cruel. A row of facts laid out. “It doesn’t replicate. The mechanism doesn’t make sense. If we *could* read those numbers, the most likely thing is they’d just confirm he was chasing something that wasn’t there.” She looked at the ruined folder. “That’s the most likely thing.”

“Yes,” Anastasia said.

“That’s still a loss,” Naila said.

“Yes.”

Nobody added anything to that. Frik, who had been holding himself very still by the door the way he did when a place mattered too much to risk his throat in it, let out a breath that was almost a word and then wasn’t. The radiator ticked. Outside the window the light was going the flat Siberian grey that is not sunset, only subtraction.

They sat with it. That was all there was to do, and so they did it — the crew of people the world had no use for, and the woman who had no use the world could see for nineteen years of keeping a dead man's basement boxes, all of them around a table with a brown ruin in the middle that had once been a column of numbers that would have settled everything. The cold came up through the floor. Edith did not lift the camera again. There are things you photograph and things you simply attend, and the discipline she was learning, all this book, was the second one — when to be a witness with your hands empty.

Tomi reached out, at the very end, and did not touch the dissolved page, but laid two fingers on the table beside it, the way you set your hand near someone in a hospital bed without taking their hand, because the touching is for you and not for them.

They left with the lunar data.

Anastasia copied the 1958 plates for them onto a drive with a label written in Cyrillic in a hand that might have been her father's — the spectra, the venting, the real and published and citable thing, the smaller and better thing. *The Moon breathed and a man caught it on glass*. That part of him was safe; that part of him was in Sky and Telescope and would outlast all of them. She put the drive in Edith's hand specifically, and Edith understood the choice — the one who'd kept the camera down was the one trusted with the picture.

"You will not write the other thing," Anastasia said. Not a question.

"What other thing," Edith said.

The keeper almost smiled. She walked them to the door of the room and no further, staying with her boxes as the crew filed back into the corridor where the heating, marginally, remembered to exist. Tomi went last and stopped on the threshold and looked back at the table.

"What was it for," Tomi asked. Not the machine. The whole of it — the mirrors, the silence, the kept and rotting folder. The question that was Jakobus's question, that the crew had been built to ask, though Jakobus was three time zones away minding the Beast and would have, had he been there, said nothing at all and let the moment go to the woman who owned it.

Anastasia thought about it.

“For him,” she said. “Only for him. He built a thing to tell himself the years were real.” She put her glove flat on the lid of the box. “Some things are not for us. They are not a puzzle. They are a door, and the man closed it, and then the water came, and now it is shut from both sides.”

She turned back to her cold room.

The fish, riding out on Yusuf’s shoulder into the grey, looked back once at the closed door and, for the second time that day, declined to say a word.

Chapter 11 — The Speed of Lies

The conference centre carpet was the colour of a bruise that hadn't decided yet, and it ate sound the way Yusuf remembered, the way they always did — so that when a man at the next booth said *quantum coherence* into a lapel mic, the words landed soft, almost kind, the way a lie lands when the room is engineered to swallow the edges off it.

He knew this room. Not this one specifically. He knew the *kind* — the regional wellness expo in a city that wasn't quite a destination, San Diego's hotel belt, the airport shuttles, the lanyards, the pyramid of free-trade pamphlets nobody read. He'd worked rooms like this from the inside for eleven years. He could still feel the muscle of it in his face: the open, frictionless smile that said *I am not selling you anything, I am sharing something*.

"You've gone quiet," Edith said. She had the small camera down at her hip, not raised. She kept it low in places like this, on purpose. "That's new."

"I'm reading the floor."

"You're reading your CV."

He didn't argue with her. Across the aisle, a stand the colour of seafoam advertised the **TachyoLife Pendant** — *Restructures your body's biofield using stored tachyonic energy*. A woman in white linen pressed one against a volunteer's spine and the volunteer's arm, when pushed, held firm where before it had folded. Applied kinesiology. The oldest trick in the carnival, dressed in a word that had cost

real physicists real arguments to define.

Naila had stopped six feet back, where she could see the whole booth without being seen by it. She had her phone out, reading.

“The word’s real,” she said. “That’s the thing that bothers me. *Tachyon*. Gerald Feinberg, 1967, he coins it for a hypothetical particle that only ever travels faster than light — never slower, never crosses c , because c is a wall you can be born on either side of but never climb over.” She said it the way other people recite a hymn they don’t believe but love the shape of. “It’s a serious idea. People still publish on it. And the Higgs field — before it settles — the equations describe it as *tachyonic*. Imaginary mass. It’s in the textbooks.”

“And that,” Yusuf said, nodding at the seafoam booth, “has nothing to do with that.”

“Nothing.” Naila’s mouth went flat. “There’s no tachyon in the pendant. There’s no tachyon stored in anything. You can’t store a particle that can’t stop. The word got picked up off the physics floor like a dropped wallet and spent.”

Frik, beside her, had been quiet a long time, which with Frik was a weather system. He was watching the linen woman’s hands. When the volunteer’s arm held strong and the small crowd made its small impressed sound, Frik’s head jerked once and he said, very clearly, into the engineered hush:

“Push down. Push up. Liar’s choice.”

It carried. The linen woman looked over, found him, found the thing in his face that wasn’t aggression, and smiled the professional smile back. Yusuf had used that smile to glide past hecklers a hundred times. *Bless you, sir. Move the eyes back to the mark.*

“He’s right, though,” Naila murmured, only for them. “Watch her wrist. She changes the vector. Pushes *out* on the held one, *down* on the failed one. The body can’t resist a sideways pull the same as a vertical one. It’s not the pendant. It’s geometry.”

“It’s the oldest geometry there is,” Yusuf said.

The official story of this whole wing of the expo was clean, and the book of it could be written in an afternoon. *Tachyon energy products do not work. There is no stored tachyonic energy. There is no biofield the pendant restructures. The studies cited are self-published, uncontrolled, or fabricated. The word is a marketing veneer over the placebo effect, and the placebo effect, while real, is not what's on the price tag.* Skeptics had said it for thirty years. The FTC had said versions of it with subpoenas attached. It was true. Yusuf, of all the men alive, knew exactly how true it was, because he had once stood on the seafoam side of an aisle and made the same firm-arm magic with the same kind wrist.

The potato wasn't the lie. The lie was easy. Anyone could see the lie.

The potato was how *good* it was. How much labour went into the frictionless floor. The lanyards were laminated. The pamphlets had citations — real journal names, real volume numbers, attached to studies that, when you pulled them, dissolved into conference proceedings nobody peer-reviewed. The pendant had a holographic sticker with a serial number, because a serial number is a tiny promise of accountability, and the promise was the product. The linen woman believed it. That was the part the smug version never got. *Most* of them believed it, which was what made the few who didn't so hard to find.

"There," Yusuf said.

Three booths down, a man in a good grey jacket was not selling anything. He was looking. He had a leather folio under his arm and a way of standing slightly square to each booth, the way you stand when you're pricing a thing rather than buying it.

"Adversary?" Edith asked, low.

"Buyer." Yusuf had seen the jacket before — not the man, the *grammar* of him. "He's not here for pendants. He's here for the companies that make the pendants. Watch — he's not testing the products. He's reading the booth signage for an LLC name."

"Why would you buy a company that sells a thing that doesn't work?"

"Because the thing isn't the asset." Yusuf felt the old vocabulary load itself into his mouth, clean and bitter. "The *claim* is the asset. The trademark. The 'tachyon energy' wordmark, the customer list, the regulatory file that says *we got a warning letter and survived it*. You

buy six of these little outfits, you own a category. You own the word in the marketplace, whatever it means in a physics department. And then—” he stopped.

“And then what?” Naila said.

“And then you can do whatever you want with it. Sell it cleaner. Or sell it louder. Or sit on it.” He watched the grey jacket make a small note. “Or buy the ones near a real patent and blur the line so nobody downstream can tell the wallet they picked up off the floor from a wallet that’s actually got something in it.”

Naila went still in the particular way she did when two things she’d been holding in separate hands turned out to be the same object. “That’s the whole game. That’s why they’re ahead of us at every node. They don’t care if it works. We keep checking if it works. They’re buying the *quiet* around whether it works.”

Nobody said anything to that. Edith’s camera stayed down at her hip. Somewhere behind them a speaker began, amplified, to explain that water has memory.

The man Yusuf actually came to find was running the smallest booth on the floor, in the corner by the loading doors, where the air moved and the carpet gave up. The sign said **CLARITY DIAGNOSTICS** in a font that was trying to be a hospital. There was a machine on the table the size of a bread maker, with two brass handholds and a dial, and a banner behind it that said it measured *cellular voltage and bio-coherence*.

The woman behind it was maybe sixty, with reading glasses pushed up into grey hair and the patient, exhausted bearing of someone who had explained the same thing to the same kind of person for a very long time. Her lanyard said **D. ARAGÓN**. When she saw Yusuf coming she watched him the way the Pogue keeper had watched them in Winnipeg, the way Naila watched booths — full attention, no warmth spent until earned.

“You want a demo,” she said. It wasn’t a question. “Hands on the brass.”

Yusuf put his hands on the brass. The dial swung. A green light came

up the side of the box.

"You're coherent," Aragón said, deadpan. "Congratulations."

"What's it actually measuring?"

"Skin conductance. Mostly how sweaty your palms are." She let the green light sit there, accusing both of them. "I built the firmware. Eleven years ago. For a man who told me it would be a wellness *aid* — a feedback toy, like a mood ring with a chip. I wrote it honest. Skin conductance, displayed pretty." She tapped the banner behind her. *Cellular voltage. Bio-coherence.* "I didn't write that. The marketing wrote that. Then the marketing got bought, and the new owners kept the firmware and changed the manual, and now there's a clinic in three states telling cancer patients their *cellular voltage* is recovering."

Frik made a sound, a small swallowed click, and then said, quiet and exact, "*That's not a mood ring.*"

"No," Aragón said. "It is not." She looked at Frik a moment longer than was comfortable, and something in her face unclenched a degree, the way it does when one careful person recognises another. "You see it too. Good."

Yusuf knew the next part before it came, because he'd been on the other end of it once, a younger man getting a quiet warning he'd ignored. He said, gently, "You went to someone."

"I went to everyone." Aragón's voice didn't rise. That was the worst of it — that it didn't rise. "Three years. FTC takes a complaint, sends a letter, the company restructures, the letter's attached to a shell that dissolves, the asset moves on. The asset is the word and the customer list. You know this." She'd clocked him entirely. "You've worn the jacket."

"I've worn the jacket."

"Then you know there's a man on this floor today buying the word." Her eyes went, just once, to the grey jacket three aisles over. "He came to my booth this morning. Offered to acquire Clarity Diagnostics. Firmware, trademark, files. He was very polite. He said — and I wrote it down, because I write everything down now — he said the *therapeutic positioning had matured into a defensible category.*"

Naila said, almost to herself, “He wants the regulatory file. The warning letters. He wants to own a thing that’s already survived being investigated.”

“He wants to own the *quiet*,” Aragón said, and Yusuf felt that word land in his chest like a hand. “He wants every loud, embarrassing, easy-to-debunk little outfit in this category folded into one shell that’s too boring to sue. And once it’s boring, nobody looks. And once nobody looks—” She stopped. She pushed the glasses down out of her hair and looked at Yusuf through them, the demo machine purring its green lie between them. “You’re not buyers.”

“No.”

“You’re the other kind. The ones who turn it into a podcast.” Her mouth thinned. “*SUPPRESSED. THEY DON’T WANT YOU TO KNOW.* You’ll make the word *bigger*. You’ll make him richer.”

“That’s not what we do,” Edith said, and lifted the camera then, slowly, so Aragón could see the lens before it saw her, and lowered it again before it was raised. A choice, made in front of her. “I don’t take the heroic angle. I don’t take the crank angle either.”

“Everybody takes an angle, honey.”

“She takes the boring one,” Naila said. “On purpose. So it stays true.”

Aragón looked at the four of them a long beat — the woman who lowered the camera, the man who read the booth grammar like a native because he was one, the one reading the wrist physics on her phone, the one whose mouth said the true thing the room had built itself to muffle. The demo box ticked over. The green light died and re-lit, coherent, coherent, coherent.

She reached under the table and brought up the leather folio Yusuf had not seen her holding, because she’d had it the whole time, the way the careful ones keep the real thing close to the body.

“I kept two sets,” she said. “Eleven years. The firmware I wrote, and the manual they printed. Side by side. Every revision. Every time the honest thing got a dishonest sentence stapled to it, I logged the date and who signed off.” She set the folio on the brass, on top of the machine, where the demo had been. “He offered me money for this folder this morning, you understand. Not the company. *This*. He

knew it existed. He wanted to buy the one record that shows the join.”

“And you said no,” Yusuf said.

“I said I’d think about it.” She looked at him, and for the first time there was something underneath the exhaustion, something he recognised because it was the thing that had eventually walked him out of the jacket for good. “I’ve been thinking about it for six hours, sitting at this booth telling people they’re coherent. Then you walked up reading the floor instead of the products.” She turned the folio a half-inch toward him, not letting go of it yet. The brass handholds gleamed. Behind them the loading doors opened and the corridor air came through, smelling of exhaust and rain off the lot.

“Come and see the real books,” Doña Aragón said. “Both of them. While there’s still two.”

Chapter 12 — The Laundered Word

The unit smelled of old carpet tile and the inside of a kettle. Somewhere above the drop ceiling a fluorescent tube was dying in stutters, throwing the room into and out of a faint green that made everybody look slightly drowned. Yusuf had worked out of rooms like this. He knew the carpet tile, the partition walls that didn't reach the ceiling, the laminate desk with one leg shimmed by a folded coaster. The geography of a place that wanted you to believe a great deal was happening and could be folded into a Transit van in forty minutes.

"This was the call centre," said Marlene. She had a lanyard from a job she no longer had and a ring binder she held against her chest like a child holds a lunch tray. "Six headsets. Two of them worked. We took inbound, mostly. People who'd seen the testimonials."

"Inbound," Yusuf said. He liked the word in her mouth. It was an honest word for the thing. "You didn't cold-call."

"We didn't have to." She set the binder on the laminate. "They came to us."

The product had been called Tachy-Ion at first, then — after a letter from somewhere — TachyLife, then a wellness line whose name Yusuf had already half-forgotten, the way you forget the third repaint of a betting shop. Pendants. A sticker for the back of your phone. A bottle of water that had been "charged." A device the size of a hockey puck that you set on your router and which did, electrically, nothing at all, because Naila had opened one in the van the night before and found a coin cell, a blue LED, and a disc of laminated card with a spiral printed on it.

“The disc,” Naila said now. She’d brought the opened puck and set it on the desk, two halves and the little spiral lying face up between them. “What was the disc supposed to be?”

“Orgone alignment.” Marlene said it without flinching and without belief, the way you’d read out a horoscope to a room. “That was the second script. The first script said it ‘harnessed faster-than-light particles to restore the body’s natural frequency.’ Legal made us change it.”

“To orgone.”

“Orgone wasn’t a regulated claim.” Marlene almost smiled. “Faster-than-light was. Somebody decided.”

Edith had her camera down by her thigh, not raised. She’d been keeping it there since they came in, and Yusuf had clocked the discipline of it, the way you’d clock a man keeping his hand off a glass. The room wanted to be photographed. SUPPRESSED, the room wanted to say, or VICTIMS, depending who held the camera. Edith wasn’t giving it either.

Yusuf opened the binder.

He’d asked to, and Marlene had let him, which told him more than the binder would. Inside was the real architecture, and he read it the way Naila read a patent — as a thing that had been *made*, by people, on purpose, with skill. Three companies. The trading name that took the money. A holding company in a jurisdiction that didn’t make you say who owned it. A third that owned the trademark on the word and licensed it to the first for a fee that happened to equal the first company’s annual profit, so that the first company, the one with the customers and the refund liability, never quite had any money in it.

“That’s clean,” Yusuf said. He hadn’t meant to say it out loud, and Frik, by the door, went very still, which was how Frik went still before he didn’t. “I mean the structure. Whoever built this knew what they were doing.”

“Was it you?” Marlene asked. Not unkindly. She’d had the crew described to her, evidently, or worked it out — the reformed one, the fixer. “Yusuf. You were in this.”

“Adjacent to this,” Yusuf said. “Different word. Same shape.” He turned a page. The licensing agreement for the trademark. The word itself, *tachyon*, sitting in a box on a form, valued — actually given a number, a figure with a currency symbol, the way you’d value a brand of crisp. Somebody had paid an accountant to put a price on a word. “The whole machine is built to move the word from one pocket to another and never let the money sit anywhere a refund could catch it.”

“What’s the word doing,” Naila said.

It wasn’t a question for Marlene. It was Naila talking out loud, which she did when a thing was wrong in a way that hummed at her. She had the patent-reading face on, the one she’d had over Pogue’s four real ones, except there were no patents here, there was a trademark, and the difference was the whole story.

“It’s a real word,” Naila went on. “Tachyon. It’s in the physics. A hypothetical particle with imaginary mass that, if it existed, would always travel faster than light. It was a serious idea. The Higgs field was once described as tachyonic — that’s a real, technical statement about an unstable field rolling to a lower energy, it’s in the textbooks, it doesn’t mean anything goes faster than light.” She touched the spiral disc, didn’t pick it up. “And in 2011 a real experiment, OPERA, measured neutrinos arriving early, and for six months it looked like the rules might bend, and it was a loose fibre-optic cable. They found the cable. They published the cable. That’s the whole life of the word in science. A clean idea, a beautiful mistake, a loose cable, a correction.”

“And then,” Yusuf said.

“And then somebody took the word out of the building.” Naila looked up. “It still had the smell of the physics on it. That’s what they bought. Not the particle. The smell.”

The fluorescent tube stuttered. Marlene didn’t turn it off. Maybe the switch was somewhere she’d have had to ask about, in a building she didn’t own anymore.

Yusuf turned to the page he’d been working toward without admitting it. The marketing deck. Slide after slide of it, and not one equation, not one circuit, not one measurement, not one number that meant

a physical thing. *Frequency. Resonance. Field. Alignment. Restore. Natural. Quantum.* Words that had each been carried, like the central word, out of a real building where they meant something exact, and set down here where they meant *buy this*. There was a slide with a photograph of a galaxy on it. There was a slide that said WHAT THEY DON'T WANT YOU TO KNOW, and Yusuf felt the old shape of it in his own hands, the cover that was also the pitch, the suppression-story that did double duty: it explained why there was no evidence (*suppressed*) and it flattered the buyer (*you're one of the ones who knows*). He had built that slide, in his life, with a different word on it. He recognised the kerning.

"There's no physics in here," he said.

"There was never any physics in it," Marlene said. "That was the thing I didn't —" She stopped. Started again, flatter. "I thought there was, at the start. There was a man we called the Doctor. He had a coat. He said things about waveforms. I believed there was a basement somewhere with the real device in it and the pucks were just, you know, the consumer version." She looked at the two halves of the puck and the coin cell. "There was no basement. The Doctor's PhD was honorary, from a thing he'd founded. The waveforms were a screensaver."

Yusuf had known a Doctor. Several. The coat was always real; the coat was the cheapest true thing in the operation.

Frik made a sound. Not a word yet — the intake, the click in the throat that the crew had all learned to leave room around, the way you leave room for a wave to land. Marlene didn't know to leave room. She looked at him.

"It's fine," Edith said quietly, to Marlene, not to Frik. *Leave it.*

"The only — *the only real thing is the word,*" Frik said. The blurt came out level and clean, the way they did when they were also true, which was the part nobody warned you about; the way Frik's wiring sometimes reached past the room and put its hand flat on the actual thing. "*The word's the only real thing in the building.*" He blinked, came back to himself, looked faintly stricken the way he always did at the door of one. "Sorry. Sorry. But it — it is, isn't it. Everything else is a coin cell. The word's the only thing that was ever real and they stole it off the physicists."

Nobody contradicted him. Naila had the puck halves in her two hands like she was weighing them against each other and finding them equal, equally nothing.

"How many people," Naila said. To Marlene. The question she'd been circling. "Who came to you. Inbound."

Marlene didn't answer that. Instead she opened the binder to the back, past the company structure, past the deck, to a section held by a tab she'd made herself out of a folded Post-it, gone tacky and grey at the fold. And here, Yusuf saw, the geography changed. The clean shapes stopped. What was in the back of the binder was not architecture. It was handwriting.

"You asked the wrong man if I built it," Yusuf said. The thing in his chest had moved while she turned the pages, and he let it. "I want to ask the right one. Did you know? While you were taking the calls. Did you know it was a coin cell."

"No," Marlene said. "And then yes. There's a day. There's an actual day where it stopped being no and started being yes, and I worked another five weeks past it because I had a lease and a kid in Year Nine, and I'm not going to dress that up for your camera." She looked at Edith. "You can put that in. That's true."

Edith did not raise the camera. She'd lowered it further, if anything, hooked the strap over the back of a chair, so that it sat there being a thing that wasn't pointed at anyone. It was the loudest thing she'd done all day.

"He bought you," Yusuf said. He'd seen it coming in the structure, in the way the third company's trademark licence had a clause about *change of control*, and he wanted to hear her say it, because she'd earned the saying of it. "Whoever's ahead of us. They didn't shut you down. They bought the word off you."

Marlene turned the binder around so it faced him, open at the handwriting, and tapped one page, then slid a single sheet of A4 out from behind it — a real letter, on real letterhead, the letterhead Yusuf had been watching for across four graves now, the outfit that kept arriving one room before them and leaving with everything that could be filed away clean.

"They offered for the trademark," she said. "Just the word. Not the

customers, not the lease, not the refund liability — they were very clear, the offer was for the intellectual property only. They wanted the word with the physics-smell still on it and none of the people it had been used on. They said they were *retiring the brand*.” Her finger stayed on the letter. “Retiring it into a drawer. Where nobody could ever dig it back up and read the deck and find out it was a coin cell. The word goes quiet. Everyone who built the lie gets paid to forget the lie. That’s what *retiring the brand* is.”

“And you,” Yusuf said. “Did you take it.”

“I’m not the trademark holder. I was a manager. Nobody offers the manager anything.” She almost smiled again, the bad version. “But I had this.” She turned the binder fully toward them, both hands flat on the open pages, and now Yusuf could read what the handwriting was.

It was a list. Columns, ruled with a biro against the edge of something straight. A name. A date. An amount. A tick, or a blank.

“When the calls started going the other way,” Marlene said. “When they started phoning to ask for it back. There was no system for that. There was never going to be a system for that, the whole structure was built so there wouldn’t be. So I did it on paper. The ones I could push through before the money moved. The ones I caught.” She turned a page, and there were more, two hundred lines, three, the biro pressure changing where a different pen had taken over, where a day had ended and another begun. Ticks and blanks. More blanks than ticks, toward the end, where the money had got faster than she had. “This is the only true record in the building. Everything in the front of this folder is a machine for moving a word. This is the only part that counts anything real. It counts the people I could give it back to.”

She slid the binder the last inch across the laminate, under the dying green light, into the space in front of Yusuf, and took her hands off it.

“You read the men,” she said. “That’s what the dead-quiet astronomer’s people told me you do, the ones you saw before us — you read the men and not the machines. There’s no man to read here. There was no genius, there was a coat and a screensaver. So read this instead.” Her finger came back down, once, on the column of ticks and blanks. “If you’re going to put any of us anywhere, put the

list. Not the word. They've already bought the word."

Yusuf looked at the page. At a tick. At the name beside it, a stranger's name, somebody who'd seen a testimonial and phoned the number and then, later, phoned it again, and had got, against the whole architecture of the thing, eleven pounds back.

Frik, by the door, said it very softly, almost to himself, and for once it was not a blurt at all but a man choosing the word: "*Honest*." Then, the worry, the reaching-back: "Sorry. Is that — that's allowed. That's the right word."

"Yeah," Yusuf said. "That's the right word."

He did not pick up the list yet. He let it lie there in the green stutter, under the smell of kettle and carpet tile, the front of the folder full of clean shapes that moved a word and the back of it full of a manager's biro counting out the people she could reach. Outside, the letterhead was already a room ahead of them, carrying the word off into a drawer to go quiet.

He left his hand flat on the binder. He didn't close it.

Chapter 13 — The Grift and the Frame

The storage unit smelled of cardboard and the inside of an old television — that hot dust smell of electronics that have been switched on and off for thirty years and never once cleaned. Edith stood in the roller-door light with the camera dead in her hands and looked at the word.

It was on everything. Printed on the boxes in a font that wanted to be the future and had landed somewhere near a bottled-water label. Embossed on the lids of little vials. Silk-screened on a banner, rolled and leaned in the corner like a sleeping thing: **TACHYON**. Below it, smaller, the promise — *charged energy for body & home*.

“Faster than light,” Yusuf said, from the floor. He had a box open across his knees and he was holding a single vial up to the strip-light, turning it, the way you’d check a wine for sediment. “That’s what it means. That’s the whole of what it means. A tachyon is a particle that — if it existed — would only ever go faster than light. Couldn’t slow down to our speed. Couldn’t stop. Couldn’t sit in a plastic tube on a shelf in Phoenix and cost you forty dollars.”

“So it’s nothing,” said Bashir. He had the spreadsheet open on the laptop, the customer list the keeper had let them photograph, eleven thousand rows of it. “It’s a tube of nothing.”

“It’s a tube of resin.” Yusuf tipped the vial so the light caught the little fleck inside it. “Cast acrylic. They’ll tell you it’s been *restructured at the subatomic level*. You can’t restructure resin at the subatomic level. You can melt it and pour it into a mould, which is what this is.”

Edith lifted the camera halfway and stopped.

The man whose unit this was — the man they had driven nine hours to read — was not dead. That was the strange part of this node, the part that sat differently in her chest. Four times now they had stood in the rooms of the gone, with the keeper holding the dead man's papers like a relic. Here the maker was alive, and he had sold the company, and he was sitting in a coffee shop two miles away because he had said, *I'll talk, but not in there. I can't be in there.*

His daughter was in there instead. She stood by the boxes with her arms folded, watching them photograph her father's life's work, and she had the look Edith had come to know in keepers — the one that measured you for the headline you were going to write.

The official story was clean and it was correct.

There is no such thing as tachyon energy. The word was lifted, whole and warm, out of physics, where it has a real and specific and entirely theoretical home — a faster-than-light particle hypothesised in the 1960s, never observed, a mathematical *what if* that physicists keep on a high shelf and take down to argue about. For one bright stretch the Higgs field's lowest energy state was modelled as *tachyonic* — a genuine, published technical term, a sign in the equations that the vacuum was sitting in the wrong place and wanted to roll downhill. The word means something. It means it to people who can read the maths.

And then it walked out of the journals and into the wellness aisle, and now it meant a tube of resin that cured your insomnia for forty dollars plus shipping.

"This is the cleanest one," Naila said. She was sitting on a stack of cartons with the patent — there was no patent — with the *literature*, the glossy fold-out that came in each box, and she was reading it the way she read everything, flatly, all the way down, missing nothing. "There's no machine. With Pogue there was a real carburetor that really clawed back fuel. With the vacuum there's a real force you can weigh. Here there's —" she turned the fold-out over, found the back equally empty of anything — a word. They sold the word. There was never anything under it."

“So we burn it.” Bashir didn’t look up from the spreadsheet. “Eleven thousand people. We’ve got the list. We name the company, we name the price, we run the line about resin, and we let the comments do the rest. It writes itself. *We bought tachyon energy so you don’t have to.*” A small, hard smile. “It’d be the most-watched thing we’ve done.”

The fish, if the fish were here, would have had a field day with the banner. *Charged energy for body and home* — heckle that for a week.

Edith watched the daughter’s arms tighten.

She had been to the coffee shop first. That was her own rule, the one she’d made somewhere around Winnipeg and kept since: meet the face before you frame it.

The maker was a soft-spoken man in his sixties with reading glasses pushed up into grey hair, and he had not defended the product. That was what undid her. He hadn’t said *but it works for people, it really does*. He had wrapped both hands around a paper cup and said, “I believed it when I started. That’s the thing nobody puts in the story. I read about the vacuum energy, the real physics, the — there’s a force, you know, they’ve measured it, you can hold a number for it.” He’d looked at her, checking she knew it was real, and she’d nodded, because Naila had taught her it was. “I thought I’d found a way to — gather it. I thought I’d found the edge of something. And by the time I understood I hadn’t, I had a warehouse and a payroll and four thousand customers who told me they slept better.” He turned the cup a quarter-turn. “And I couldn’t tell which thing I was anymore. The man who’d found the edge of something. Or the man with the warehouse.”

That was the sentence Edith couldn’t film. Not because it would harm him. Because it was true, and the truth of it would die the instant it became a clip.

“We don’t burn it,” she said.

Bashir looked up. Naila kept reading. Yusuf set the vial down very carefully on the concrete, label up, and waited, because Yusuf had sold a laundered word himself once, in another life, and Yusuf knew

which way this went.

“The eleven thousand,” Edith said. “If we run it your way — the resin, the price, the comments — what happens to them?”

“They learn they got conned. That’s a service.”

“They learn they’re idiots. In public. Eleven thousand people who were tired, or sick, or scared, or just wanted something to *hold* —” she stopped, because she was getting near the line the themes file forbade, the one nobody in this crew would ever say aloud, the one about the world filing people away as housekeeping. She came at it from the side instead. “I’ve shot people at the worst hour of their lives. You know what the difference is, between a photograph that documents and a photograph that sells?”

Nobody answered. They knew she’d say it.

“The selling one needs them not to look back at you.” She turned the camera in her hands, the lens pointing at the floor. “The documenting one survives them looking back.”

Frik had been quiet in the doorway, half in the desert light, doing the thing he did when a room got tight — counting the boxes, lips moving, a small private liturgy of inventory. He said, suddenly, too loud, the way it always came, “*frame the mark* — ” and then his hand went to his mouth and his face did the thing, the apology that wasn’t an apology, the flinch.

Nobody laughed. That was the rule and they kept it.

But the words sat on the air. *Frame the mark*. And Bashir’s smile was gone.

Yusuf stood up. He went to the daughter. He did it the way he did everything that mattered now, slowly, hands visible.

“Your dad,” he said. “Did he sell, or did he get bought?”

She studied him. “What’s the difference.”

“All of it.” Yusuf gestured at the boxes, the banner, the eleven thousand rows glowing on the laptop. “If he sold, he took a cheque and walked away from a thing he didn’t believe in anymore. That’s a man

cutting his losses. If he got bought —” He didn’t finish. He let her finish it.

She unfolded her arms. She went past him to a filing cabinet wedged between the cartons, opened the second drawer, and took out a folder, and she held it the way the Winnipeg woman had held the patents — at the chest, deciding.

“They came in March,” she said. “Group out of San Diego. Said they were consolidating the wellness-tech space. Made him an offer for the brand, the formula — there’s no formula — the customer list, the trademark. All of it.” She opened the folder herself, this time, didn’t hand it over; the read was hers to give. “Dad would’ve signed for the money. He was done. But there was a clause. Page nine.”

Edith came close enough to see, and did not raise the camera.

It was a non-disparagement clause, and below it a non-disclosure, and they were ordinary, the way these things are ordinary, lawyers’ boilerplate. But the daughter’s finger was on the third paragraph.

The Seller shall not, in perpetuity, publish, assist in the publication of, or provide source materials toward any account, critical or otherwise, of the scientific basis, efficacy, or development history of the Products or the underlying claims.

“They didn’t just buy the company,” the daughter said. “They bought his right to ever explain it. To anyone. To say *I was wrong*, even. To say *it’s resin*. He signs that and the truth dies with the trademark, and the trademark’s theirs now, and they can sell it forever or shelve it forever and either way—” She closed the folder. “Either way nobody ever asks the question again.”

Edith felt the floor shift under the chapter she thought she was in.

Bashir had been right that it wrote itself — the resin, the price, the conned eleven thousand. That was the story the genre would tell, the exposé, the sneer with a public-service face. And the San Diego group had read that story too. They had read it cold and they had built around it. Because if the crew burned the brand, the brand would die and the question would die with it — *was there ever an edge of something, and where exactly did the man cross from finding it to*

selling it? — and if the crew didn't burn it, the clause buried the man who could answer. Either way the asking stopped.

They weren't buying machines. There was no machine; there had never been a machine. They were buying the moment a man stopped being able to talk. They were buying the quiet.

"It's not the product they want gone," Yusuf said, very low, and Edith understood he'd seen it a beat before she had, from the inside, the way he saw all of this. "It's the *history of the product*. The development record. The — the trail back to the real physics, the part where he wasn't lying, the part where he believed it. That's the only dangerous thing in here. The honest beginning." He looked at the boxes, the banner, the resin. "The fraud's worthless. They're not protecting the fraud. They're erasing the fact that it started as a hope."

Naila had stopped reading. She had the fold-out flat in her lap and her thumb on the one true sentence printed anywhere in it, the only fact the company had ever told its customers, buried in a footnote in four-point grey: *Based on the demonstrated reality of zero-point vacuum energy*.

True. Maddeningly, exactly true. The vacuum isn't empty; the force is real; you can weigh it. And you cannot run a single bulb off it, and the word *tachyon* on the box above means a particle that cannot stop, and between those two real facts a man had built a warehouse and lost the ability to tell which one he was standing on.

The daughter was watching the door, where the desert light lay flat and white across the threshold. Her father was two miles away with both hands around a cold cup, deciding whether to sign page nine.

"He hasn't signed yet," Edith said.

"They want it by Friday." The daughter put the folder back in the drawer and did not close the drawer. "There's a man flying in. Closes the deal in person." She looked at Edith, and at the camera pointed safely at the concrete. "You came to take his picture and put it on the internet."

"Yes."

"And now?"

Edith looked at the lens. At the eleven thousand rows. At the word

that meant something to people who could read the maths and nothing at all to the tired and the scared, and at the soft-spoken man who could no longer tell which of those rooms he lived in.

“Now I think there’s a question in here somebody flew across the country to kill,” she said, “and I’d like to know what it’s worth to them. Before Friday.”

Outside, somewhere past the rows of identical roller doors, an engine that was not the Beast slowed, idled, and stopped.

The daughter heard it too.

She left the drawer open.

Chapter 14 — The Man Who Would Not Let Go

The roller door came up on a sound like a long breath, and then the cold of the unit reached out and touched them all at once.

Tomi stood at the threshold with her hands in her sleeves. Behind her the industrial estate ran flat and grey to a ring road — units the size of aircraft hangars, a tyre place, a man in a hi-vis vest eating a pasty out of a paper bag and watching them without much interest. Unit 47. Someone had painted the number on by hand and then, a long time ago, the white had gone the colour of a tooth.

“Mind the step,” said the keeper. “He never fixed it. Said he’d get round to it.”

His name was Len. He was somewhere past seventy, in a fleece zipped to the chin and reading glasses pushed up into hair that had given up being any particular colour. He had a key on a fob shaped like a little plastic football and he’d taken a long time getting the padlock open, not because his hands shook but because he was watching them do it, the way you watch a stranger pet a dog that bites.

Inside, the light was the kind that hums. Two long strips, one of them stuttering. Shelving down both walls, the industrial steel sort, and on the shelves the ruins of a man’s whole argument with the universe.

“This is it,” Len said. “The lot. What there is.”

Tomi went in slow.

The official story she could have recited in her sleep, by now; they all could. **John Roy Roe Searl. Born 1932. The Searl Effect Genera-**

tor — concentric rings of segmented magnets, set spinning, that were said to produce more electrical power than was put in, and at a certain point to lose weight and lift off the bench entirely.

A disc, in the famous telling. Discs, plural. Flown over Mortimer in the sixties. Photographed, supposedly. The energy crisis solved in a lock-up exactly like this one, if only the world had listened.

And the world's answer, stated dead straight, was the cleanest answer in the whole book: *no*. Not "unlikely." *No*. A device that puts out more than it takes in is not a device, it is a definition of the thing physics is built to forbid. There was no SEG in any independent laboratory, no measurement by anyone who didn't already believe, no photograph that survived a second look. There was a Danish engineer named Heerfordt who had gone looking, properly, in good faith, for one — *one* — witness who had actually seen a disc fly, and had found exactly none. Not a sceptic's count. A believer's, near enough, counting and coming up empty.

That was the floor. They'd walked all the way down to it. Four men above them on the ladder, each dismissal a little harder to argue with than the last, and here at the bottom the dismissal was simply correct.

So Tomi had come down the ladder expecting nothing, and she'd been wrong, because nothing isn't a smell.

It smelled of cold oil and old paper and, underneath, faintly, of a man — the way a coat smells of someone after they've stopped wearing it.

Don't, she told herself. *That's the trap. That's the exact trap.*

"Can I—" she started.

"You can look," Len said. "Hands behind your back."

Edith had come in behind her with the camera down at her thigh, lens cap on. She'd put it on in the van, the cap, a small deliberate thing, and Len had clocked that too. Naila stood just inside the door reading the shelves left to right like a page. Yusuf had stayed by the Beast with his phone, watching the road; he hadn't said why, and none of them had asked, because they all knew why. Somebody else was coming. Somebody always was, now.

The shelves.

There were the diagrams first, and the diagrams were beautiful. Tomi hadn't expected beautiful. Big rolled sheets, some of them, unrolled and weighted at the corners with whatever had been to hand — a spanner, a mug, a fist-sized lump of something that turned out to be a magnet when she leaned close and felt the air go thick. The rings drawn out in pencil and then inked, segment by segment, each little block of the rotor labelled in a tiny upright hand. Numbers running round the rims. A scheme of magnetisation — north, south, the angles — annotated like a man arguing with himself across years, a later pen correcting an earlier one, *no — try at 90 — see notebook 11*.

"He wrote everything down," Tomi said, before she could help it.

"Everything." Len said it flat, but his eyes had gone somewhere. "Forty years of it. You'd find him at four in the morning out here with the heater off because the heater cost money, doing the maths again. Same maths. He'd done it a thousand times. He'd do it once more."

There was love in it. There was also a thing she recognised and didn't want to name, the thing the diagrams had — the going-round-and-round of it, the same calculation worked and reworked because the working was a place to live.

Then the models.

These were not beautiful. These were heartbreak.

Rings of a kind, yes — actual rings, machined or cast, she couldn't tell, segmented round their circumference, each segment a little bar of magnet potted into the steel. But broken. One rotor cracked clean across, the break old, the metal gone matte where it had sat. A stator ring with half its rollers missing and the other half seized, furred with a grey bloom. A smaller set, maybe a demonstration model, the size of a dinner plate, mounted on a wooden base that had warped so the whole assembly tilted, as if it had been caught in the act of giving up.

She bent to the dinner-plate one. The rollers were the bits that were meant to run round the outside, the orbiting magnets, the part that in the telling went faster and faster until the thing wanted to leave the planet. She could nudge one with a thought of a touch and it wouldn't move. Rust had married it to its track.

The potato of it sat right there under the humming strip light, and it was a sadder potato than any garage carburettor.

Because here was the impossible part made physical, made *cheap*. You could see, in the broken plate-sized model, exactly how easy the idea was to draw and exactly how hard it was to make a magnet hold its temper, to keep a roller true, to stop a ring cracking when you span it, to even get the thing to turn at all under its own weight without a hand. The legend said *it lifts*. The shelf said: it would not run. The legend said *over-unity*. The shelf said: one of these has been seized to its bearing for thirty years and another is in two pieces.

And the maths, the lovely terrible maths, said what it always said, the thing Naila had said quietly in the van that morning with no heat in it at all: you can spin magnets till the cows come home and the bookkeeping closes to zero. Energy in, energy out, friction takes its cut, and the bench stays on the floor. There is no entry in the ledger for *and then more*. There never has been. The men who keep finding one are not finding one. They are finding a place to live.

Tomi straightened up and her eyes were stinging and she hated it.

“He really thought,” she said, and stopped.

“He didn’t think,” Len said. “He knew. There’s a difference, and it’s the whole difference.” He’d folded his arms. “You lot. You always want to do the bit where you go *bless him, he believed*. That’s the kind one. The cruel one and the kind one come out the same place.”

Naila had stopped by the far shelf. Tomi knew the stillness — Naila reading something all the way through before she’d let herself speak. There was a box there, a banker’s box, the cardboard soft at the corners with damp. Naila wasn’t touching it. Hands behind her back, like she’d been told, but her head was tipped, reading the spine of the lid.

“These are court papers,” Naila said.

“Some of them.”

“Nineteen eighty-three.” Naila read it off the lid without leaning in. “Theft of electricity.”

That one was real. Tomi had read it on a microfiche in a county library two weeks ago and not quite believed her eyes. *The Times*, among

others. The man who said he made power for free, the man with the disc that needed nothing, convicted and jailed for taking electricity off the mains without paying — a bypass round the meter at the house, the supply boards' inspectors at the door, the bench in the kitchen and the wires going the wrong way. The genre had three ways to read it and Tomi had heard all three in the same week. *He was framed — they cut his power to stop him.* That was the believers. *He was a fraud and there's your proof — the free-energy man stealing from the grid.* That was the sneer. And the third, the only one she could stand, was no reading at all, just the facts sitting there refusing to resolve: a man who could not pay his bill and would not stop building, and a meter, and a court, and a sentence.

"He did it," Len said. Tomi turned. The old man was looking at the box like it owed him money. "I'm not going to stand here and tell you he didn't. He took the power. He'd lost the job, lost the house, the boards were chasing him, and he had the bench to feed and no money to feed it, so he fed it the only way there was." His jaw worked. "You want to know the worst of it. He said to me once — years after, this — he said, *Len, I made them all the power they'd ever need, and they put me away for borrowing a bit while I finished.* And he believed that. Right to the end. That he'd already given them the answer and they jailed him for the postage."

Nobody said anything. The strip light stuttered. Outside, a long way off, a lorry changed down through its gears.

Tomi looked at the broken plate, the seized rollers, the diagrams gone the colour of weak tea. She looked at the box of papers that proved the worst version and somehow didn't close it, didn't end it, only made the not-ending heavier.

"What was it for," she said. It came out small. She hadn't meant to say it out loud — it was the question they always got round to, Jakobus's question, the one that turned the whole thing if you could ever get it answered. *What is a made thing for.* Only she'd never asked it of a thing that didn't work.

Len's head came up.

And the kindness went out of his face, all at once, like a light switched at the wall.

"There it is," he said.

"I didn't—"

"You did. That's the question, isn't it. That's the one you came for. *What's it for*. Like he was a sum you could finish for him." He took a step, not toward her, just away from the shelf, putting himself between them and the box without seeming to decide to. "I've had your sort here before. Different jackets. The man last year — proper company, lovely manners, cheque-book voice — he stood where you're standing and he asked me what it was *for*, and what he meant was, what's it *worth*, and what he wanted was to put the lot in a van and write the story himself. *Suppressed Genius*. Or the other one — *Tragic Crank*. Either one sells. He didn't mind which."

Tomi's mouth was dry. Behind her she heard Edith not raise the camera, heard it the way you hear someone deliberately not move.

"You've come a long way," Len went on, quieter now, which was worse. "You've got the van and the girl who reads and the one who won't film till she's allowed. I'll give you that, the not-filming. That's more than most." He stopped. He looked at each of them, taking his time, and his eyes were wet and absolutely steady, and Tomi felt herself being weighed the way the padlock had been watched. "But I'm not opening that box for you on the strength of a question. He's not a mystery for you to solve so you can feel something. He was a man, and he's dead, and I'm what's left to mind him, and I've minded off cleverer than you."

The roller door was still up behind them. The cold came in over the warped wood and the cracked rings and the soft cardboard with the year on the lid.

Len held the little plastic football fob in his fist.

"So," he said. "Before I let any of you say one more word about what John was *for*." He looked, of all of them, at Tomi, because she'd asked it, because she was the one who'd wanted it to mean something. "Tell me what you want it to be. The truth. Not the kind answer. Not the clever one." A breath. "Did you come here hoping he was right — or hoping he was sad? Because they're not the same, and I'll know which from your face before you finish."

Chapter 15 — The Made Thing That Isn't There

The lock-up smelled of cold concrete and the inside of an old kettle — limescale and dead heat. Somebody, decades ago, had run a single bar fire off an extension lead and it had blistered the paint on the breeze-block in a fan shape, and the fan was still there, and the fire was long gone, and that was the most honest thing in the room.

Tomi had the notebook open on her knees because the trestle table was full.

It was full of magnets.

Not lab magnets. Hardware-shop ferrites, the kind that come glued to the back of a promotional bottle-opener, and a tin of curtain-ring magnets, and a few of the big black arcs you'd lever out of a dead loudspeaker, and one beautiful machined ring of something that had clearly cost a man his groceries one week in 1971. They were laid out in the order Searl's diagram laid them out — a stator ring, then rollers around it, each roller a stack of plates: copper, then a layer the notes called *the imprinted neither*, then aluminium, then nylon. The word *imprinted* had a square drawn around it in biro, the kind of box you draw around a thing you cannot explain to yourself but need to keep.

"The plates have to be magnetised in a pattern," Naila said. She was on the floor, cross-legged, with the photocopies fanned around her like she was the centre of a sundial. "A sine wave printed into the metal. He calls it the layer of the squares — magnetic squares, alternating, so the rollers feel a series of little hills and they're always rolling downhill." She tilted her head. "It's a lovely idea. You'd never stop rolling downhill. You'd roll downhill forever."

"That's the part," Yusuf said, from the doorway where the light was, "that doesn't happen."

"That's the part that doesn't happen," Naila agreed, without heat. She said it the way you'd confirm a train time.

Tomi looked at the diagram in the notebook on her knees. It was a good diagram. That was the thing nobody warned you about — it was *competent*. The hand was sure. The man knew what a bearing was, knew how to draw a section, knew to put the little hatching on the cut faces. You could build a roller assembly from this. You could turn the rollers. They would turn because you pushed them, and then they would stop, because everything stops.

She found she was angry and didn't know who at.

The official story is clean, and you should hear it clean, because it deserves that.

John Roy Robert Searl, born Berkshire, an orphanage childhood, a man who said that at the age of four or five he had a recurring dream of rotating rings of numbers and that when he grew up he built the thing the dream had been showing him. He called it the Searl Effect Generator. A set of magnetised rings and rollers that, once you got it spinning, would keep spinning, faster, drawing energy out of — the word changes depending on the decade — the air, the ether, the vacuum, the *neither*. And then, past a certain speed, the whole disc would lose weight, ionise the air around it pink, and rise.

A flying generator that powered itself.

It is, and the kindest skeptics in the world will tell you this is the kindest framing, a *perpetual-motion machine*. Not fringe-but-open. Not awaiting-better-instruments. It asks a closed system to put out more than goes in, forever, and that door has been bolted since Joule, and the bolt has held against every single thing anyone has ever pushed against it. There is no measured residue here. There is no Casimir force weighed to within a percent. There is no crater on the Moon with his name on it. There is a man, and a dream of rings, and sixty years of building.

"Read it," Tomi said to Jakobus. She hadn't meant it to come out like

a dare.

He'd been standing near the fan-shaped scorch with his hands in his jacket the way he stood when he was waiting to be useful and didn't want to take up the air. He came over slow. He picked up the loudspeaker arc, turned it, set it down exactly where it had been. He picked up the machined ring — the grocery ring — and held it to the bar of light from the door, and turned it, and Tomi watched him do the thing he did, the thing they'd flown across an ocean for him to do, the reading of a made thing: *what is this for, what hand made it, what did the hand want.*

He held the ring a long time.

He put it down.

"It's a ring," he said.

"Jakobus."

"It's a beautiful ring. Somebody turned it on a lathe and they cared." He wasn't deflecting; she could hear him not deflecting. "A carburetor tells you what it's for. You hold it and it tells you — the heat goes here, the fuel sits there, here's where a man was clawing back something real. A sightline points. A water map remembers a river." He spread his hands over the trestle, over all of it, the bottle-opener ferries and the curtain rings and the grocery ring. "This doesn't point at anything. There's nothing under it. It's not a machine that does a hidden thing. It's a—" and he stopped, because he didn't sneer, it wasn't in him, "—it's a shape somebody needed to be true."

Frik, who had been very still in the corner, said, "*there's nothing there,*" in the flat blurted voice that wasn't his choosing, and went rigid with the misery of having said it, and Naila reached up off the floor without looking and put her hand flat on his shin, *I know, it's fine, it's true and you didn't mean us.*

It was true. That was the thing. The blurt had named it and the room had gone quiet around the naming.

The keeper had said almost nothing all morning.

Her name was Ruth and she was not a granddaughter and not a solic-

itor; she had simply been there, for the last eleven years, the woman who'd answered an advert for a part-time carer and stayed past the wage. She had the only key. She sat now on an upturned milk crate by the roller door with a mug she wasn't drinking from, and she'd let them in, and she'd watched them lay the magnets out, and she'd watched Jakobus pick up the ring, and she'd watched him put it down.

Edith hadn't lifted the camera once.

That was the loudest thing in the room and Tomi only half-logged it — Edith standing by the wall with the camera *down*, at her hip, the lens cap off but the thing pointed at the floor, like a person at a graveside who has decided that some things you do not take.

Because of the photos.

They were in a box file, slid out across one cleared corner of the trestle: the famous ones, the ones that had been on every poster, every late-night upload, *SUPPRESSED — THE MAN WHO BEAT GRAVITY*. A disc, mid-air, over a hedge. A disc above a Warminster field at dusk with a pink corona round it.

Naila had spent forty minutes with a loupe and the desk lamp.

"The horizon line runs behind the disc here," she said, "and in front of it here. Same photograph." She set the loupe down. "And the corona's painted. It's airbrush. You can see where it goes under the edge of the disc instead of round it. The light source on the disc is bottom-left; the light source on the field is top-right." She said it gently, gently, the way you'd tell someone a sum had come out wrong. "It's a model on a sheet of glass. A good one. A loved one. But it didn't fly. Nothing flew."

Heerfordt — the Danish engineer who'd spent years trying to find one honest witness to a working Searl disc, the patient kind of skeptic, the kind who *wanted* to be wrong and went looking — had got, at the end of all of it, the cleanest result in the whole subject. Not one. Not one witness to a working device. Not one.

"So that's it," Tomi said.

She heard her own voice do something she didn't like.

Here is the cost, and it landed on her, and she didn't see it coming because you never do see the one that's shaped like you.

She'd come down the ladder all this way being the sober one. She'd watched Naila say *the physics is against it* over the carburetor and agreed. She'd watched the Russian paper dissolve in her hands and held the maybe only where the residue earned it — Pogue's refused number, Kozyrev's breathing Moon, a real force weighed real. She'd been *good* about it.

And she realised — no. She didn't realise anything. What happened was she reached for the loupe to look at the corona herself, to find the seam Naila had found, to *check*, and her hand stopped above it, and didn't pick it up, and she understood from the not-picking-it-up that she didn't want to see the airbrush. She wanted the disc to have hung over that hedge. After everything. After being so careful. She'd been saving this one. She'd walked the whole descent telling herself she only held the maybe where it was earned, and the truth was she'd been carrying a small private hope all the way down the bottom of the ladder that *here*, the indefensible one, the one with nothing under it — here was where it would finally, against all her own sums, turn out to be real.

She'd needed it to be real.

She put her hand in her lap.

"Tomi," Naila said, quiet, not looking up, because Naila always knew where everyone's hands were.

"I'm fine."

"You're allowed to want it."

"I know the physics."

"I didn't say you don't know the physics."

Ruth set the mug down on the concrete. The small clink of it turned the room.

"He showed me a working one," she said.

Everyone went still.

"In the kitchen. Two years before he died." She wasn't performing it; she said it the way you mention what someone had for breakfast. "He had a little roller set on a tray and he spun it with his thumb and it kept going. Round and round. He looked at me to see I'd seen. And I'd seen — it was on good bearings and he'd given it a real push and it ran down slow, the way a good well-oiled thing runs down slow, a long time, you know, the way a bicycle wheel does." She picked the mug back up. "And I said, *isn't that something, John*. Because it was. It was something."

She looked at the magnets on the trestle, all of them, the bottle-opener ferrites and the grocery ring.

"You want to know what it's *for*." She'd heard them, then, all morning. "I'll tell you what it was for. He was four years old in a place that had three hundred children in it and nobody who was only his. And he had a dream about rings of numbers that fit together so perfectly nothing could stop them turning. They held each other up forever. Nothing fell down. Nothing got thrown away." She nodded once, at the table. "He spent sixty years building the dream where nothing gets thrown away. He believed it because he needed it more than he needed it to be true. And one of those is real, isn't it. The needing's real. The needing is the realest thing in this room."

Nobody said anything. Frik had both hands pressed over his own mouth, holding something in or holding it together.

"That's not a machine you can read, your man," Ruth said to Jakobus, not unkind. "There's no answer in it for you to find. I'm sorry. I know that's what you came for." She almost smiled. "I had the same disappointment, eleven years ago. You get over it. You stay anyway."

Jakobus inclined his head, the small grave bow he gave to anyone who'd just out-read him, and stepped back, and gave her the room. It was hers. It had always been hers.

Tomi looked at the loupe she hadn't picked up. She left it where it was.

The roller door went up with a bang of its own counterweight, and the grey afternoon came in, and a man came in with it.

He was in a good coat and he was holding a leather folder the way you hold a thing you've practised handing over, and behind him a woman in driving gloves stood by a clean silver estate car that had not been hired from anywhere the crew could afford. He was not from a cabal. He had a lanyard from a real foundation that endowed real chairs at a real university and ran, among other things, an archive-acquisitions programme, and he'd flown in from somewhere with a per diem, and he was, Yusuf would say later, *exactly the man I used to be, just better dressed*.

"Ms—" he checked the folder, "—Ruth. Hello. I'm sorry to interrupt." His eyes went over the trestle, the magnets, the box of photographs, and stopped on the photographs with the briefest flicker of a man pricing a thing. "I represent the estate's, ah, an interested party. We've made a formal offer for the complete archive." He held the folder out toward her, over the magnets, careful not to disturb them. "Notes, prototypes, photographs, the lot. It's generous. Considerably above the assessed value." A beat, and then the part he'd come to say, the clean-handed part, the part that wasn't a threat because it didn't need to be: "Our condition is custody. The full collection, sealed, under our stewardship. Withdrawn from circulation. For his sake, frankly. To stop the — the posters." He glanced at the corona photo, the airbrushed pink. "To stop people building a man into something he wasn't. He deserves the quiet."

He deserves the quiet.

Edith lifted the camera.

For the first time all day, she lifted it — not at Ruth, not at the magnets, not at the dead man's airbrushed disc.

At the folder, held out over the table, and the gloved hand on the door of the silver car behind it, and the offer that wanted to buy the silence and seal it shut forever.

The shutter went once.

Chapter 16 — A Prison Door Closes

The clipping was the size of a postcard and Edith would not pick it up.

She had a rule about that — had carried it out of a dozen worse rooms than this one. You do not touch the photograph until you have looked at it without your hands in the frame. So she stood over the trestle table in the lock-up with her arms folded and read the column of newsprint the keeper had laid out, the way you read a face before you raise the camera.

The Times. The date in the corner gone soft and brown. **MAN GUILTY OF ELECTRICITY THEFT**. And then the small paragraph that the world had decided was the whole of him.

“They left the meter alone and ran the house off the main below it,” the keeper said. Geraint was his name, a broad man with a draughtsman’s careful hands, and he kept the Searl papers the way other men kept their fathers’ tools — oiled, labelled, used. “Bypass. Crude as you like. He didn’t deny it. Said the Board had cut him off and a man has to boil a kettle.”

Naila, by the second box, had the court report on her phone, cross-checking the way she could not help. “Conviction’s real,” she said, flat and gentle at once. “Stoke Newington. Early eighties. There’s no argument about the wiring.”

“None,” said Geraint. “He stole it. He’ll tell you himself he stole it — told the magistrate, near enough.” He smoothed the clipping with two fingers, not to clean it, just to touch it. “The funny part’s only

funny if you're the sort who finds it funny."

The fish, riding the dashboard back in the Beast where Yusuf had left the door open, supplied the obvious. *Man who invented free energy. Gets done for nicking the leccy. Stop, you're killing me.*

Nobody laughed. The joke was true and that was the whole weight of it.

Edith made herself look at the man instead of the headline.

There was a press photo with the report — a long-faced man, sixties, in a jacket that had been good once, coming down some steps with his chin up. Not a fugitive's chin. A lecturer's. The chin of a man who still believed that if he were only given the floor for twenty minutes the misunderstanding would resolve itself. She had photographed that exact posture in three countries and it always meant the same thing. It meant the man had not yet understood that the floor was not going to be given.

"He thought the trial was where he'd finally get to explain the disc," she said. It came out before she'd decided to say it.

Geraint glanced at her. Reassessing. "Yes," he said. "That's exactly what he thought. He brought drawings. The brief asked him not to bring the drawings."

The official story sat on the table between them, sober and complete, and Edith respected it. A man wired around his electricity meter. A man who, for thirty years, had said he'd built a disc — the Searl Effect Generator, magnetised rollers on a ring, spun up till it lifted and threw off power for nothing — and who, in all that time, had never once put it in front of a meter, a camera, or a witness who'd stand behind it. Heerfordt had gone looking, later, for one person who had seen the thing run. The dossier in Naila's other hand was specific about the result of that search. *Not one witness.*

"It doesn't fly," Naila said, to the box, to the room, to herself. "The machine. It can't. The energy has to come from somewhere and there's nowhere for it to come from. I want to be — I want that on the record before anything else gets said in here. The disc does not fly."

"I know it doesn't," Geraint said. He didn't say it sadly. He said it the way you'd confirm a man's height. "I've had the rollers in my hands. Lovely engineering, the segments. Beautiful, even. And it doesn't fly."

So that was settled, the way it had been settled at every grave they'd stood over coming down this ladder. The machine was wrong. The book they were all standing inside was honest about that. And then the room did the thing the whole descent had been teaching Edith to wait for: it failed to close.

Because the man had still gone to prison.

Not for the disc. For the kettle. For the part of the story that was true.

"Tell us about after," Yusuf said. He had come in quiet from the Beast and was standing where he could see the roller door, which was how he stood everywhere now, since the last node, since he'd learned what the other outfit's people looked like.

Geraint sat. The lock-up had two chairs and he took the one that faced the boxes, the way a man sits with a coffin.

"After," he said. "Right. So you've got the trial, and the trial's a story — papers love it, the energy crank done for energy. And then the story stops, because a man coming out of an eighteen-month stretch in his sixties isn't a story. He's just a man with no money and a name everyone's heard and nobody believes." He turned one box a quarter-turn, lining it with the table edge. "He went on. That's the thing nobody puts in the documentaries, because there's nothing to film. He went *on*. Made more rollers. Wrote to the Russians, the Americans, anyone. Did the conference circuit when somebody paid the train fare, which got rarer. Sat in front rooms in front of six people and four cameras held by men who'd already decided what they were filming."

Edith's stomach did the thing it did. She knew those men. She had, on her worst days, been those men.

"For how long," she said.

"Decades. He died in 2018." Geraint let that sit. "Do the sum. From the trial. That's thirty-five years of going on."

“And in the thirty-five years,” Yusuf said, careful, “how many people came to actually—”

“To help him? Properly? With nothing in it for themselves?” Geraint looked at the boxes rather than at any of them. “There were the believers. There’s always believers, and God love them, some of them sent money they didn’t have. But they came to be near the *miracle*. They wanted the disc to fly because of what it would mean for them. That’s not the same as coming for the man.” He drew a breath. “And the clever ones, the ones who could’ve told whether there was anything in it — they came once. Took a look. Wrote the article. *Crank*. And the article was right, mind. That’s what I can’t get past, even now. They were *right*. So who’s going to drive to Stoke Newington a second time to sit with a man who’s right about being wrong?”

The lock-up was very quiet. Out past the roller door a forklift beeped somewhere in the industrial estate, businesslike, indifferent.

Edith looked at the press photo again. The chin. She understood now what it cost to keep a chin up for thirty-five years in front of cameras that were only there to confirm what they’d brought.

It was Frik who said it, and Frik who couldn’t have stopped himself if his life had hung on it.

He’d been quiet against the wall, doing the thing he did when a room got too full of grief, going very still and breathing through it, keeping the rope short. Edith had watched him manage it across four nodes now and she knew the cost of the managing. She knew he’d lost.

“There’s no machine to save,” Frik said. The blurt came out cracked down the middle, half his own voice and half the other thing that lived under it. “There’s nothing — *nothing* — there’s nothing left but the man.”

Then his hand came up to his mouth, too late, the way it was always too late, and his eyes went to Geraint, braced for the room to turn on him.

The room didn’t turn.

Geraint looked at Frik for a long moment. Something in the broad careful face came loose.

“Yes,” Geraint said. “That’s it. That’s the whole of it. There was never a machine to save. There was only ever him.” He pressed the back of his hand to his own mouth, mirroring Frik without knowing it, and brought it down again. “I’ve been minding these boxes eleven years waiting for someone to say it that plain.”

Naila had gone still in her own way — not Frik’s stillness, the other one, the one where the world arranged itself into something she could finally bear. She set the dossier down face-down. The disc does not fly was on the record. It could stay on the record and not be the last thing said.

Edith took up the camera.

She did not photograph the headline. She did not photograph the rollers — Geraint had them, beautiful and inert, in the smaller box, and a lesser version of her would have made them gleam, made them *suggest*. She photographed Geraint’s hand flat on the closed lid of a box, the labelled spine, the draughtsman’s nails. She photographed the man who’d stayed. That was the photograph the descent had been walking her toward, the hardest version of her own discipline — not *suppressed genius*, not *crank*, just: someone kept these. Someone is still here.

Yusuf’s phone went, and his face did the thing.

He stepped to the roller door, took it outside, came back in thirty seconds with the temperature of him changed.

“That’s the offer,” he said. “Properly, this time. They’ve stopped pretending it’s a museum acquisition.” He kept his voice level for Geraint’s sake and it cost him. “They want the lot. The papers, the rollers, the correspondence, the trial record, the lot — and a non-disclosure with it, the kind that means the boxes don’t come out again, not to a researcher, not to a documentary, not to anyone. They’re not buying it to *show* it.” He looked at the table. “They’re buying it to make it go quiet. There’s good money in him staying a crank forever. Costs them more if anyone ever sits down with this and writes it straight.”

The adversary had a face and a name and a budget, and the face was, in the end, the most ordinary one — a man doing arithmetic. Searl’s

silence had a market value, and the market preferred it intact.

“You could take it,” Yusuf said to Geraint, and Edith heard him make himself say it, because it was true and Geraint deserved the truth. “It’s real money. It’d be his name out of your hands and into a vault, and you’d never have to mind the boxes again. You’re allowed to take it. Nobody here’d blame you.”

Geraint sat with his hand still flat on the lid.

The decision was his. Edith felt the whole crew feel it — the firewall they carried like a spine. Not their boxes. Not their man. Not their answer. They had read the pattern across five graves; Geraint had read the one person, alone, for eleven years, and what was done with him now was his to give or to withhold and theirs only to witness. She kept her hands off the frame. She kept the camera down.

“What’s it *for*,” Geraint said. Not asking them. Asking the boxes, the way Jakobus asked things — Jakobus who’d shipped the Beast and stepped back out of the doorway hours ago and let this be theirs. “That’s the question your fixer keeps putting, isn’t it. What was it all *for*.” He moved his hand off the lid at last. “Wasn’t a power station. We’ve established that. The disc never lit a bulb.”

He stood. He went to the roller door and put his hand on the chain hoist, and he looked back at them, and the long-faced obstinacy in him was suddenly and exactly the chin in the press photo, thirty-five years on, kept up.

“It’s for this,” he said. “It’s so that when they come to make him quiet for good — and they’ve come — there’s somebody in the room who knew he stole the electricity *and* stayed. Both. At once. That’s what the boxes are for. Witnessing.” He pulled, and the chain ran, and the roller door began to come down on the daylight and the indifferent forklift and the man’s people waiting somewhere in the estate with their pen. “Tell them no. Yusuf — tell them no, in whatever words they understand.”

“Done,” Yusuf said, already typing.

“And the rest of you.” The door was halfway, the light going amber, then less. Geraint looked at Edith, at the camera she still held low. “Stay. Don’t film it for anyone but him. Just — be the people who were here when it closed.”

The door met the concrete with a sound like a sentence ending.

In the dim, the rollers in their box did not fly, and never had, and the man who'd staked his one life on it was twelve years dead with nobody to vouch that he'd ever made it spin. None of that had moved an inch. And Edith stood in the dark with her crew and the keeper who'd out-stayed them all, and held the camera down, and witnessed.

Chapter 17 — The Quiet

The roller door went up on its rust and the morning came in sideways, low and grey, the way January light comes into a place that was never built to be looked at. The storage unit smelled of dust and a faint sweetness Tomi couldn't place until she saw the open tin of three-in-one oil on the shelf — uncapped, dried to amber, left there years ago by a man who meant to come back the next day.

The unit was nearly empty. That was the thing she hadn't braced for. She'd built it up in her head as a treasure-room — a workshop frozen mid-thought, magnets in their cradles, the disc on its stand under a sheet you'd pull away to a gasp. The chase had run on that picture. Yusuf had spent two weeks reading the company that wanted it, the one that called itself a "heritage-energy acquisitions vehicle" on its own letterhead, which was the most honest thing about it. They'd flown a man in. He was outside now, in a hired Audi, drinking a service-station coffee, waiting to make an offer to a woman who had already decided.

The woman was Pamela. Sixty-odd, a cardigan the colour of wet slate, hands that knew how to lift things. John Searl's — well. Not his daughter. The relationships around the man had always been complicated and the press had always got them wrong, and she'd stopped correcting it. *I kept the unit*, she'd said on the phone, and that was the whole of her claim, and it was enough.

"You can put your camera down," Pamela said. She didn't look at Edith when she said it. She was looking at a tea-chest of papers. "I don't mean stop. I mean — you've been holding it up like a shield since you came in. You can put it down."

Edith lowered the camera. Held it at her hip, both hands, the way you

hold something you'd rather not set on the floor.

"There," Pamela said. "Better."

Here is the official story, and it is not unkind to tell it straight.

John Roy Robert Searl said he built a generator that needed no fuel. The Searl Effect Generator: concentric rings of segmented magnets, rollers running round them, and out of the spinning — electricity, lift, more energy than you put in. Discs that rose. Discs that *flew*, he said; flew off into the sky over Mortimer in the 1960s and had to be tethered. A craft that could take you anywhere. Clean power for every home on Earth.

It cannot work. The crew had said so to each other in the van without anyone needing to argue it. You cannot get more energy out of a closed cycle of magnets and motion than you put in to move them; the rotor would slow, not spin up; "over-unity" is the polite dressing-gown that perpetual motion puts on to be let into the party. No independent witness ever measured the excess. No photograph ever showed the flight that anyone could date and place and hold. Heerfordt, the Danish researcher who'd gone looking, sympathetic, *wanting* it — found not one witness to a working machine. Not one.

And there is a court record. In 1982 the South of England Electricity Board prosecuted Searl for abstracting electricity — for *stealing* it, dressing the meter, running his house off the grid he said he'd made obsolete. *The Times* carried it. He went to prison. The man who said he'd freed the world from the power company was convicted of robbing the power company. You can stand the whole legend up against that one fact and watch it sway.

Naila had laid that out in the reading room in a flat, exact voice, no cruelty in it, the way she laid out the patents in Winnipeg and the Casimir force in the lock-up and Kozyrev's gold medal beside his crater. She read the dismissal and found it sound. Four times down the ladder she had found it sound. This was the fifth and lowest rung, and there was no kernel left at the bottom. No real patent that worked modestly. No force weighed to a percent. Just a man, and a conviction, and an empty room.

So the question the book was built to ask had nowhere to go. *What*

is it for. Jakobus's question. Aim it at the disc and it dissolves in your hand.

"This is the last of it," Pamela said.

There were six boxes. A man's life in six boxes was not unusual; Tomi had learned that this winter. She'd learned the standard weight of a written-off life and it was about what one person could carry in three trips.

Frik was crouched by the open tea-chest, very still. He got still when a thing came too close. He'd been still in Siberia when the keeper described the room they'd thrown Kozyrev's name out of, and he'd been still in the San Diego industrial park when the word-thief explained, smiling, how you launder a real piece of physics into a wellness product. He was still now. Tomi watched the muscle work in his jaw and waited, because the wait was the kindest thing she could give him.

"He believed it," Frik said. Quiet. No blurt in it. "*Fraud*—" the word came out of him sideways, involuntary, and he rode it down with his breath the way he'd taught himself, "—is the wrong word. You can be wrong with your whole heart. That's not the same."

"No," Pamela said. "It isn't."

She'd been waiting to see if they'd say it. Tomi understood that now. The whole morning had been a test the way Winnipeg had been a test — not of cleverness, of *temperature*. The acquisitions man outside thought he was the buyer. He'd come to take the thing. Pamela had been deciding, all morning, whether the people who'd come to *read* it could be trusted to hold it without turning it into either a martyr or a joke. SUPPRESSED GENIUS or LOL CRANK. The two cheapest frames in the shop, and she'd watched both of them be hung on John for forty years.

"Here," she said.

She reached into the second box and lifted out a disc.

Not a whole one. A segment — a curve of the outer ring, maybe a third of an arc, the magnet material dull and chipped, one edge sheared clean where it had broken. It was heavier than it looked. It was also, Tomi saw, made with enormous care. Whatever else was true,

the *making* was not a lie. The segmentation was even. The surface had been worked. Someone had spent days of his one life on the geometry of this curve, believing the geometry would lift the world.

Pamela put it in Tomi's hands.

"Some things you hold," she said, "because someone had to try."

Tomi held it.

The cold of it came up through her palms. She did not say it lifted, because it didn't, and she did not say it never could, because that was the man outside's job and he was welcome to it. She held a broken third of a circle that a poor boy from a children's home had built out of the conviction that he'd been *shown* the answer in a dream, twice, the same numbers, and had spent the rest of his life trying to make the world see the dream.

The fish in her — the heckle she'd carried since the cold open, the one that had said *the physics is against you* to every single rung — the fish opened its mouth and found it had nothing to heckle. You can't heckle a man for being wrong with his whole heart. You can only heckle the *claim*, and the claim wasn't in her hands. The claim was a court record and a missing photograph and a closed file. What was in her hands was the *try*. And there is no debunk for a try.

"He's not on the patents," Naila said. She'd come up beside Tomi, looking at the segment, reading it the only way she could read anything, which was honestly. "Pogue had four. Real ones, on the record, you can pull them up. This—" she touched the sheared edge, light, "—there's no patent that works. There's no measurement that holds. There's nothing under it but him."

"I know," Pamela said.

"I'm not saying it to be cruel."

"I know you're not. You're the one who isn't." Pamela took the segment back, gently, and set it on the tea-chest, and looked at it the way you look at a photograph of someone gone. "That's why you can hold it. The cruel ones can't. They have to make it big or make it small. You're letting it just — be the size it is."

A car door, outside. The acquisitions man, deciding he'd waited politely long enough.

He came in under the roller door without ducking, though he was tall enough to need to, and that told you something about how he saw the place — as ceiling, not as somebody's last room. He had a folder. They always had a folder. He said the name of his heritage-energy acquisitions vehicle and Yusuf, by the door, said it back to him before he'd finished, with the dates, with the name of the holding company above it and the name above *that*, the one that owned a lot of dead men's silences in three countries and had a board meeting in April where this would be a line item.

The man recalibrated. You could watch him do it. He'd come to buy a credulous old woman's treasure. He found a fixer who'd already read his books and a crew standing around six boxes like a wake.

"There's interest," he said, to Pamela, "in preserving the legacy. Properly. Resourced. The collection deserves—"

"It's going to the museum in Mortimer," Pamela said. "The local one. It's small. They've got a case for it and a label that says what's true and what isn't. I wrote the label myself."

"We could offer rather more than a—"

"I know you could." She wasn't unkind to him either. That was the thing about Pamela; she'd run out of unkindness somewhere, used it all up, the way you do. "You'd put it in a vault. Or you'd put it on a stage. Either way it goes quiet. I've had enough of him being quiet. The museum's open Tuesday to Saturday and a child can walk in off the street and read what he tried to do." She looked at the man, mild. "You don't want the disc. You want it gone. Those aren't the same and I won't pretend they are for your money."

Tomi watched the offer die in his folder. He didn't argue long. People who buy silence don't have a script for someone who's decided to stop being silent; the leverage is all in the wanting-it-gone, and Pamela wanted it *seen*. He left a card on the box nearest the door, which nobody picked up, and he ducked, this time, going out, and the Audi pulled away over the gravel and that was the chase. That was the well-funded outfit racing to take it. It just drove off. The descent

ended not with a seizure but with a man giving up because there was nothing here he could use, which was its own kind of verdict on the whole bottom rung of the ladder. There was nothing here anyone could *use*. That had always been the trouble with John.

They boxed the last of it. Edith took her photographs — not the SUPPRESSED-GENIUS shot, not the crank's-clutter shot, just the boxes, just Pamela's hands on the lid, just the dried oil tin on the shelf — and she took them quietly and showed Pamela each one on the back of the camera before she kept it, which was a thing she'd started doing in Winnipeg and hadn't stopped.

"That one," Pamela said, of the oil tin. "He always meant to come back the next day. Forty years of next days." She wasn't crying. She'd done that already, probably, a long time ago, the way you do it before strangers arrive. "You can use that one."

The roller door came down. The museum van would come Thursday. Pamela shook each of their hands, and held Frik's a second longer, because she'd watched him go still and known what it meant even if she didn't have the word for it, and Frik let her, which he didn't always.

In the Beast, the heater ticking, the windscreen fogging with the six of them, nobody started the engine for a while.

Yusuf had the acquisitions man's card and was tearing it into small even squares for something to do with his hands. Naila had her notebook open on the patents page — Pogue's four, the real ones, top of the ladder — and she wasn't writing, just looking at where the book had started, the rung you could stand on, the long way down from there to a tin of dried oil. Edith had the camera in her lap, lens cap on, both hands. Frik had his eyes closed.

"There's no number," Tomi said. Not to anyone. The fish-thing, finding its voice last. "With Pogue there was a number he wouldn't say. You could hold the place where it should be. Here there's not even that. There's nothing he wouldn't tell you. He'd tell you everything. It just isn't *there*."

“No,” Naila said.

“So what was it for.”

Jakobus’s question, in Tomi’s mouth, in the fogged van, with the man it was meant for thirty years dead and the disc going to a glass case in Mortimer under a label a tired woman wrote so a child could read it. *What is it for.* Aim it at the machine: nothing. The machine is for nothing; the machine does not work; the book has said so plainly and means it.

Aim it at the man, and it doesn’t dissolve. It just doesn’t answer.

Nobody answered it. That was the thing they could do, in the van, that the man outside couldn’t and the SUPPRESSED-GENIUS crowd couldn’t and the LOL-CRANK crowd couldn’t — they could leave the question standing in the warm fogged air and not knock it down with a verdict it didn’t earn. Frik’s hand found the door handle, not to open it, just to hold something. Edith reached over without looking and put her hand on top of his, the camera hand, the one she kept being asked to lower. Yusuf stopped tearing the card.

Tomi started the engine. The wipers cleared two arcs of grey Berkshire morning and the gravel lot and the lowered door of the unit where a man had meant, for forty years, to come back the next day.

She didn’t drive yet. She let it run. The heater pushed warm air around the six of them, and the question sat in the middle of them, unanswered, unanswerable, and it was the warmest thing in the van, and it was all they got to keep.

She put it in gear.

Chapter 18 — What Is It For?

The Beast smelled of rain and cold coffee and the inside of a man who had been driving since four.

Jakobus had the wheel. Nobody had asked him to; he'd simply been in the driver's seat when they straggled out of the lock-up in Mortimer, and the road north out of Newquay had taken him without comment. He drove the way he did everything — leaning slightly back from it, letting the thing tell him what it wanted. The wipers laid down their two arcs and the Cornish hedges streamed past green and dripping, and in the back the crew had gone the particular quiet that came after a long day of being careful with someone.

Yusuf had the Searl folder open on his knees and was not reading it.

“Pogue first,” said Naila, from the window seat, to nobody. She had her knees up and her chin on them, and she was watching the wing mirror, where the road they'd already driven kept arriving and leaving. “He wanted his number to be true.”

“They all wanted their number to be true,” Yusuf said.

“No.” She turned the word over, found it accurate, kept it. “Pogue wanted it true and *small*. He had it. High twenties. A thirsty V8 in Winnipeg in 1933 coughing its way up toward thirty miles to the gallon off the exhaust heat it was already throwing away — that's *real*, that works, Jakobus stood there and told us what it was for.” She didn't look at the front seat. None of them looked at the front seat, which was how the front seat liked it. “Claw back the fuel the engine's wasting. A made thing that does a modest, honest job. And

then somebody printed two hundred, and the number ate the man.”

“The editor printed two hundred,” said Edith. She had the small camera in her lap, lens cap on, the way she carried it when she wasn’t going to use it. “Pepper. And then Pepper said it was *entirely misleading* — his own paper, his own word — and it didn’t matter. The number was already loose.”

“And in fifty-three he was asked.” Naila’s voice didn’t change but something underneath it did. “Did you claim two hundred. And he said no — *or even half that*. And then they asked him the real one. The one he actually got. The honest high-twenties one that would have made him a clever mechanic instead of a fraud or a martyr.”

“And he wouldn’t say it,” said Frik, from the back, gently, and then his shoulder jerked and the next word came out of him sideways, not chosen: “*Coward—*” He shut his eyes. Opened them. “Not. Not him. The word’s not for him.”

“I know,” said Naila. “It’s never for them.”

The wipers went. Jakobus changed down for a bend and the Beast settled into it.

What is it for, then. The refusal. A man holds the one number that would clear him and he takes it into the ground. Naila put her forehead against the cold glass and let the question sit in her hands the way you’d hold something that might still be alive. Not *fraud* — fraud names it and closes the file, and the file wouldn’t close, four real patents lying on the granddaughter’s table refusing to vanish when you called him a crank. Not *martyr* either; nothing proved. Somewhere a man decided that the true small number was worth less to him than the silence, and made oil filters, and died. The dignity of the man who will not be measured on the world’s terms even to save himself.

“He kept it,” she said finally. “The number. That was what it was for. To keep.”

Nobody answered that, because it was right.

The road widened into A-road and Jakobus let the speed come up. Yusuf closed the Searl folder and opened the laptop instead, and

on the screen for a second was the thing he kept coming back to — the Casimir plates, Lamoreaux's measurement, the force weighed between two pieces of metal a hair's breadth apart, agreement with theory inside a percent. Real. Measured. *In the literature.*

"The vacuum thing," he said. "That's the one I can't put down."

"Because it's true," said Naila.

"Because it's true *and* it's useless." He turned the screen so Bashir could see it, and Bashir nodded the way he nodded at things he'd already filed. "The vacuum isn't empty. You can weigh it. Two plates, close enough, and there's a push you can put a number on — and the number's *right*. And then every garage in California sells you a box that says ZERO POINT ENERGY on the side and it's a battery and a wire and a lie. The realest physics on the slate, wearing the cheapest word."

"So what's it for," said Edith. Quietly. She'd started saying it, this trip. Jakobus's question, passed hand to hand down the van like a flask. "The real one. Not the box."

Yusuf thought about it longer than Yusuf usually thought about anything. He'd sold the laundered word, once, in another life — *tachyon*, *quantum*, *zero-point* — sold it to people who wanted to believe and had money, and he knew exactly the shape of the wanting because he'd built tools for it.

"To say you don't know yet," he said. "That's what the real one's for. The Casimir force is the universe admitting the floor isn't where you put it. Empty isn't empty. That's not an answer, it's permission to keep asking." He shut the laptop. "The frauds sell you the answer. The physics sells you the question. People can't tell them apart because they sound the same out loud."

"They don't sound the same," said Frik, and waited, and the rest of the sentence was his own this time. "Not if you're listening for who's asking you for money."

Edith laughed, once, surprised out of her. Even Jakobus's eyes moved in the mirror.

Bodmin Moor came up grey and the rain thinned to mist and they stopped at a layby because Mira needed to stand on ground that wasn't moving. They all got out. Nobody said *Kozyrev* but it was *Kozyrev's* turn and they stood in the wet with the moor going on forever and let him come.

The astronomer. The real one — the gold medal, the crater on the far side of the Moon with his name on it, no charity in that, you don't name a crater after a crank. The Gulag that really took ten years of him. And the night in November of fifty-eight when he sat at the Crimean telescope and watched a reddish gas vent out of the central peak of Alphonsus, and photographed it, and the world eventually, grudgingly, agreed: something had breathed off the Moon.

"He was right about the thing nobody believed," Mira said, to the moor. "The breathing. He caught it. And then he was wrong about everything he built on it for the rest of his life."

"Time having a current." Naila, beside her. "Torsion. The mirror. None of it replicated. Unhideable, that — sixty years and nobody's caught time flowing in a lab."

"So he gets buried for being right and then he buries himself by reaching too far." Mira pulled her sleeve over her hand. "What was it for. The reaching. After the camp. After they took the ten years."

The mist moved over the moor and didn't answer and that was the only honest thing on the moor.

"Grief," said Edith. She had the camera up now, but she wasn't pointing it at any of them; she was pointing it at the empty moor, the nothing, the grey. *Click*. "He lost his wife while he was in there. He lost ten years. He came out and the universe owed him something it was never going to pay, and he spent the rest of his life trying to prove that time had a direction you could *feel* — that the dead weren't only behind you, that the current ran." She lowered it. "I don't know if torsion fields are real. They're not, the literature says they're not. But I know what it's *for* to need time to have a current after the State has stolen ten years of yours and your wife is dead. That's not physics. That's a man building a machine to hold his grief in."

Mira was crying, a little, in the practical way she did everything. Frik put his hand on her shoulder and his face did its thing and he didn't

fight it this time, just let the tic run and kept the hand where it was, which was its own kind of sentence.

Back in the Beast. The tachyon node went fast because the tachyon node was the cheapest — Yusuf's own old country, the word stolen clean. The Higgs had been *tachyonic* in the maths once, a real technical term, a sign in an equation, and the OPERA neutrinos had famously, briefly, run faster than light until somebody found the loose fibre-optic cable and the whole thing collapsed into a connector that wasn't screwed in tight.

"A loose cable," Yusuf said, and shook his head, almost fond. "Sixty nanoseconds of the most exciting result in physics and it was a *cable*. And the people who'd already printed the t-shirts didn't take them off the racks."

"What was it for," said Naila. The question was just theirs now, the family word.

"Language." Yusuf, surer here than anywhere, because this was the con he used to run. "Tachyon's a real word that means a real thing in one equation and *nothing* in a product. They stole the word because the word does the believing for you. You hear it and your brain goes: *physics, hard, true, suppressed*. It's a machine made of one word." He looked out the window. "That's what it was for. To borrow a real man's vocabulary so you didn't have to build a real man's machine."

"He learned that one the expensive way," Frik said, and this time it was kind, and it was chosen, and Yusuf took it like a hand on the shoulder.

The motorway, then, and the dark coming on, and the Searl folder unopened on Yusuf's knee because Searl was the floor and they'd been on the floor all week and nobody wanted to go down there again with words.

The disc was in the footwell.

Not *the* disc — there was no the disc, there never had been a flying one, no photograph, no witness, Heerfordt's careful and devastating *not one witness*. The conviction was real — *The Times*, eighty-two,

eighty-three, the electricity meter run backward, the man who said he made power for free caught stealing the very power he claimed to give away. That was real and the book said so and it didn't flinch.

But the keeper had given them a piece of one. A segment. A wedge of ring with the little magnets seated in it, hand-wound, the windings done by a man's actual fingers in an actual lock-up over actual years, and it had never flown and it would never fly and it sat in the footwell of the Beast catching the motorway lights, broken, beautiful, *made*.

Tomi reached down and picked it up.

Held it in both hands. The magnets had a faint pull, you could feel them want each other and refuse, the way magnets do, the way the whole thing did — wanting to spin, refusing to spin, forever. Tomi turned it over. Found the place where one winding had come loose and been re-soldered, twice, three times, you could see the layered solder, a man going back to the same failure again and again with the iron, *here, this time, here*.

What is it for.

They'd asked it five times. They'd answered it five times and every answer had been a real thing, a physical thing, a function you could hold — claw back the wasted fuel, weigh the empty that isn't empty, photograph the Moon breathing, borrow the real man's word, hold the grief in a mirror. Jakobus's question always had an answer and the answer was always grounded and the answer always, every time, re-opened the thing instead of closing it.

And this one wouldn't answer.

Tomi turned the broken wedge in both hands. It wasn't for flight; nothing flew. It wasn't for free power; the meter ran backward, the man stole what he said he gave. It wasn't for the world, which had tried him and jailed him and was right to. The little magnets pulled and refused. The solder went back and back to the same wound.

In the front seat Jakobus said nothing at all. He didn't turn around. He was the road and the road doesn't tell you where it ends; he'd handed every answer all week to the crew and the keepers and he handed this one too, by saying nothing, by keeping his eyes on the wet lit lane.

Tomi held it. *What is it for.*

And let it stay unanswered. Set the wedge gently back in the footwell, where it caught the lights, where it would not fly. The hum of the Beast. Edith's lens cap on. Naila watching the road arrive and leave in the mirror. Five men reaching, and the world doing its housekeeping, and one piece of a thing that never worked riding north in the dark in the hands of people the world had also tried to sweep away.

Some things you get to keep without being able to say what they're for.

The motorway unspooled. Nobody filled the quiet. It was, for once, the good kind.

Chapter 19 — The Van Between the Graves

The Beast ran best at night, which Frik had stopped trying to explain to anyone. Something about the cold air and the diesel and the way Yusuf drove when there was nothing to overtake — a long, even pressure on the throttle, the road unspooling under the headlights like cassette tape pulled from a broken case. Edith had the passenger seat with her boots on the dash and her camera asleep in her lap. Naila was in the second row with a printout of the OPERA timing diagram she had read forty times and would read again. Bashir slept against the window with his mouth open. Mira had her headphones half-on, half-off, the way she did when she wanted the music and the talk both.

They were three hundred kilometres from the storage unit in San Diego where a man's whole afterlife sat behind a roller door, and four days from finding out whether the people with the chequebook had got there first. Nobody was talking about that. They were talking about the carburetor.

"High twenties," Naila said, to the window. "That's the honest number. A vaporiser on a thirsty V8, you claw back the fuel that used to go out the pipe unburned. High twenties, maybe thirty on a good day with the wind behind you." She turned the printout over. "It's a *good* device. It does a real thing."

"And the two hundred?" Mira asked, headphones down now.

"A gallon of petrol holds about thirty-three kilowatt-hours." Naila said the number the way other people said *amen*. "That's fixed. God doesn't negotiate it. A carburetor only mixes — it doesn't add energy,

it can't add energy, there's no energy *in* it to add. So you can burn it cleaner, you can stop wasting it, and modern engines already burn ninety-seven percent of it, which is why your fancy hybrid still can't do two hundred." A pause. "The man built something real and the world only wanted the lie about it."

"BUFFALO," said Frik.

It came out of him sideways, the way they did, a word with no owner. Bashir didn't wake. Edith's mouth moved in the dark, almost a smile. Nobody flinched anymore, and the not-flinching was a thing Frik would have driven through the night just to sit inside of.

"He never said two hundred," Frik went on, when the word had finished using his throat. "That's the part. Nineteen fifty-three. A reporter found him — old man by then, making oil filters — and asked him straight. And he said he never claimed two hundred miles to the gallon." He watched the white lines come. "He said he never claimed *half* that."

"Half of two hundred is a hundred," Mira said.

"A hundred's also impossible," Naila said. "Same physics. Doesn't matter."

"Right. But he wouldn't say the real one." Frik felt the thing the way he always felt it, like a stone in a shoe you've walked on so long it's a friend. "He had the floor. He could've said *twenty-eight*, and twenty-eight is *true*, and twenty-eight is *good* — and he just. Wouldn't. Took the real number into the ground."

The van held that for a while. The heater ticked.

"Maybe twenty-eight didn't feel like enough anymore," Edith said, not turning around. "After they'd written two hundred on him. Maybe once the world tells you you're a miracle, the truth feels like a confession."

The fish — they all called it the fish, the little voice that lived in the dashboard speaker and in the back of Frik's own skull, the one that heckled the *claims* and never the people — the fish said what the fish would have said: *Or maybe he just liked being mysterious. Cheaper than being right.* Frik let it. The fish was allowed. The fish kept them honest. You needed something in the van that didn't get sad.

They came off the interstate near two in the morning because Yusuf wanted fuel and Mira wanted to stand on ground that wasn't moving, and because the brown sign at the off-ramp said something none of them expected to see in the dark in a part of the country none of them had a reason to be in.

It wasn't grand. A turnout, a gravel apron, a single low light on a pole that buzzed. A bench. And a stone — granite, machine-cut, the kind a small historical society puts up when one of theirs gets a paragraph in a book forty years too late. There was a plaque, and the plaque had a name they'd been saying for nineteen chapters of their own lives, and under it the dates, and under the dates the words the society had agreed on after what must have been a long meeting: *INVENTOR. He believed it would help everyone.*

"Huh," said Yusuf. He'd cut the engine. The silence after the Beast's silence was enormous.

Edith was out first, no camera. That was the tell — she'd lowered the lens before her boots hit gravel. She walked to the stone the way she walked into rooms where something had happened, slow, weight on the outside of the foot.

The plaque got the patents right. That mattered to Naila, who came and stood with the printout still folded in her hand and read the numbers off the granite under her breath — the real ones, the ones that lived on Google Patents, the ones that didn't vanish when you called the man a fool. No typo. The society had checked. Someone here had cared enough to check, and that was the warmest thing Frik had seen in a week.

There was nothing about two hundred miles to the gallon. There was nothing about suppression, no oil barons, no break-ins, no shadowed men in good cars — the things the man himself had said, late, uncorroborated, into a country that had stopped listening. The plaque didn't claim it and didn't sneer it. *He believed it would help everyone.* You could stand on that. It was the one sentence about him that nobody, on any side, could prove wrong.

"They got the maybe right," Edith said, very quietly, and that was as close to preaching as any of them would come, and she knew it, and

she shut her mouth.

It started with Bashir, who had woken when the engine died and come out blinking and found them all standing at the stone. He didn't ask what it was. He read it. Then he went to the van and came back with a single hex nut from the parts tin — a real one, threaded, the kind that holds something to something — and set it on the base of the granite, square, the way you'd torque it.

"For the device that worked," he said. "The one he actually built."

Nobody told him not to. The historical society could sweep it tomorrow. That wasn't the point and everyone knew the point.

Mira left a guitar pick. She didn't explain it and Frik didn't need her to; she'd told him once, at three in the morning in a different country, that a pick was a thing you held to make a true sound out of a dead object, and if that wasn't this whole book she didn't know what was. She wedged it in a seam of the stone where the wind wouldn't take it.

Naila stood the longest, and what she finally left she left badly, hands not sure of themselves — she folded the OPERA printout into a tight square and tucked it under the hex nut. The wrong inventor's evidence at the right inventor's grave. "It doesn't matter that it's the wrong one," she said, before anyone could say it was. "It's the right *thing*. It's a number somebody refused to fake." She straightened. Her face did the work her voice wouldn't. "He'd have understood that."

Edith took a photograph then. Just one. Not of the stone — of the nut and the pick and the folded paper, the small junk of people who'd decided to care, lit by a buzzing pole-light at two in the morning. She'd spent thirty years deciding what was worth the shutter and what wasn't, and she'd told Frik once that the war taught her the camera could either witness a man or use him, and that there wasn't a third option, and that most of her career had been the second one. This was the first. She lowered it again fast, like it had got hot.

Frik's turn. He didn't have a thing. He'd checked his pockets twice. He had a receipt and a guitar capo that was Mira's and a USB stick of nothing.

“FRAUD,” he said, to the stone, loud, in the dark, and the word went out across the gravel and the off-ramp and the empty country, and it was the worst possible word and the worst possible time and he felt the old shame come up his neck — and then nobody laughed at him and nobody flinched and Bashir just put a hand flat on his shoulder, and the word hung there and changed shape while it hung.

Because that *was* the word, wasn’t it. That was the word they’d hung on the man. That was the file the world had closed. Frik’s throat had reached into the museum of everything anyone ever said about Charles Nelson Pogue and pulled out the one tag they’d stapled to him and read it aloud at his grave so the man could hear how stupid it sounded in the open air.

“That’s what I leave,” Frik said, when he had his own voice back. “I leave it here. He doesn’t have to carry it anymore.”

The fish, in the dashboard, fifty metres away, through closed doors, said nothing at all. The fish knew when to shut up. The fish had always known.

They sat on the bench, all six, which it was not built for, pressed together against the cold, and for a while the only sound was the pole-light and somebody’s stomach.

“They’ll get to San Diego,” Yusuf said, eventually. He was the one who could say it, because he’d been the one who used to sell the silence. “The lawyers. The chequebook. Day after tomorrow, maybe sooner. They’ll buy the lock-up off the family and the family will be relieved, because the family is *tired*, because being the keeper of a man the world calls a crank is a job nobody applied for.” He scrubbed his face with both hands. “And then it’s gone. Behind an NDA. Tidied. Quiet.”

“So we read it first,” Mira said.

“We read it first.” Yusuf nodded at the dark. “But that’s the part I keep — they’re not stealing a *machine*, you understand. The disc doesn’t fly. There’s no machine. There’s a man’s *papers*, a man’s *answer*, a man’s *quiet* — and they’re buying the quiet, because the quiet is the only thing that ever had any value, because the quiet is what kept everyone from having to be wrong about him.” He laughed, short, not happy. “Cheapest acquisition in the world. Nobody else is bidding.”

“We are,” said Bashir.

“We can’t outbid them.”

“We’re not bidding money.”

And that landed in the cold and sat there, true, and Frik felt it the way he felt the good words, the rare clean ones that came up right: they weren’t here to *own* the man. They couldn’t and shouldn’t and the keeper would tell them so, the keeper always out-read them, that was the whole grammar of every grave they’d stood at — the keeper knew the *person* and the crew only ever got to read the *pattern*, and the answer of what it was *for* was the keeper’s to hand over or to keep. They weren’t racing the chequebook to *take*. They were racing it to be there. To witness. To put a hex nut on the stone before the roller door came down for good.

“What if she won’t open the file?” Mira asked. The keeper in San Diego. The one who’d already buried her people once. “What if she’s tired too. What if she just wants the cheque and the quiet.”

“Then she keeps it,” Edith said. “Then it stays hers. Then we drove a long way to sit with someone and not get the answer, and that’s allowed.” She had her camera in her lap again. She did not raise it. “That’s the job. Most of the job is the not-getting.”

Naila had gone very still beside Frik, the way she went still when a thing was assembling itself in her, plate by plate.

“He took the number into the ground,” she said. “Pogue. He had it and he kept it and the keeping was the only power he had left, and we’re never getting it, it’s *gone*, it’s been gone since nineteen eighty-five.” She turned and looked down the length of the bench at all of them, which cost her, eye contact always cost her and she spent it anyway when it mattered. “And it doesn’t make me want to stop. It makes me want to read the next one harder. Because somebody refused to lie even when the truth was less than the legend, and the world filed that under *fraud*, and *that’s wrong*, and I can’t fix it, but I can *not look away*.”

“BUFFALO,” said Frik, gently, which this time meant nothing and everything, and Naila almost smiled.

The pole-light buzzed. Somewhere out past the apron a truck went

by on the interstate, doppler-low, gone. Six people on a bench built for three, between a granite stone and a parked van, in the dark, in the cold, with two days' lead on the people who buy the quiet and no way to win except to get there.

Yusuf stood up first. He always stood up first; somebody had to be the one who started the engine. He looked at the stone, at the nut and the pick and the folded paper Naila had left, the small evidence of caring that the wind would test and the society would sweep and that had still, for one night, existed.

"San Diego," he said. It wasn't a question.

They got in. The Beast caught on the second try, the way it did when the air was cold, the diesel finding itself. Edith put her boots back on the dash. Naila found her diagram. Bashir was asleep before they reached the on-ramp.

Frik watched the stone go small in the wing mirror — the buzzing light, the bench, the name nobody would have to carry alone tonight — and then the dark took it, the way the dark takes everything, and the road came up white under the headlights, unspooling, and there was another grave at the end of it, and they were going to go and read it whether the world cared or not.

The fish, when Yusuf flicked the speaker on, said: *You know they probably won't even let you in.*

"We know," said five voices, more or less at once, and somebody — Frik, it was Frik, it was always going to be Frik — started to laugh, and it ran down the bench like current, and the van drove on between the graves in the warm dark, full of the unhirable, going the wrong way for all the right reasons, and not one of them would have been anywhere else.

Chapter 20 — The File Never Closes

The Beast came back wrong.

Not broken — Jakobus would have heard a broken thing a county off. Wrong the way a coat is wrong after a funeral: still the coat, still your arms in the sleeves, but every fold remembering a room. Edith sat in the back with the camera in her lap, lens cap on, and watched the home road come up under the wheels the way it always did — the same gravel, the same listing fence post that had been listing since the first book, the same dog two farms out that ran the boundary line barking at nothing and meaning it.

She did not lift the camera. She had a list, by now, of the things she no longer photographed, and the homecoming was on it.

They unloaded in the long light. Nobody hurried. Yusuf carried the document boxes two at a time the way a man carries boxes he has decided not to feel, and Edith catalogued the decision in the set of his shoulders and let it alone. Bashir backed the trolley up to the roller door. Mira propped it with the brick that lived for exactly that. Naila stood in the middle of the yard with one box held against her chest, not moving, reading the label she had written herself in Winnipeg a lifetime of nodes ago — POGUE — EST. PAPERS — COPIES ONLY — in her own square hand, and not setting it down.

“Naila.” Mira, gentle. “Trolley’s free.”

Naila set it down. The cardboard kissed the concrete. She stepped back from it like you step back from a sleeping thing.

Inside, the base did the merciful work bases do. The kettle. The wrong mugs. The hum of the chest freezer that nobody had ever managed to make stop and everybody had stopped hearing. They laid the books along the trestle in descent order without anyone saying descent order — Pogue's folders first, fattest, the photocopied patents with the case numbers Naila could still recite if you woke her at three: one-seven-five-zero, three-five-four; not the typo, never the typo, she'd corrected a podcaster's comments section about it once and felt better for a week. Then the ZPE node, thinner, the Casimir reprints with Naila's marginalia getting smaller and smaller down the page. Kozyrev. The tachyon folder, which Yusuf had labelled simply WORD and underlined twice. And at the end of the trestle, in a flat archival box that had cost more than the rest of the trip's stationery combined, the Searl materials. The court report. The photograph of a man standing next to a thing that was not in the photograph because there was no photograph of the thing.

And the prototypes. What they had of them. A length of Pogue's exhaust manifold, real iron, real soot in the seams, the made thing that genuinely did the modest work it was made to do — claw a thirsty old V8 up into the high twenties, no further, never further, the gallon holding its fixed ration of energy the way a gallon always has and always will. A coil from a man who'd believed it pulled current out of nothing. A segment of disc, aluminium and magnet, that did precisely what aluminium and magnets do, which is sit on a table.

Edith photographed the table once. The whole length of it. The descent laid out like a staircase you could only walk down.

She did not photograph the faces. That was the other discipline, the one she'd learned in this book and not the wars: that there is a kind of looking that takes, and the keepers had taught her, every one of them, the cost of being taken. The Winnipeg woman with her arms crossed over the folders until she believed them. The man in San Diego with his quiet *not one witness* and his refusal to make his friend a saint or a fool. They had out-read her, all of them. They had read the band before the band read the bench. They had known, looking at the camera, exactly which book they were in danger of becoming.

SUPPRESSED GENIUS! the worst version went. CRANK, said the other worst version. Both of them frames. Both of them taking.

She lowered the camera.

“I’m doing the files,” Frik said, from the corner with the good light, and the way he said it — flat, careful, the voice he used for sentences he needed to come out whole — meant *leave me to it* and meant *stay near*. Both. The crew had learned to hear the both.

He had the laptop and he had the index cards, because Frik wrote everything twice, once in the machine that could be searched and once in the hand that could be held, and nobody had ever asked him why and he had never offered. Edith brought him tea and put it down out of his elbow’s arc and sat across the room where he could feel her there without it being a thing.

The case files were the band’s, the way the telling was always the band’s — never the place, never the bench, never the man. They owned the writing-up. They did not own the answer. That line had been drawn in the seed of them before Edith ever picked up a camera, and the whole book had been an exercise in not crossing it, five times, while everything in the trade pulled the other way.

Frik wrote.

NODE 1 — POGUE. He typed the patent numbers and then read them aloud, low, to check them against the hand-card, and the room let him. *Four real patents. Exhaust-heated vaporiser. Works — modestly. Editor’s retraction: “entirely misleading.”*

His head ticked, once, hard, to the left. Nobody looked. You learned not to make the not-looking a performance; that was its own kind of looking. He breathed and came back to the keys.

1953, he typed. Pogue denied 200 mpg “or even half that.”

And then the thing his body did at the worst and truest moments, the blurt that arrived sideways and landed dead centre — not loud, this time, barely above the kettle:

“— *wouldn’t say the number.*”

It came out of him the way the truth comes out of a confession nobody wanted to make. The room held very still. Because that was it.

That was the whole node in three words his throat had thrown without asking him. The man had four patents and a working — *modestly* working — machine and the press of the entire continent at his back and he had stood there in 1953 and refused, to his death, to name the figure. Not the impossible figure. *The real one*. The high-twenties truth that would have been a small honest triumph. He'd buried it with the lie, on purpose, and taken the reason down with him.

Naila, by the trestle, had gone still the way Frik's body had gone still — the recognition that passed between the two of them without a word, the one the crew never said out loud because saying it killed it. A man who would not give the world the number. A girl who had spent her life being handed holes where numbers should be and asked to be grateful for the quiet. Edith watched it pass and did not photograph it and felt the not-photographing as the deepest thing she'd done all year.

"Leave it open," Naila said. Not to Frik. To the room. "Don't close that one."

"I'm not going to close any of them," Frik said.

He did them all, the long way, over hours. The chest freezer hummed. Outside the light went from gold to the grey-blue that meant nine o'clock in this part of the world, and Bashir made the second round of toast because in this house grief was eaten, always, it was a rule older than any of them. Yusuf came and stood behind Frik's chair when he got to the tachyon node and read over his shoulder the word he'd sold once, in another life, to people who'd paid for the borrowed shine of real physics and got nothing under it. *No mechanism. No evidence. A real word, stolen*. Frik typed it clean and Yusuf put a hand on the back of the chair and took it away again and went to do the dishes that did not need doing.

The Searl box stayed shut until last. When Frik reached for it his hand stopped over the lid.

"The court report's real," he said, like he needed it confirmed out loud. The Times. The conviction. The dates. The dull bureaucratic fact of a man and a meter and a sentence, all of it documentable, none of it the thing that mattered. And then the other real thing, the keeper's,

the one the San Diego man had given them at the end with his hands open and empty: *not one witness*. For all the years and all the films and all the people who swore they'd seen the disc rise, when you went looking for the one human being who had stood in the room when it left the ground — there was no one. The keeper had said it without triumph and without contempt. He'd said it the way you say a thing about a friend you love and will not lie for.

Frik opened the box. Read it. Typed the node up to the last line.

And then he did the thing Edith had been waiting for him to do without knowing she was waiting.

He left the last page blank.

Not empty by accident. Blank on purpose — a clean sheet at the end of the file, a card with nothing on it slid behind the rest, the white of it deliberate as a held breath. He squared the cards. He did not write *and maybe*. He did not write *some things we are not meant to know*, which would have been a lie about the whole point of them, who were built to want to know and went to the ends of the earth to find out. He did not write the comforting sentence in any of its costumes. He left the page blank and set the pen down beside it where the next thing could go, if a next thing ever came, and if it never came the blank would sit there being exactly as honest as the rest.

"Done," he said.

His voice did the small tic and then it was just his voice.

Hecklefish, had Hecklefish been there, and Hecklefish was always there, would have surfaced from the kitchen tap or the dark behind the freezer and said something insufferable. *Oh, you left a page blank. Very deep. Very brave. You want a medal, or you want to admit the disc doesn't fly?* And he'd have been right, the fish, he was right about the machines every single time — the disc doesn't fly, the coil doesn't pull current from nothing, the gallon holds what the gallon holds. The fish heckled the *claims* and the fish was correct. The fish just never had a thing to say about the men. That was the one room the heckling didn't go into. Even a fish knew that much.

Edith stood up. Her knees told her how long she'd been sitting. She

walked the length of the trestle one more time, past the descent, the staircase of it, down from Pogue's good iron to the segment of disc that was only ever aluminium and magnet doing what they do. Five lives. Five quiets. Five men the world had filed under *crank* with — and this was the part that undid her, the part the whole book had been built to make her feel in her own hands — with *good reason*, four times out of five, maybe five times out of five, the dismissal so usually correct that it functioned like housekeeping, like sweeping, like the necessary daily work of a world that cannot afford to take every garage prophet at his word.

And underneath the correct dismissals: a man who would not say a true number. A girl who recognised him. A page left white.

She did not lift the camera. There was nothing here that taking would not break.

Jakobus had been outside with the Beast, doing the thing he did at the end of every book that none of them watched — running his hand along a made thing, reading what it was for, deferring the reading to whoever it belonged to and finding, at the end, that the Beast belonged to no one and so it was alright to listen to her himself. He came in with the cold on his coat and the patience of a man who has never once in twenty years been the lead and has never once wanted to be.

He picked up the last envelope. The Pogue patents, the originals-of-the-copies, one-seven-five-zero-three-five-four and its three brothers, the four real things, the documentable floor on which the whole descending house had stood. He weighed the envelope in his hand the way he weighed everything, asking it the only question he ever asked.

What is it for.

A vaporiser to claw back unburned fuel. A water map. A lunar almanac. A key. A made thing always made *for* something, grounded and physical and re-opening, every time, the door the debunk had shut. The patent in his hand was for a real and modest miracle and a refused number and a quiet a man took to his grave, and Jakobus did not say any of that, because it was not his to say, and because

the saying would have been a taking too.

He crossed to the long flat drawer where the finished books went to sleep. He slid the envelope in among the others — Pogue at the front, where the floor always is. He pressed it level with two fingers.

He left the drawer open a hand's width.

Then he went to put the kettle on, because the toast was gone and somebody would want tea, and behind him the file lay in its drawer not closed, the last page white, the number unsaid, the door at exactly the width a door has to be for a question to keep breathing through it.

The freezer hummed. Outside the dog ran its line and barked at the dark and meant it.

Nobody answered.

Chapter 21 — Let the Quiet Stand

The museum was a converted grain office on the edge of a Manitoba town whose name Tomi had already half-forgotten, two rooms and a corridor that smelled of floor wax and old radiators. A hand-lettered sign by the door read LOCAL HISTORY & INDUSTRY, the ampersand drawn by someone who had run out of confidence halfway through. Outside, the prairie went flat and gold to a horizon that did not bother with hills. Inside, the heating ticked.

They had not meant to come back. That was the thing Tomi kept turning over as he stamped snow off his boots in the entryway. The chase was done — Yusuf had the receipts filed, the adversary had bought his silence somewhere far south of here and gone home a poorer kind of rich — and there was no node left to read. The box was just a box now. It had a home. They had driven four hours off the route to look at a box that had a home.

Naila went straight for the case, the way she always did, hands behind her back so she wouldn't touch the glass. Edith hung in the doorway with the camera down at her hip, not raised, the strap looped twice around her wrist. Frik stood very still by the radiator with the particular stillness Tomi had learned to read as *working*, the throat doing its quiet arithmetic.

The case held four things and a card.

The four things: a brass-bodied carburetor, green-black with age, its float bowl unscrewed and set beside it so you could see down into the works. A coil of copper line. A photostat of a patent drawing, the lines gone soft grey on the copy. And a single page of typescript

under non-reflective glass, a letter, the signature at the bottom too faint to read upside down.

The card said: *Charles Nelson Pogue (1897–1985), Winnipeg. Inventor. His vapour carburetor was the subject of much excitement in 1933.*

That was all. *Much excitement.* Tomi felt something in his chest do a small complicated thing at *much excitement* — at the discipline of it, the keeper who had stood at this case with a pen and decided that two words would carry what a thousand breathless ones had failed to.

“High-twenties,” Naila said, not loudly, to the glass. “Maybe. On a V8 that drank like the thirties drank. You bubble the air through the fuel, you heat the chamber off the exhaust, you get the mix finer than the old jets could, you claw back a little of what was going out the pipe unburned.” She tipped her head. “It’s *clever*. The float-bowl placement is clever.”

“It’s clever,” Tomi agreed.

“It is not two hundred miles a gallon.” She said it the way she said all the impossible numbers, with no scorn at all, the way you’d correct a child’s arithmetic — kindly, because the child had tried. “A gallon holds what it holds. Thirty-three and a bit kilowatt-hours. The carburetor only *mixes*. You can’t mix your way past what’s in the can.” She leaned a half-inch closer to the brass. “It does the modest thing well. The modest thing is real.”

Behind them, the floor creaked.

The keeper was a woman of perhaps seventy, in a cardigan the colour of weak tea, with reading glasses pushed up into grey hair and a name badge that said GWEN and, under that, VOLUNTEER. Tomi had spoken to her on the phone — she was the granddaughter of a man who had worked at the Pogue shop on Sherbrook Street, or had run errands for it as a boy, the lineage thin and honest and exactly the kind the world doesn’t keep. She had been wary on the phone. He understood the wariness. She had met their kind before, the ones who came with a tripod and a thesis and left a man’s name spelled SUPPRESSED in capital letters.

“You found it, then,” Gwen said, and Tomi couldn’t tell if she meant

the museum or the case.

“We did.”

She came and stood with them at the glass, hands folded, proprietary in the good way, the way of a person who dusts a thing weekly. She did not perform for them. That was the first gift. She looked at the carburetor the way Naila looked at it, like a thing that had a job.

“People still come for the two hundred miles a gallon,” she said. “Once a month, maybe. A fellow last spring wanted to know where the suppressed plans were kept.” A small dry sound that wasn’t quite a laugh. “I told him I keep them in the float bowl, and would he like a coffee.”

Edith made a sound — surprised, warm. The camera stayed at her hip.

“There were no suppressed plans,” Naila said. “There are four patents. They’re on the internet. Anybody can read them. That’s —” and here her voice did the thing it did when the world had handed her another hole where a number should be, a flatness with grief inside it, “ — that’s the part nobody likes. There’s no vault. There’s just the patents, and the press, and then the editor took it back.”

“Pepper,” Gwen said.

“Pepper.” Naila nodded, almost grateful to be met. “*Canadian Automotive Trade*. He ran the two-hundred-mile story and then he called his own report *entirely misleading*. The man who *printed* it. And then in fifty-three—”

“And then Charlie said he never claimed two hundred.” Gwen finished it gently, took it out of Naila’s hands the way you take a hot dish from someone, because it was hers to hold. “Nor even half that, he said. My granddad was at that — well, not at it, he heard it after. Charlie wouldn’t say the real number. They asked him straight and he wouldn’t say it.” She looked at the brass. “Took it to the grave. Eighty-five he died. Made oil filters, the last while. Good ones, I’m told.”

The radiator ticked. Frik exhaled through his nose, slow, the stillness holding.

There it was — the hole, the only real one, sitting in a converted grain

office under non-reflective glass. Not *the oil companies hid it*. Not *they killed the genius*. Just a man who had once, briefly, been the most exciting thing on the prairie, and who had then, for thirty-two years, been asked the one question and had declined, every time, to answer it. The number was in him and he had walked it out the door and let the door close.

Tomi had spent a lot of the last weeks wanting that number. He could admit it now, standing here. He'd wanted it the way you want the last word of a sentence someone died halfway through. *High-twenties* was Naila's honest guess and it was probably right and it was not the same as knowing, and Pogue had known, and Pogue had said nothing, and the nothing was the truest thing the man ever left behind.

The bell over the front door went.

A woman came in stamping snow, towing a kid of maybe eight by the cuff of his coat, the kid already gone limp with the boredom of being somewhere educational. "Five minutes," the mother told him, "and then the pie place, I promise," and went to read the panel about the grain elevator with the air of a person discharging a duty.

The kid drifted. Kids drift to the shiny thing. He fetched up against the glass case at about Naila's elbow and looked in at the brass and said, to nobody, to everybody, "What's that funny carburetor?"

Tomi watched it happen.

He watched Naila's mouth open — saw the whole true, careful, devastating paragraph load behind her teeth, the thirty-three kilowatt-hours and the exhaust heat and the float-bowl cleverness and the *no, not two hundred, never two hundred*. He watched Edith's hand twitch toward the camera and stop. He watched himself want to be the one who answered, want it badly, the old hunger to be useful in a room by knowing the thing.

And he watched all three of them step back.

It was not decided out loud. There was no glance that passed, no signal. It was the kind of thing the van had taught them somewhere around the second year, the choreography of *not yours* — Naila's mouth closing, Edith's weight shifting back onto her heels, Tomi finding the radiator suddenly interesting. Frik did not move because Frik had never moved. They peeled their attention off the glass and gave

the case to the boy and the boy to Gwen, who came around the corner of the display with her hands still folded and crouched, with the small grunt of seventy, to be at his height.

"That," Gwen said, "belonged to a man named Charlie Pogue. He was a mechanic right here, in Winnipeg, a long time ago. He built it himself in a shop with his own hands."

"What's it do?"

"It mixes the gasoline with air so the engine can burn it. Every car has one — well, had one, your mum's car does it with computers now. But Charlie's was special. Charlie heated his up with the engine's own hot breath, the exhaust, so the gasoline turned to a sort of fog before it went in. Finer. Cleaner." She tapped the glass, once, over the unscrewed float bowl. "So the engine wasted a little less."

"My grandpa said it could go two hundred miles on one gallon," the kid announced, the way kids deliver the legend, head-on, delighted.

Tomi held his breath. He didn't know he was going to until he was.

"A lot of people said that," Gwen said. "It was in all the papers. It was very exciting." She didn't sneer. She didn't wink, either. She just laid it down level, the way you'd lay down a heavy thing you'd carried a long way. "But it couldn't, not really. There's only so much go in a gallon of gas, the way there's only so much water in a glass. You can be careful with it. You can not spill any. But you can't pour out more than you put in."

The kid considered this with the frown of a person being handed, for possibly the first time, the shape of a real limit in the world.

"So it didn't work?"

"It worked," Gwen said. "It just didn't do the *magic* thing. It did a smaller, real thing. It made an old thirsty engine a little less thirsty. That's a good thing to build. That's a hard thing to build." She straightened up, hand on her own knee. "And here's the part I like best, and you can decide for yourself about it, because nobody knows. When Charlie was an old man, people kept asking him: *how good was it, really, Charlie? What was the real number?* And do you know what he did?"

The kid shook his head. So, fractionally, did Tomi.

“He never told them.” Gwen smiled, and it was not a mysterious smile, it was a fond one, the smile you keep for a difficult relative you have decided to love anyway. “Not once. Right to the end. He took the real number with him.” She looked down at the boy. “What do you suppose it was?”

“A million,” the kid said immediately.

“Could be,” said Gwen, who knew exactly what it could not be, and let the boy have his million the way you let a child have a coin to throw in a fountain — not because the wish will come true, but because the throwing is the point.

“*Million*,” Frik said, from over by the radiator. Quiet, involuntary, the throat letting it out at the exact wrong-right moment — not a mockery, the opposite, an echo, the word arriving in his mouth the way it had arrived in the boy’s, undefended. The kid turned and grinned at him like he’d found an ally. Frik’s ears went red. He looked at the floor. Nobody laughed at him, which was its own old quiet kindness, and the mother called “Two minutes!” from the grain elevator, and the moment folded itself up.

The kid pressed one fingerprint onto the glass over the brass, right where it would madden Gwen later, and was towed out into the cold toward the promised pie, and the bell rang, and the museum was just the five of them and a volunteer and a box that had a home.

Gwen got a tissue from her cardigan sleeve and wiped the fingerprint off without a word.

“You didn’t help me,” she said, not turning around. There was something in it that wasn’t quite a complaint.

“It was yours to tell,” Tomi said.

She finished the glass. Folded the tissue. Looked at the four things in the case — the brass, the copper, the soft grey drawing, the letter no one could read upside down — and then at the people who had driven four hours to look at them and had then stood back and let her hand a million dollars to an eight-year-old.

“He’d have liked that,” she said. “Charlie. He’d have liked that better than the documentary.” She put the glasses down off her hair and onto her nose, which was how Tomi knew the visit was over and she

was a volunteer again, with panels to dust. “Drive safe. The 16’s bad past Portage.”

They didn’t talk much for the first while in the Beast. The heater fought the prairie cold and mostly lost. Edith had the camera in her lap, still off, thumb moving over the housing the way it did when she was deciding not to look at something. Yusuf drove. Naila sat with her forehead against the cold window and her breath fogging a small grey patch she did not wipe.

“High-twenties,” she said, eventually, to the glass, to nobody. Not a question. Closing it the only honest way it closed, which was not closed at all.

Nobody answered her. That was the answer.

Outside, the light was going the way prairie light goes, all at once and everywhere, the gold draining off the snow until the fields were just a held breath under a darkening sky. Somewhere behind them in a converted grain office a box sat in a case under a card that said *much excitement*, tended by a woman who would never sensationalise it and would never quite let it go, and the one number that would have settled everything sat where it had sat for forty years — inside a man who had made good oil filters and said nothing, and was gone, and had taken the nothing with him.

Tomi found he didn’t want it anymore. The number. Or — he wanted it the way you want the dead to come back, which is to say always and not really and not in a way you’d trade the love for.

Frik, in the back, had stopped being still. He was humming something tuneless and content, the danger past, and Edith had finally lifted the camera, not to her eye but just up, into her two hands, turning it over, looking at the dark glass of the lens like she might find the day already in there, waiting to be a thing she could carry.

“Pie place was closed,” Yusuf said, from the front, aggrieved.

“It was not closed.”

“It was *closing*.”

“That’s a different—”

The argument lifted up off the floor of the van and filled it, warm and stupid and theirs, and the road went on flat and gold-going-grey toward the next town, the next box, the next quiet someone had been glad of — and the question Charlie Pogue had carried out the door rode along with them in the cold, unanswered, unanswerable, and, for the length of that drive at least, enough.

Chapter 22 — Real / Debated / Invented: Notes and Gazetteer

Here is the part where we turn the lights on.

Every book like this one owes you a moment at the end where the magician shows you the hands. Not because the trick was a lie — most of what's in here is documented to the hilt, and you can hold the documents — but because the whole point of *The Why Files* is that **you** decide, and you can't decide honestly if you don't know which bricks are load-bearing and which are set dressing. So: the wall. Knock on every brick. We'll tell you which ones ring hollow.

We're sorting into three bins. **Real** — you can check it tonight, from your couch, on a free public database. **Debated** — a real claim with real evidence on both sides and no clean bottom. **Invented** — the people we made up to carry the true things, because the true things mostly happened to people who are dead and can't sit in a van with us.

Let's go in the order the book did. Down the ladder.

POGUE

Real. The patents. Say them out loud, because the fringe internet has been mangling one of them for forty years and we're not going to add to the pile. **US 1,750,354. US 1,938,497. US 1,997,497.**

US 2,026,798. Filed and granted between 1930 and 1936, in the name of **Charles Nelson Pogue**, of Winnipeg, Manitoba. They are on Google Patents right now. They describe, in dry patent-ese, an exhaust-heated vaporizing carburetor — air bubbled up through liquid gasoline, atomizing nozzles, a heating chamber warmed by the engine’s own waste exhaust. Real engineering. Real drawings. A real man’s real name on the line where you sign.

☒ The number you’ll see screamed across a thousand “they-buried-it” forums is **1,759,354**. That patent is not Pogue’s. Somebody fat-fingered a digit in roughly 1978 and the typo has been breeding ever since, the way typos do when nobody checks. It’s 1,750,354. We are tediously insistent about this because the whole disease of the genre is people repeating a number nobody looked up. Don’t be a vector.

Real. Charles Pogue lived (1897–1985). He was a self-taught mechanic. He later, by every account, made his living in oil filters — which is a quietly devastating fact and we let it be quiet.

Debated. The mileage. The 200 miles per gallon. A Ford V8, Winnipeg, 1933; a Winnipeg-to-Vancouver run “on 14.5 gallons.” Here’s the thing the believers and the debunkers both skate past: **the originating editor retracted it**. R. Pepper, of *Canadian Automotive Trade*, the man whose magazine put the number into the world, later called the report “**entirely misleading.**” That’s not a skeptic forty years on. That’s the source.

And then 1953: Pogue himself, alive, asked directly, denied claiming 200 mpg “or even half that.” A hundred would have made him the most important man of the century. He declined a hundred. He declined fifty.

And he refused to say what the real number was.

That refusal is real. It is in the record. It is the hole, and it does not close, and we did not close it for you. Naila is right that the physics forbids 200 — a gallon of gasoline holds a fixed amount of energy, roughly 33.4 kilowatt-hours, and a carburetor only *mixes*; it doesn’t add. Modern engines already burn north of 97% of the fuel; there is no factor-of-eight hiding in the unburnt 3%. The miracle is impossible. We say so in the book and we say so here.

The man’s silence is a separate fact from the machine’s failure. We

kept them separate. You should too.

Invented. The keeper. There is no granddaughter with a folder, no named estate solicitor, no specific prairie-museum volunteer in this book who is a real person. We built a keeper to hold the real papers, because the real papers exist in the world without a single guardian we could honestly point a camera at, and the book needed a hand to put them in. The patents are real. The hand is ours. The break-ins and threats Pogue is said to have described in 1953 — uncorroborated, in his mouth, never the narrator's. We let a character say it. We did not let the book believe it.

ZPE / THE CASIMIR FORCE

Real, and this is the one that can take any punch a skeptic throws. The vacuum is not empty. This is not fringe; this is the bench. In 1948 Hendrik Casimir predicted that two uncharged metal plates, brought close together in a vacuum, would feel a tiny attractive force arising from the quantum fields between them. In 1997 **Steve Lamoreaux measured it** — to within about 5% of prediction. Later experiments tightened that to under 1%. The force is real. It has been weighed. You can find the papers in *Physical Review Letters* with your library card.

The hole the book leans on is the gap between that real measurement and the garage. The vacuum has energy in it. This is true. It does **not** follow that you can run a lightbulb off it, and the book says — flatly, no wink, no “and yet” — that you cannot. The “zero-point energy generator” sold to you online fails on conservation of energy, which is not a guideline. The same word, *zero-point*, sits on a Nobel-grade experiment and on a \$400 crystal pendant. That collision is the whole chapter. We did not invent it; physics named the real thing and the grift stole the name.

Invented. Nothing about the physics. Names of any present-day “free-energy” promoters depicted are ours unless stated; the device categories are real, the specific characters peddling them in-scene are composite.

KOZYREV

This is the one where you should sit up, because the dismissal-bin is where most of him belongs and the residue-bin is heavier than you'd think.

Real. **Nikolai Aleksandrovich Kozyrev** (1908–1983), Soviet astronomer. The Gulag is real. He was arrested in 1936 and spent roughly a decade in the camps; the conditions, the cold, the survival are documented in Soviet biographical record and in his own later accounts. The man was buried by the State, literally, for years, and dug himself back out into astronomy.

Real. The gold medal. The International Academy of Astronautics awarded him a gold medal. There is a **crater on the Moon named Kozyrev**, on the far side. The IAU does not name craters after cranks as a courtesy.

Debated — and this is the famous one. On the night of **3 November 1958**, Kozyrev, at the Crimean Astrophysical Observatory, pointed a spectrograph at the lunar crater **Alphonsus** and recorded what he interpreted as **gas emission — the Moon, in his reading, breathing**. A transient outgassing event on a body everyone had filed as geologically dead. The spectrogram exists. The interpretation was fought over for decades. Some took it seriously enough that it fed into later study of lunar transient phenomena. Others read the plate differently. We did not adjudicate it. We are not qualified to and neither, frankly, is anyone with a strong opinion and no plate.

Fringe, labelled fringe. Kozyrev's "torsion field" / "time has a current" theory — his second life, the one he spent trying to prove time was a physical flow you could detect with a gyroscope — is **unreplicated** and sits outside accepted physics. The Novosibirsk parapsychology that orbited it: unreplicated. We let the grief drive the man; we did not let the theory become true in the narration. A real genius the State really buried, who really caught something on a plate in 1958, and then spent the rest of his life reaching for a thing that didn't hold. All four of those clauses are real and they live in tension and the tension is the man.

Invented. The Siberian institute with the heating off, as a scene — composite. The keeper who walks the crew through his papers — ours.

TACHYON

Real. The word *tachyon* is a real term for a hypothetical faster-than-light particle. Here's the lovely true thing: **the Higgs field, in the standard mathematics, is described as starting in a "tachyonic" state** — that's real theoretical physics vocabulary, not woo. The word has a clean home.

Real. The **OPERA experiment**, 2011: a measurement that appeared to show neutrinos travelling faster than light. It made headlines. And then they found it — **a loose fiber-optic cable**. A bad connection. The faster-than-light neutrino was a plug not seated all the way. OPERA's own team ran it down and announced it. That is science working exactly as advertised, and it is also the perfect, almost unbearable parable for this whole book: the most exciting result of the decade was a cable.

Fringe, labelled fringe, with feeling. "Tachyon energy" products — tachyonized water, tachyon pendants, tachyon healing chambers — have **no mechanism and no evidence**. The word was laundered out of physics and into the wellness aisle. Yusuf knows this from the inside because, in our story, Yusuf sold it. That's invented. The word-theft is real.

Invented. Yusuf's specific history; the specific outfit; the chase logistics.

SEARL

The bottom of the ladder. The hardest chapter. The place where the book had to decide what it was for.

Real. **John Roy Robert Searl** existed. The Searl Effect Generator

— the disc that was supposed to fly and make free power — is, on the merits, **perpetual motion**, which violates conservation of energy, which means it is not “fringe-but-open,” it is simply wrong, and the book says so.

Real. The conviction. This is the documented fact we will not soften: Searl was **convicted** in the early 1980s — reported in *The Times* (1982/83) — in connection with **abstracting electricity**, drawing power off the mains illegally. The man who said he had built a machine that makes electricity for free was found to have been **stealing it from the grid**. That is real and it is in the newspaper of record and it is the saddest sentence in this book.

Debated — barely. Whether the disc ever flew. There is **no photograph** of it flying that survives scrutiny. The engineer **Heerfordt**, who looked hard, found **not one witness** to a flight that held up. The book does not pretend otherwise.

Invented — and here we owe you the most honesty. The keeper at Searl is the most fictional person in the book. We built someone to love him, because the real record gives you a conviction and a non-flying disc and a man who went quiet, and it does not give you the thing the chapter needed, which was tenderness. We are telling you plainly: that tenderness is constructed. The conviction is not. We did not invent a single exonerating fact. We invented a person who could sit in the room with the un-exonerated ones.

JAKOBUS

Invented. The road, never the lead. You know this.

A SHORT GAZETTEER, FOR THE RABBIT-HOLERS

- **Google Patents.** Free. Type in the four Pogue numbers. Read the actual claims. Notice how modest the real device is.

- **Lamoreaux, S.K., *Physical Review Letters* (1997)** — the Casimir measurement. And the follow-ups under 1%.
 - **The IAU Gazetteer of Planetary Nomenclature** — search “Kozyrev.” It’s there, on the far side.
 - **The OPERA collaboration’s own retraction** — the cable. Read how cleanly they ate it. That’s the good stuff.
 - **Snopes; Straight Dope; the relevant Wikipedia talk pages** on Pogue — for the “oil companies buried it” claim. They concur: **no evidence**. We concur with them.
 - **The 1953 Pogue interview** for the refusal of the number. The hole, in his own voice.
-

So. The quiet is real.

That’s the one thing in this whole book that lands in the **Real** bin without an asterisk. Not the disc — the disc never flew. Not the 200 miles — the editor took it back. Not the breathing Moon, not the current in time, not the energy in the empty jar you can buy with a pendant.

The quiet. The moment each of these five men stopped talking. Pogue, who held real patents and refused, to his death, to say one number, and went and made oil filters. Kozyrev, behind the Academy’s verdict. Searl, behind a prison door. Five men the world filed under *crank* with a tidy little click — and the world was mostly right, which is exactly what made the filing feel so good, and exactly what makes it so hard to see the once or twice it might have buried something with the body.

We are not going to tell you it buried something. We checked. The machines don’t work. We told you that on page one and we’re telling you here.

But the silence — somebody, sometimes, was glad of the silence. And we are not going to tell you who, or whether it matters, or what it was for.

That part’s yours.

Hecklefish has spent the whole book waiting to say the machines don’t fly. They don’t. He’s right.

He's also, for once, got nothing to say about the men.

Neither do we.

That's the door. Mind the step on your way out.

With thanks

This book was written in the spirit of a curiosity I learned from someone else: the discipline of taking the official story seriously, telling it straight and with real respect, finding the one honest place it does not quite close — and then trusting the reader to decide.

With genuine thanks to AJ Gentile, whose work lit that way of looking for me. The stories here are my own, the sites and the people invented or freshly drawn; what I borrowed is only the *posture* — eyes open, sneer off, wonder intact. I am grateful for it, and I hope I have done it honour.



Klaus

Custos, non Conditor